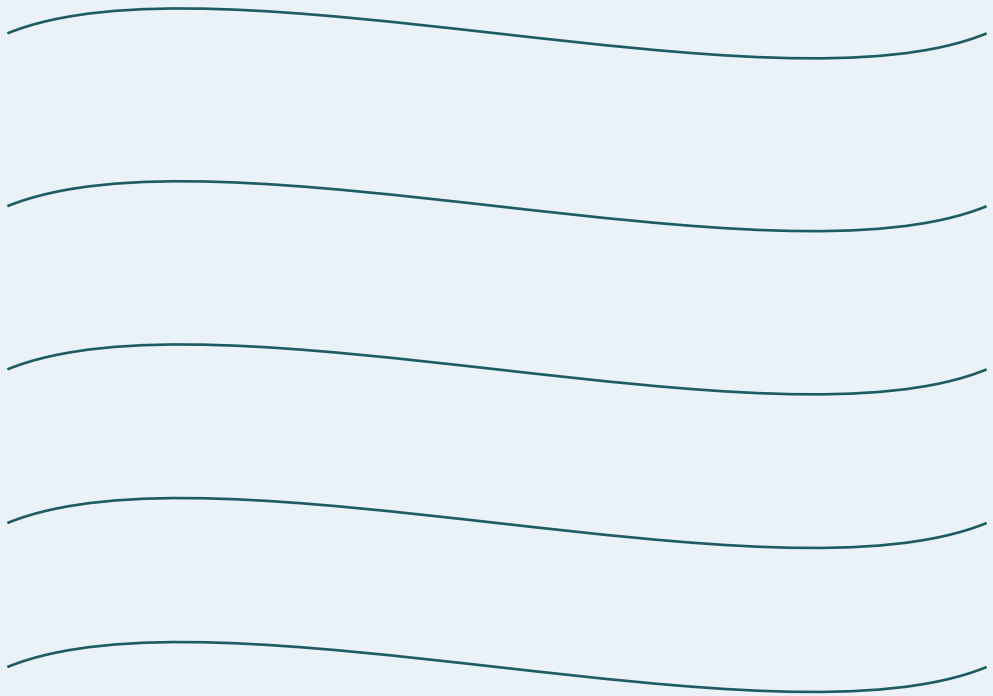

Transseries for Mere Mortals

A step-by-step tutorial from ordinary
asymptotic series to exponentials beyond all orders



Formal logarithmic-exponential transseries, differential equations, Borel summation, Stokes phenomena, resurgence, Hardy fields, and surreal numbers

Prepared as a self-contained mathematical tutorial

This tutorial is an original synthesis based on the source materials supplied with the request and on the literature listed in the bibliography. Its principal expository sources include Edgar's introductory papers, van der Hoeven's treatment of transseries and real differential algebra, the asymptotic differential algebra of Aschenbrenner–van den Dries–van der Hoeven, and the resurgent-transseries literature of Costin, Dorigoni, and Aniceto–Başar–Schiappa.

The word *transseries* is used in several overlapping traditions. This book states explicitly which formal or analytic structure is being used at each point. It also distinguishes formal identities from claims about actual functions.

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Preface

Power series are among the first genuinely magical objects in mathematics. A finite formula can be replaced by an infinite list of coefficients; differentiation and integration become mechanical; and local behavior becomes algebra. But ordinary power series are only the first rung of a much taller ladder.

Suppose that a function has an expansion in inverse powers of a large variable,

$$f(x) \sim \frac{a_0}{x} + \frac{a_1}{x^2} + \frac{a_2}{x^3} + \cdots, \quad x \rightarrow +\infty.$$

The expansion may diverge. Even when it is useful, it cannot detect a term such as e^{-x} , because e^{-x} is smaller than every power x^{-N} . Two different functions can therefore have exactly the same power-series asymptotics. If a differential equation has a free constant multiplying e^{-x} , a power series literally loses one degree of freedom.

A transseries repairs this loss by allowing a hierarchy of scales:

$$x^{-1}, x^{-2}, \dots; \quad e^{-x}, e^{-x}x^{-1}, \dots; \quad e^{-2x}, \dots; \quad \log x, e^{\sqrt{x}}, e^{-e^x}, \dots$$

The result is not merely a longer series. It is a formal language with an ordering, arithmetic, exponentials, logarithms, differentiation, integration, and composition. In analytic problems it is also the natural bookkeeping device for Borel summation, Stokes jumps, and resurgence.

This book has three goals.

1. Build the central ideas from calculus and ordinary asymptotic expansions, with no assumption that the reader already knows ordered fields, valuations, or resurgence.
2. Show enough of the formal construction that the algebra does not look like hand-waving with infinitely many symbols.
3. Work through actual calculations: factorial divergence, Lambert's W function, formal integration, algebraic equations, Riccati equations, WKB, Borel transforms, and Stokes discontinuities.

The intended reader is a mathematically mature undergraduate who has completed a standard calculus sequence and has seen elementary differential equations and power series. Complex analysis is helpful in the chapters on Borel summation, but the necessary contour ideas are reviewed informally.

How to read the book

Chapters 1–3 are the shortest route to the main motivation. Chapters 4–9 build the formal field. Chapters 11–15 explain its calculus. Chapters 17–20 apply it to equations. Chapters 22–24 explain analytic reconstruction and resurgence. The final part places transseries among Hardy fields, surreal numbers, and model theory.

A reader mainly interested in applications may read Chapters 1–3, skim Chapters 4–9, and continue at Chapter 17. A reader interested in foundations should follow the order given.

A common trap

A formal transseries is not automatically the expansion of a function, and an asymptotic expansion is not automatically convergent. Throughout the book, a hat such as $\hat{y}$ often emphasizes a formal object. Symbols without hats may denote actual functions when the analytic setting has been specified.

Notation and standing conventions

Unless stated otherwise, $x \rightarrow +\infty$. We compare positive quantities by eventual size. The notation

$$f \prec g \quad \text{means} \quad \frac{f}{g} \longrightarrow 0,$$

and $f \asymp g$ means that f/g tends to a nonzero finite constant or, in a purely formal setting, that f and g have the same leading monomial. The familiar notation $f \sim g$ is reserved for asymptotic equivalence $f/g \rightarrow 1$ when confusion is possible.

For a transseries T , $\text{supp } T$ is its support, $\text{lm}(T)$ its leading monomial, and $\text{lc}(T)$ the coefficient of that monomial. We write $T \prec 1$ for an infinitesimal transseries and $T \succ 1$ for an infinitely large one.

The real logarithm is used on positive real arguments. In complex-analytic chapters, Log denotes a chosen branch. Formal exponentials and logarithms are algebraic operations defined by canonical decompositions; they do not require numerical convergence.

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Part I

Why ordinary series are not enough

Scales, orders, and asymptotic expansions

1.1 An expansion needs a scale

When we write a Taylor series near $x = 0$, the scale is understood:

$$1, x, x^2, x^3, \dots$$

For asymptotics at infinity it is often more natural to use

$$1, x^{-1}, x^{-2}, x^{-3}, \dots$$

The important feature is not that these objects are powers. It is that each one is much smaller than the preceding one.

Definition 1.1 (Asymptotic scale). A sequence $\phi_0, \phi_1, \phi_2, \dots$ is an *asymptotic scale* as $x \rightarrow +\infty$ if

$$\phi_{n+1}(x) \prec \phi_n(x) \quad (n = 0, 1, 2, \dots).$$

Equivalently, $\phi_{n+1}/\phi_n \rightarrow 0$.

Examples include

$$1, \frac{1}{x}, \frac{1}{x^2}, \dots; \quad x, \log x, \log \log x, 1, \frac{1}{\log \log x}, \dots;$$

and

$$1, e^{-x}, e^{-2x}, e^{-3x}, \dots$$

The scales can be mixed. For instance,

$$1, \frac{1}{x}, \frac{1}{x^2}, \dots, e^{-x}, \frac{e^{-x}}{x}, \frac{e^{-x}}{x^2}, \dots, e^{-2x}, \dots$$

is ordered by decreasing size, although it is not naturally indexed by a single integer without some bookkeeping.

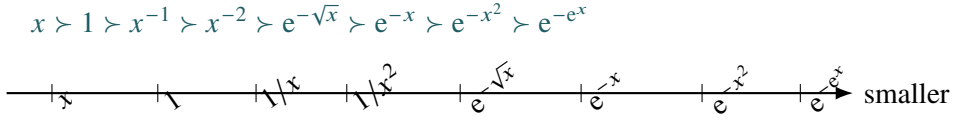


Figure 1.1: A tiny fragment of the hierarchy of asymptotic scales. There is no last scale.

1.2 Poincaré asymptotic expansions

Definition 1.2 (Poincaré expansion). Let $\phi_0, \phi_1, \dots$ be an asymptotic scale. We write

$$f(x) \sim \sum_{n=0}^{\infty} a_n \phi_n(x)$$

if, for every fixed $N \geq 0$,

$$f(x) - \sum_{n=0}^N a_n \phi_n(x) = o(\phi_N(x)) \quad (x \rightarrow +\infty).$$

The infinite sum is notation for the whole sequence of finite remainder statements. It need not converge for any x .

There are two ideas to absorb.

First, N is fixed before $x \rightarrow \infty$. We are not asserting that an actual infinite numerical sum equals $f(x)$. Second, a later term improves the error relative to the preceding scale, even when the coefficient sequence grows so quickly that the series diverges.

Example 1.1. For $x > 0$,

$$\frac{1}{x+1} = \frac{1}{x} \frac{1}{1+x^{-1}}.$$

Expanding the geometric factor gives

$$\frac{1}{x+1} \sim \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3} - \dots.$$

Here the series actually converges whenever $x > 1$, but convergence is stronger than asymptoticity and is not part of the definition.

Example 1.2. Repeated integration by parts gives

$$\int_x^\infty e^{-t} t^{-1/2} dt \sim e^{-x} x^{-1/2} \left(1 - \frac{1}{2x} + \frac{3}{4x^2} - \frac{15}{8x^3} + \dots \right).$$

This is an expansion on the mixed scale $e^{-x} x^{-1/2}, e^{-x} x^{-3/2}, \dots$. The exponential factor is part of every monomial in the scale.

1.3 How coefficients are extracted

If the scale is fixed, the coefficients are determined one at a time. Suppose

$$f \sim a_0\phi_0 + a_1\phi_1 + a_2\phi_2 + \cdots.$$

Then

$$a_0 = \lim_{x \rightarrow \infty} \frac{f(x)}{\phi_0(x)},$$

provided the normalization of ϕ_0 makes this limit meaningful. After subtracting the first term,

$$a_1 = \lim_{x \rightarrow \infty} \frac{f(x) - a_0\phi_0(x)}{\phi_1(x)},$$

and so on. This is the asymptotic analogue of extracting Taylor coefficients, but it uses successive limits rather than derivatives at a point.

Proposition 1.1 (Uniqueness relative to a scale). *If $\phi_{n+1} \prec \phi_n$ and*

$$f \sim \sum_{n \geq 0} a_n \phi_n \quad \text{and} \quad f \sim \sum_{n \geq 0} b_n \phi_n,$$

then $a_n = b_n$ for every n .

Proof. If the coefficient lists differ, let k be the first index with $a_k \neq b_k$. Subtract the two remainder statements through index k . The difference is

$$(a_k - b_k)\phi_k = o(\phi_k),$$

which is impossible because $(a_k - b_k)\phi_k/\phi_k$ is a nonzero constant. $\square$

1.4 The invisible-function problem

Uniqueness of coefficients does *not* mean uniqueness of the function. For every integer $N \geq 0$,

$$e^{-x} = o(x^{-N}).$$

Therefore

$$0 \sim 0 + \frac{0}{x} + \frac{0}{x^2} + \cdots \quad \text{and} \quad e^{-x} \sim 0 + \frac{0}{x} + \frac{0}{x^2} + \cdots.$$

More generally, if

$$f(x) \sim \sum_{n \geq 0} a_n x^{-n},$$

then for every constant C ,

$$f(x) + Ce^{-x} \sim \sum_{n \geq 0} a_n x^{-n}.$$

No amount of inverse-power data distinguishes these functions.

Intuition

An ordinary inverse-power expansion is like a measuring instrument with a finite family of resolutions. An exponentially small quantity falls below every one of those resolutions. Transseries enlarge the instrument by adding new scales rather than pretending the invisible quantity is zero.

The same phenomenon repeats at every level. Relative to powers of e^{-x} , the term e^{-x^2} is beyond all orders:

$$e^{-x^2} = o(e^{-Nx}) \quad \text{for every fixed } N.$$

Relative to powers of e^{-x^2} , the term e^{-e^x} is beyond all orders again. There is no final universal list of scales.

1.5 Logarithms create another hierarchy

For every fixed $a > 0$ and $b \in \mathbb{R}$,

$$\log x \prec x^a, \quad (\log x)^b \prec x^a, \quad x^{-a} \prec (\log x)^{-b} \quad (b > 0).$$

Thus a more refined region of the scale hierarchy is

$$x \succ x^{1/2} \succ (\log x)^{100} \succ \log x \succ \log \log x \succ 1 \succ \frac{1}{\log \log x} \succ \frac{1}{\log x} \succ \frac{1}{x}.$$

The comparison can usually be made by taking logarithms. For positive f, g ,

$$f \prec g \iff \log f - \log g \longrightarrow -\infty.$$

For example,

$$\log \left(\frac{e^{\sqrt{x}}}{x^{1000}} \right) = \sqrt{x} - 1000 \log x \longrightarrow +\infty,$$

so $e^{\sqrt{x}} \succ x^{1000}$.

1.6 A practical comparison algorithm

For expressions built from positive powers, exponentials, and logarithms, the following informal procedure often works.

Algorithm 1.1 (Comparing two positive scales). To compare f and g as $x \rightarrow +\infty$:

1. Form the quotient f/g .
2. Simplify its logarithm.
3. If the logarithm tends to $+\infty$, then $f \succ g$; if it tends to $-\infty$, then $f \prec g$.
4. If it tends to a finite number, remove the corresponding constant factor and compare the next correction.

Worked calculation

Compare

$$f(x) = e^{x+\sqrt{x}}x^{-7} \quad \text{and} \quad g(x) = e^x x^{100}.$$

Their quotient is

$$\frac{f}{g} = e^{\sqrt{x}}x^{-107},$$

and

$$\log(f/g) = \sqrt{x} - 107 \log x \longrightarrow +\infty.$$

Therefore $f \succ g$. The apparently large factor x^{107} loses to the smaller exponential $e^{\sqrt{x}}$.

1.7 Exercises

Exercise 1.1. Arrange the following in decreasing order as $x \rightarrow +\infty$:

$$x^{-10}, \quad e^{-\sqrt{x}}, \quad (\log x)^{-3}, \quad e^{-x}, \quad x^{-1/100}, \quad e^{-x/\log x}.$$

Exercise 1.2. Show that if $f \sim \sum_{n \geq 0} a_n x^{-n}$, then $f + e^{-\sqrt{x}} \sim \sum_{n \geq 0} a_n x^{-n}$. Explain why the same conclusion need not hold after adding $(\log x)^{-1}$.

Exercise 1.3. Suppose $f(x) = \sqrt{x^2 + x}$. Obtain the first four nonzero terms of its expansion in descending powers of x as $x \rightarrow +\infty$.

Chapter summary

An asymptotic expansion is defined relative to an ordered scale. Its coefficients are unique, but the represented function is not: terms smaller than every monomial in the chosen scale are invisible. Exponentials and iterated logarithms force us to work with a much richer hierarchy than ordinary powers.

Factorial divergence and the missing constant

The simplest useful doorway into transseries is a first-order differential equation whose formal power expansion diverges and loses its arbitrary constant.

2.1 The Euler equation

Consider

$$y'(x) + y(x) = \frac{1}{x}, \quad x \rightarrow +\infty. \quad (2.1)$$

We first look for a formal inverse-power series

$$\widehat{y}_0(x) = \sum_{n=0}^{\infty} a_n x^{-n-1}. \quad (2.2)$$

Differentiating term by term gives

$$\widehat{y}'_0 = - \sum_{n=0}^{\infty} (n+1) a_n x^{-n-2} = - \sum_{n=1}^{\infty} n a_{n-1} x^{-n-1}.$$

Substitution into Equation (2.1) yields

$$a_0 x^{-1} + \sum_{n=1}^{\infty} (a_n - n a_{n-1}) x^{-n-1} = x^{-1}.$$

Hence

$$a_0 = 1, \quad a_n = n a_{n-1} \quad (n \geq 1),$$

so

$$\widehat{y}_0(x) = \frac{1}{x} + \frac{1}{x^2} + \frac{2!}{x^3} + \frac{3!}{x^4} + \frac{4!}{x^5} + \cdots = \sum_{n=0}^{\infty} \frac{n!}{x^{n+1}}. \quad (2.3)$$

2.2 The series diverges for every fixed x

The ratio of consecutive terms is

$$\frac{(n+1)!x^{-n-2}}{n!x^{-n-1}} = \frac{n+1}{x}.$$

For fixed x , this ratio eventually exceeds 1, so the terms do not even tend to zero. The series cannot converge.

That does not make it useless. For a large x , the first several terms decrease before the factorial growth wins. At $x = 20$, for example, the term sizes decrease until n is close to 19 or 20. Truncating near the smallest term gives an excellent approximation.

Definition 2.1 (Optimal truncation). For a divergent asymptotic series whose terms first decrease and then increase, *optimal truncation* means stopping at or near the term of smallest magnitude.

For Equation (2.3), the smallest term occurs near $n = x - 1$. Stirling's formula gives, for $n \approx x$,

$$\frac{n!}{x^{n+1}} \approx \frac{\sqrt{2\pi x}(x/e)^x}{x^{x+1}} = \sqrt{\frac{2\pi}{x}} e^{-x}.$$

The best attainable error from the inverse-power series is therefore on an exponentially small scale. The divergence is telling us where the next scale lives.

Intuition

Divergence is often information, not failure. The late terms of the power series grow in precisely the way needed to announce an omitted contribution of order e^{-x} .

2.3 The exact family of solutions

Multiplying Equation (2.1) by the integrating factor e^x gives

$$(e^x y)' = \frac{e^x}{x}.$$

On any interval not crossing $x = 0$,

$$y_C(x) = e^{-x} \left(C + \int_{x_0}^x \frac{e^t}{t} dt \right), \quad (2.4)$$

where C is arbitrary. Equivalently, after absorbing a constant into C ,

$$y_C(x) = e^{-x} \text{Ei}(x) + Ce^{-x}.$$

All members of this one-parameter family have the same formal expansion Equation (2.3), because

$$Ce^{-x} = o(x^{-N})$$

for every N .

The ordinary power series has forgotten the constant of integration. The smallest correction needed to restore it is exactly Ce^{-x} .

2.4 The first transseries solution

We now write

$$\widehat{y}(x; C) = \sum_{n=0}^{\infty} \frac{n!}{x^{n+1}} + Ce^{-x}. \quad (2.5)$$

This is a very small transseries: one divergent power-series sector plus one exponentially small sector. Nevertheless it already does three jobs that the power series alone cannot do:

1. It retains the arbitrary constant of the differential equation.
2. It records the natural scale of the optimally truncated error.
3. It anticipates the ambiguity that appears when the divergent series is Borel summed across its Stokes direction.

Because Equation (2.1) is linear, no powers C^2e^{-2x} , C^3e^{-3x} , $\dots$ are generated. Nonlinear equations usually produce an entire tower of such sectors.

2.5 Checking the formal identity

A formal identity is checked coefficientwise. Differentiate Equation (2.5):

$$\frac{d}{dx} (Ce^{-x}) = -Ce^{-x},$$

so this term cancels when $y' + y$ is formed. The factorial series telescopes:

$$\begin{aligned} \left(\sum_{n \geq 0} \frac{n!}{x^{n+1}} \right)' + \sum_{n \geq 0} \frac{n!}{x^{n+1}} &= - \sum_{n \geq 0} \frac{(n+1)!}{x^{n+2}} + \sum_{n \geq 0} \frac{n!}{x^{n+1}} \\ &= \frac{1}{x}. \end{aligned}$$

Thus the transseries solves the equation formally, with no analytic summation yet assumed.

2.6 Remainders by integration by parts

The factorial coefficients can also be derived from the exact integral. Formally integrate by parts:

$$\int^x \frac{e^t}{t} dt = \frac{e^x}{x} + \int^x \frac{e^t}{t^2} dt.$$

Repeating N times gives

$$\int^x \frac{e^t}{t} dt = e^x \sum_{n=0}^{N-1} \frac{n!}{x^{n+1}} + N! \int^x \frac{e^t}{t^{N+1}} dt + \text{constant}.$$

After multiplication by e^{-x} ,

$$y_C(x) = \sum_{n=0}^{N-1} \frac{n!}{x^{n+1}} + N!e^{-x} \int^x \frac{e^t}{t^{N+1}} dt + Ce^{-x}.$$

For fixed N , the integral remainder is $O(x^{-N-1})$, as required. When N grows with x , the same expression leads toward exponentially improved asymptotics, but that requires more care than ordinary Poincaré asymptotics.

2.7 A numerical feel for optimal truncation

At $x = 10$, the terms $n!/10^{n+1}$ decrease through approximately $n = 9$ and then grow. The partial sum through $n = 9$ is

$$0.1 + 0.01 + 0.002 + 0.0006 + \cdots + \frac{9!}{10^{10}},$$

and its least term has size about 3.6×10^{-5} . The exponential scale is

$$e^{-10} \approx 4.54 \times 10^{-5}.$$

The close agreement is not accidental; it is the Stirling estimate above.

A common trap

For a divergent asymptotic series, adding more terms after the least term generally makes the numerical approximation worse. “More terms” is not the same as “more accuracy.”

2.8 Exercises

Exercise 2.1. For the equation

$$y' - y = \frac{1}{x},$$

find a formal inverse-power solution and the exponentially small or large homogeneous correction. Which correction is small as $x \rightarrow +\infty$?

Exercise 2.2. Consider $y' + ay = x^{-1}$ with $a > 0$. Derive the formal coefficients and identify the missing homogeneous term.

Exercise 2.3. Use Stirling’s formula to show more carefully that the least term of $\sum n!x^{-n-1}$ is of order $x^{-1/2}e^{-x}$.

Chapter summary

The Euler equation produces a factorially divergent series. Its least term is exponentially small, and the exact differential equation has a free term Ce^{-x} invisible to all inverse powers. The smallest useful transseries already combines a perturbative sector with a beyond-all-orders sector.

What a transseries is—and is not

There is no single one-line definition that covers every object called a transseries in the literature. The word names a family of closely related formal expansions. Two traditions dominate this tutorial.

3.1 The logarithmic-exponential viewpoint

In the formal-algebraic tradition, a transseries is a generalized series whose monomials can involve powers, exponentials, and logarithms, arranged in a well-controlled dominance order. Typical examples look like

$$T(x) = e^{x^2 + \sqrt{x}} \left(1 + \frac{3}{x} + \frac{7 \log x}{x^2} + \dots \right) + 5x^\pi e^{-x} \left(1 - \frac{1}{\log x} + \dots \right) + e^{-e^x} + \dots$$

The aim is to form an ordered differential field closed under a large collection of operations:

$$+, -, \times, /, \exp, \log, \frac{d}{dx}, \int, \text{ and suitable composition.}$$

This is the setting developed from generalized power series and logarithmic-exponential series, and presented accessibly by Edgar [22, 23] and systematically by van der Hoeven [38].

3.2 The resurgent viewpoint

In singular perturbation theory and mathematical physics, one often writes a transseries in a small parameter z as

$$\widehat{F}(z; \sigma) = \sum_{k=0}^{\infty} \sigma^k e^{-kA/z} z^{k\beta} \sum_{n=0}^{\infty} a_{k,n} z^n. \quad (3.1)$$

Here:

- k labels nonperturbative sectors;
- A is an action or singulant;

- σ is a transseries parameter, often encoding a boundary condition;
- each inner power series may diverge;
- Borel singularities of one sector are related to other sectors.

The formal expression is only half the story. One studies directional Borel sums, their jumps across Stokes rays, and the web of large-order relations called *resurgence*. Costin’s work gives rigorous summability results for broad classes of differential equations [12, 13, 15]; modern pedagogical accounts include Dorigoni [19] and Aniceto–Başar–Schiappa [1].

3.3 How the viewpoints fit together

Both viewpoints organize many asymptotic scales in one formal object. Both need a rule ensuring that sums and products are meaningful. Both distinguish a leading part from successively smaller corrections. But their emphases differ.

	Formal logarithmic-exponential field	Resurgent transseries in analysis
Typical variable	$x \rightarrow +\infty$	$z \rightarrow 0$ in the complex plane
Typical monomials	$x^a (\log x)^b e^L$	$e^{-kA/z} z^\beta (\log z)^m$
Central structure	Ordered Hahn field, derivation, composition	Borel plane, Stokes automorphisms, alien derivatives
Main question	Which formal operations are well defined?	Which functions are reconstructed, and how do sums jump?
Convergence required?	re- No	Usually replaced by summability

A concrete transseries may belong to both worlds. For example,

$$\sum_{k \geq 0} C^k e^{-kx} \sum_{n \geq 0} a_{k,n} x^{-n}$$

is a logarithmic-exponential formal series in x and also has the sector form used in resurgence.

A common trap

Do not infer analytic facts from formal closure alone. A formal transseries can be differentiated and multiplied without being assigned an actual numerical value. Conversely, two different summation prescriptions may assign different functions to the same divergent formal sector along a Stokes direction.

3.4 A minimal mental model

Before constructing the full field, it is useful to picture a transseries as an ordered dictionary

$$\text{monomial} \mapsto \text{coefficient}.$$

For example,

$$T = 3x^2 - 7 + 4x^{-1} + 5e^{-x} + 2xe^{-2x}$$

corresponds to

$$x^2 \mapsto 3, \quad 1 \mapsto -7, \quad x^{-1} \mapsto 4, \quad e^{-x} \mapsto 5, \quad xe^{-2x} \mapsto 2.$$

The monomials are ordered

$$x^2 \succ 1 \succ x^{-1} \succ e^{-x} \succ xe^{-2x}.$$

The leading monomial is x^2 and the leading coefficient is 3.

What makes transseries nontrivial is that the dictionary may be infinite, the keys form a sophisticated multiplicative group, and operations such as $\exp(T)$ generate new keys. The support conditions in Chapters 5 and 6 prevent an infinite amount of algebra from accumulating at a single monomial.

3.5 A toy nonlinear transseries

Consider the logistic-type equation

$$y' = -y + y^2. \quad (3.2)$$

Near the zero solution as $x \rightarrow +\infty$, the linearized equation is $y' = -y$, suggesting a leading correction Ce^{-x} . Write

$$y = \sum_{k=1}^{\infty} a_k C^k e^{-kx}.$$

Then

$$y' = - \sum_{k \geq 1} k a_k C^k e^{-kx},$$

and

$$y^2 = \sum_{k \geq 2} \left(\sum_{i=1}^{k-1} a_i a_{k-i} \right) C^k e^{-kx}.$$

Equating the coefficient of $C^k e^{-kx}$ gives

$$(1-k)a_k = \sum_{i=1}^{k-1} a_i a_{k-i} \quad (k \geq 2),$$

while a_1 is a normalization; take $a_1 = 1$. The recurrence yields

$$a_2 = -1, \quad a_3 = 1, \quad a_4 = -1, \quad \dots,$$

so

$$\widehat{y} = Ce^{-x} - C^2 e^{-2x} + C^3 e^{-3x} - \dots. \quad (3.3)$$

In this unusually friendly example, the transseries converges geometrically when $|Ce^{-x}| < 1$ and sums to

$$y(x) = \frac{Ce^{-x}}{1 + Ce^{-x}}, \quad (3.4)$$

which indeed solves Equation (3.2).

This example displays the standard nonlinear mechanism: one exponentially small seed generates all multiples of its exponential scale. In harder problems each sector carries its own divergent power series.

3.6 What is normally excluded

The standard logarithmic-exponential field is enormous, but it is not the class of all imaginable formal expressions. Typical exclusions include:

- supports that contain infinitely increasing monomials, such as $1 + x + x^2 + x^3 + \cdots$ when $x \succ 1$;
- uncontrolled infinite nesting, such as an infinite tower $\exp(\exp(\exp(\cdots)))$;
- infinite descent through logarithmic depth without a support discipline;
- arbitrary rearrangements of formal sums.

These are not aesthetic restrictions. Without them, the coefficient of a product monomial may receive infinitely many unrelated contributions, a “largest term” may not exist, and recursive definitions may fail to stabilize.

3.7 The promise of the formal theory

The chapters ahead will justify the following informal slogan.

Checkpoint

A logarithmic-exponential transseries behaves like a power series whose powers have been replaced by a well-ordered universe of asymptotic monomials. The universe is designed so that leading terms exist and the familiar algebraic operations can be extended recursively.

3.8 Exercises

Exercise 3.1. Verify directly that Equation (3.4) solves Equation (3.2). Expand it as a geometric series and identify the domain in which the numerical series converges.

Exercise 3.2. For

$$T = e^x + x^{10}e^{\sqrt{x}} + x^{-1} + e^{-x^2},$$

arrange the displayed monomials in decreasing order and identify the leading monomial.

Exercise 3.3. Explain in words why a nonlinear term y^3 would generate sectors $e^{-3x}, e^{-5x}, \dots$ from a leading seed e^{-x} if symmetry prevents even powers.

Chapter summary

“Transseries” names overlapping formal and analytic frameworks. Logarithmic-exponential transseries emphasize an ordered differential field of generalized monomials; resurgent transseries emphasize divergent sectors, Borel singularities, and Stokes jumps. Both extend asymptotic bookkeeping beyond ordinary powers.

Part II

The formal logarithmic-exponential field

Transmonomials and dominance

Ordinary Laurent series use monomials x^n . Transseries replace this one-dimensional list by a multiplicative group containing monomials such as

$$x^a, \quad (\log x)^b, \quad e^x, \quad e^{x+\sqrt{x}}, \quad e^{-x^2}(\log x)^7,$$

and many more. Before allowing infinite sums, we must understand the keys in our ordered dictionary.

4.1 Ordered multiplicative groups

Definition 4.1 (Ordered abelian group of monomials). A *monomial group* $\mathfrak{M}$ is an abelian group written multiplicatively, equipped with a total order $<$ that is compatible with multiplication:

$$m < n \implies mp < np \quad (p \in \mathfrak{M}).$$

Its identity is the constant monomial 1.

For asymptotics at $+\infty$, the order is chosen to match eventual growth. We write

$$m \prec n$$

when m is asymptotically smaller than n . Thus

$$e^{-x} \prec x^{-100} \prec 1 \prec \log x \prec x^{1/10} \prec e^{\sqrt{x}} \prec e^x.$$

Multiplicative compatibility says that comparisons survive multiplication by a positive monomial. For example,

$$e^{-x} \prec x^{-100} \implies x^7 e^{-x} \prec x^{-93}.$$

Every monomial has an inverse. If $m \succ 1$, then $m^{-1} \prec 1$. The group language packages this simple reciprocal relation.

4.2 Power monomials as the first model

Take

$$\mathfrak{M}_0 = \{x^a : a \in \mathbb{R}\}.$$

Multiplication corresponds to addition of exponents:

$$x^a x^b = x^{a+b}, \quad (x^a)^{-1} = x^{-a}.$$

The dominance order is

$$x^a \prec x^b \iff a < b.$$

Thus the map $a \mapsto x^a$ converts the ordered additive group $(\mathbb{R}, +, <)$ into an ordered multiplicative group.

When exponentials are introduced, their exponents themselves become ordered objects. If $L_1 - L_2 \succ 1$ and is positive, then

$$e^{L_1} \succ e^{L_2},$$

because

$$\frac{e^{L_1}}{e^{L_2}} = e^{L_1 - L_2} \succ 1.$$

This recursive use of order is the central architectural idea of logarithmic-exponential transseries.

4.3 Coefficient times monomial

A term has the form am , where $a \in \mathbb{R} \setminus \{0\}$ and $m \in \mathfrak{M}$. The coefficient does not change the monomial's dominance class. For instance,

$$10^{100}e^{-x} \prec 10^{-100}x^{-100}$$

as $x \rightarrow +\infty$, despite the enormous finite constants.

Definition 4.2 (Leading monomial and leading coefficient). For a nonzero formal sum T whose support has a largest monomial, the *leading monomial* $\text{lm}(T)$ is that largest monomial. The *leading coefficient* $\text{lc}(T)$ is its coefficient. The product

$$\text{lc}(T) \text{lm}(T)$$

is the leading term.

For

$$T = -7e^x + x^{100}e^{\sqrt{x}} + 3x^{-1} + e^{-x},$$

we have

$$\text{lm}(T) = e^x, \quad \text{lc}(T) = -7.$$

The sign of T is the sign of its leading coefficient. All smaller terms are incapable of overturning it formally, just as a lower-degree term cannot change the sign of a polynomial for sufficiently large positive x .

4.4 Dominance, comparability, and asymptotic equivalence

Three relations are useful.

Definition 4.3. For nonzero transseries S, T :

$$\begin{aligned} S \prec T &\iff \text{lm}(S) \prec \text{lm}(T), \\ S \asymp T &\iff \text{lm}(S) = \text{lm}(T), \\ S \sim T &\iff \text{lm}(S - T) \prec \text{lm}(T) \text{ and } \text{lc}(S) = \text{lc}(T). \end{aligned}$$

The last relation means $S/T = 1 + \text{an infinitesimal}$.

For example,

$$x^2 + 3x \not\asymp 7x^2 - e^x,$$

because the second expression is dominated by $-e^x$. On the other hand,

$$x^2 + 3x \asymp 7x^2 - x^{-10},$$

since these two expressions have the same leading monomial, even though their leading coefficients differ. Finally,

$$x^2 + 3x \sim x^2 - 100 + e^{-x},$$

because both have the same leading term x^2 . Exact coefficients and exact support matter in every one of these comparisons.

4.5 Large, constant, and small parts

A transseries T can be split according to whether its monomials are greater than, equal to, or less than 1:

$$T = L + c + S, \tag{4.1}$$

where

- L is *purely large*: every monomial in L is $\succ 1$;
- $c \in \mathbb{R}$ is the constant term;
- S is *small*: every monomial in S is $\prec 1$.

This decomposition will define the exponential in Chapter 9.

For example,

$$T = \underbrace{x^2 - 3 \log x}_L + \underbrace{7}_c + \underbrace{x^{-1} + e^{-x}}_S.$$

The adjective “purely” matters. A large positive transseries such as $x + 1 + x^{-1}$ is not purely large because it contains constant and small monomials.

4.6 Logarithmic derivatives

A monomial often has the form

$$m = x^a e^L.$$

Its derivative is best expressed through its logarithmic derivative:

$$\frac{m'}{m} = \frac{a}{x} + L'. \quad (4.2)$$

Thus

$$m' = \left(\frac{a}{x} + L' \right) m.$$

This simple identity is the engine behind formal differentiation and integration. Examples:

$$\begin{aligned} (x^a)' &= \frac{a}{x} x^a, \\ (e^{x^2})' &= 2x e^{x^2}, \\ (x^3 e^{-\sqrt{x}})' &= \left(\frac{3}{x} - \frac{1}{2\sqrt{x}} \right) x^3 e^{-\sqrt{x}}. \end{aligned}$$

4.7 A dominance calculation by layers

Compare

$$m_1 = x^{17} \exp\left(x + \sqrt{x} + \frac{1}{x}\right), \quad m_2 = x^{-200} \exp(x + 2 \log x).$$

Take the logarithm of their quotient:

$$\begin{aligned} \log(m_1/m_2) &= 217 \log x + \sqrt{x} + \frac{1}{x} - 2 \log x \\ &= \sqrt{x} + 215 \log x + \frac{1}{x}. \end{aligned}$$

This is positive and infinitely large, so $m_1 \succ m_2$. Notice the layered comparison:

1. The leading x in the exponential cancels.
2. The next exponential contribution $\sqrt{x}$ beats every logarithmic contribution.
3. Only if that also canceled would the powers of x decide the comparison.

Intuition

A transmonomial has something like a lexicographic address. Compare the highest exponential layer first; if it ties, move down to the next layer; only then compare powers and constants. The formal construction makes this intuition exact.

4.8 Exercises

Exercise 4.1. Find the leading monomial and leading coefficient of

$$3x^5e^{-x} + 7e^{-x/2} - 2x^{100}e^{-x} + x^{-1}.$$

Exercise 4.2. Compare

$$\exp(x + \log^2 x) \quad \text{and} \quad x^{1000}e^x.$$

Then compare $\exp(x + \sqrt{\log x})$ with the same second monomial.

Exercise 4.3. Compute the logarithmic derivative of

$$m = x^{-3}(\log x)^7 \exp(-x^2 + e^{\sqrt{x}}).$$

Chapter summary

Transmonomials form an ordered multiplicative group. A term is a real coefficient times a monomial; a transseries is ordered by its largest supported monomial. Dominance is recursive because exponentials are compared by comparing their exponents.

Hahn series: infinite sums that still behave

If we allow arbitrary sets of monomials in a formal sum, addition is harmless but multiplication can become meaningless. Hahn series provide the right support condition.

5.1 Why support matters

Consider a formal sum

$$T = \sum_{\mathfrak{m} \in \mathfrak{M}} a_{\mathfrak{m}} \mathfrak{m}.$$

Its *support* is

$$\text{supp } T = \{\mathfrak{m} \in \mathfrak{M} : a_{\mathfrak{m}} \neq 0\}.$$

For an ordinary Laurent series in x^{-1} ,

$$T = a_0 + a_1 x^{-1} + a_2 x^{-2} + \cdots,$$

the support descends

$$1 \succ x^{-1} \succ x^{-2} \succ \cdots.$$

There is a largest term, and after removing any finite initial segment there is still a largest remaining term.

By contrast,

$$x + x^2 + x^3 + \cdots$$

has no largest monomial as $x \rightarrow +\infty$. It does not describe a series proceeding from dominant terms to smaller corrections.

Definition 5.1 (Reverse well-ordered support). A subset $A \subseteq \mathfrak{M}$ is *reverse well ordered* if every nonempty subset of A has a largest element. A formal sum with reverse well-ordered support is called a *Hahn series* or a *well-based generalized series*.

Some authors order monomials in the opposite direction and say simply “well ordered.” The content is the same: the support must move one way, from larger monomials toward smaller ones, without an infinite climb back upward.

5.2 Examples and nonexamples

Example 5.1. The support

$$\{1, x^{-1}, x^{-2}, \dots\}$$

is reverse well ordered. Every nonempty subset corresponds to a nonempty set of nonnegative integers, which has a least index; that index gives the largest monomial.

Example 5.2. The support

$$\{x^{1/n} : n = 1, 2, 3, \dots\}$$

is reverse well ordered. The whole set has largest element x , and every subset has a least denominator index. The monomials approach 1 from above, but this is allowed.

Example 5.3. The support

$$\{x^{-1/n} : n = 1, 2, 3, \dots\}$$

is not reverse well ordered. It climbs

$$x^{-1} \prec x^{-1/2} \prec x^{-1/3} \prec \dots$$

toward 1 without attaining a largest member.

Example 5.4. The support

$$\{e^{-nx} : n = 0, 1, 2, \dots\}$$

is reverse well ordered, so

$$\sum_{n \geq 0} a_n e^{-nx}$$

is a valid Hahn series in the monomial group generated by e^{-x} .

5.3 The Hahn field

Given an ordered abelian group $\mathfrak{M}$ and a coefficient field K , write

$$K[[\mathfrak{M}]]$$

for the set of all formal sums with coefficients in K and reverse well-ordered support. Addition is coefficientwise:

$$\left(\sum a_m m\right) + \left(\sum b_m m\right) = \sum (a_m + b_m) m.$$

Multiplication is the Cauchy product:

$$\left(\sum a_m m\right) \left(\sum b_n n\right) = \sum_{p \in \mathfrak{M}} \left(\sum_{mn=p} a_m b_n\right) p. \quad (5.1)$$

The inner coefficient sum must be finite, and the resulting support must again be reverse well ordered. A classical ordered-group lemma ensures both facts.

Lemma 5.1 (Neumann’s lemma, informal form). *Products and finite sums of reverse well-ordered supports remain reverse well ordered. Moreover, for supports satisfying the Hahn condition, a fixed product monomial receives only finitely many contributions in the Cauchy product.*

A full proof belongs to ordered-group theory, but the basic mechanism is familiar from power series. In

$$(1 + a_1t + a_2t^2 + \cdots)(1 + b_1t + b_2t^2 + \cdots),$$

the coefficient of t^N receives only the $N + 1$ contributions with indices $i + j = N$. The support condition generalizes this finite-decomposition property.

5.4 A worked Hahn product

Let $\mu = x^{-1}$ and $\nu = e^{-x}$. Consider

$$A = 1 + 2\mu + 3\nu + 5\mu\nu, \quad B = 1 - \mu + 4\nu.$$

Then

$$\begin{aligned} AB &= 1 + \mu + (3 + 4)\nu - 2\mu^2 + (5 + 8 - 3)\mu\nu \\ &\quad + 12\nu^2 - 5\mu^2\nu + 20\mu\nu^2 \\ &= 1 + \mu + 7\nu - 2\mu^2 + 10\mu\nu + 12\nu^2 - 5\mu^2\nu + 20\mu\nu^2. \end{aligned}$$

To place this in dominance order, note that every power of x^{-1} is larger than e^{-x} . Thus the beginning is actually

$$1 + \mu - 2\mu^2 + \cdots + 7\nu + 10\mu\nu - 5\mu^2\nu + \cdots.$$

A finite display can hide a long block of monomials between two terms. The formal support, not visual adjacency, determines the order.

5.5 Formal summability of families

Transseries operations often require summing a family $(T_i)_{i \in I}$ rather than a single sequence of scalar terms.

Definition 5.2 (Summable family). A family $(T_i)_{i \in I}$ of Hahn series is *formally summable* if:

1. the union $\bigcup_i \text{supp } T_i$ is reverse well ordered;
2. each monomial belongs to $\text{supp } T_i$ for only finitely many indices i .

Then $\sum_i T_i$ is defined coefficientwise.

The second condition prevents the coefficient of one monomial from becoming an undefined infinite scalar sum. In a purely formal field we do not want to decide whether $1 - 1 + 1 - 1 + \cdots$ converges; we arrange that no such coefficient sum occurs.

5.6 Geometric inversion

Suppose $S \prec 1$. Then the family

$$1, -S, S^2, -S^3, \dots$$

is formally summable under the standard transseries support conditions, and

$$\frac{1}{1+S} = 1 - S + S^2 - S^3 + \dots \quad (5.2)$$

The proof is purely algebraic:

$$(1+S) \sum_{n=0}^N (-S)^n = 1 + (-1)^N S^{N+1}.$$

At every fixed monomial, the remainder eventually becomes too small to contribute, so the formal limit is 1.

Worked calculation

Invert

$$T = x^3 + 2x + 1.$$

Factor the leading term:

$$T = x^3(1 + 2x^{-2} + x^{-3}).$$

Set $S = 2x^{-2} + x^{-3} \prec 1$. Then

$$\begin{aligned} T^{-1} &= x^{-3}(1 - S + S^2 - S^3 + \dots) \\ &= x^{-3} - 2x^{-5} - x^{-6} + 4x^{-7} + 4x^{-8} - 7x^{-9} + \dots \end{aligned}$$

No numerical convergence is invoked. This is an identity in the Hahn field of formal Laurent series.

5.7 Truncation

A Hahn series has meaningful initial segments. If $\mathfrak{n} \in \mathfrak{M}$, define the truncation above $\mathfrak{n}$ by

$$T_{>\mathfrak{n}} = \sum_{\mathfrak{m} > \mathfrak{n}} a_{\mathfrak{m}} \mathfrak{m}.$$

Depending on the support, this truncation may be finite or infinite. Formal algorithms are often interpreted as stabilization of every truncation: once enough iterations have been performed, all coefficients above any chosen threshold stop changing.

This topology of coefficient stabilization is different from numerical convergence. It is closer to the t -adic topology of formal power series, where a sequence converges when more and more low-degree coefficients agree.

5.8 Exercises

Exercise 5.1. Classify each support as reverse well ordered or not:

$$\{x^{-n^2} : n \geq 0\}, \quad \{x^{1-1/n} : n \geq 2\}, \quad \{e^{-x/n} : n \geq 1\}, \quad \{e^{-nx} : n \geq 0\}.$$

Exercise 5.2. Using formal geometric inversion, compute $1/(1 + x^{-1} + x^{-3})$ through order x^{-6} .

Exercise 5.3. Explain why the family $T_n = x^{-n} + x^{-1}$ is not formally summable over $n \geq 1$, even though the union of its supports is reverse well ordered.

Chapter summary

Hahn series are generalized series with reverse well-ordered support. This condition gives a largest term and makes Cauchy products and coefficientwise infinite sums well defined. Formal convergence means stabilization by dominance, not convergence of numerical values.

Grids, ratio sets, and multidimensional bookkeeping

The full class of well-based supports is elegant but can be difficult to compute with. A useful and historically important subfield consists of *grid-based* transseries, whose monomials lie in a finitely generated multiplicative lattice.

6.1 Ratio monomials

Choose finitely many small monomials

$$\boldsymbol{\mu} = (\mu_1, \dots, \mu_k), \quad \mu_i < 1.$$

A product

$$\boldsymbol{\mu}^{\mathbf{n}} = \mu_1^{n_1} \cdots \mu_k^{n_k}, \quad \mathbf{n} = (n_1, \dots, n_k) \in \mathbb{Z}^k,$$

is a lattice monomial. The tuple $\boldsymbol{\mu}$ is called a *ratio set* because each generator is a small ratio between neighboring scales.

Definition 6.1 (Grid). Given a base monomial $\mathfrak{m}_0$ and a lower multi-index $\mathbf{n}_0 \in \mathbb{Z}^k$, the grid generated by $\boldsymbol{\mu}$ is

$$\mathfrak{J}(\mathfrak{m}_0, \boldsymbol{\mu}, \mathbf{n}_0) = \{\mathfrak{m}_0 \boldsymbol{\mu}^{\mathbf{n}} : \mathbf{n} \in \mathbb{Z}^k, \mathbf{n} \geq \mathbf{n}_0\},$$

where the inequality is componentwise. A series is *grid based* if its support is contained in some such grid.

The lower bound permits finitely many negative powers of the generators but forces the support eventually to move in the small directions.

6.2 The basic two-generator grid

Let

$$\mu = x^{-1}, \quad \nu = e^{-x}.$$

Then the grid consists of

$$x^{-m} e^{-nx}, \quad m, n \in \mathbb{Z}$$

above chosen lower bounds. A typical transseries is

$$T = \sum_{n=0}^{\infty} e^{-nx} \sum_{m=m_0(n)}^{\infty} a_{n,m} x^{-m}. \quad (6.1)$$

Each fixed n is an exponential sector; inside it sits an inverse-power series. The grid coordinates (m, n) make multiplication a discrete convolution.

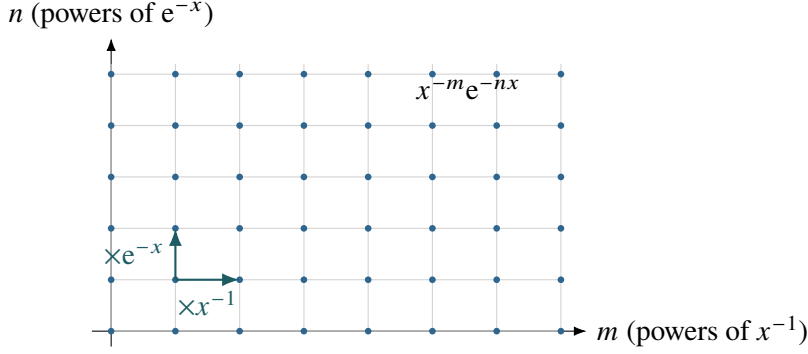


Figure 6.1: A two-dimensional grid generated by x^{-1} and e^{-x} .

Dominance is not simply lexicographic in (m, n) , but in this example the exponential coordinate decides first:

$$x^{-m}e^{-nx} \succ x^{-p}e^{-qx} \quad \text{if } n < q,$$

regardless of finite m, p . If $n = q$, then the smaller inverse-power exponent is larger.

6.3 Why grids help algorithms

On a finite-dimensional grid:

- monomials have integer coordinate vectors;
- multiplication adds vectors;
- truncation becomes a region in $\mathbb{Z}^k$;
- fixed-point iteration can be analyzed by showing that every correction moves in a positive combination of small directions;
- a computer algebra system can store a finite truncation in a sparse map keyed by integer vectors.

This is the transseries analogue of representing a multivariate power series by exponent tuples.

6.4 A bivariate geometric expansion

Let $S = 2\mu - 3\nu$, with $\mu = x^{-1}$ and $\nu = e^{-x}$. Then

$$\frac{1}{1+S} = 1 - S + S^2 - S^3 + \dots$$

Expanding by grid degree gives

$$\frac{1}{1 + 2\mu - 3\nu} = 1 - 2\mu + 3\nu + 4\mu^2 - 12\mu\nu + 9\nu^2 - 8\mu^3 + 36\mu^2\nu - 54\mu\nu^2 + 27\nu^3 + \dots$$

Every monomial corresponds to a lattice point. The coefficient of $\mu^m \nu^n$ is obtained from finitely many multinomial contributions.

6.5 Formal convergence on a grid

A sequence $T_0, T_1, T_2, \dots$ converges formally if, for every monomial threshold, its truncation above that threshold eventually stabilizes. On a grid generated by small ratios, a sufficient pattern is

$$T_{j+1} - T_j \in \mu_1^{r_{1,j}} \cdots \mu_k^{r_{k,j}} K[[\mathfrak{M}]],$$

where the correction vectors move indefinitely into the positive cone.

For the ordinary iteration

$$S_{j+1} = A + S_j^2, \quad A \prec 1,$$

starting at $S_0 = 0$, we get

$$S_1 = A, \quad S_2 = A + A^2, \quad S_3 = A + A^2 + 2A^3 + A^4, \quad \dots$$

The coefficient of any fixed power A^N eventually stops changing. This is formal convergence even if A is merely a symbol.

6.6 Grid based is not the same as well based

Every grid-based support used in the standard construction is well based, but not every well-based support fits inside a finitely generated grid. For example,

$$\{x^{1/n} : n \geq 1\}$$

is reverse well ordered, yet the exponents $1/n$ do not lie in a finitely generated additive semigroup of $\mathbb{R}$ with a common positive step. Thus

$$\sum_{n \geq 1} a_n x^{1/n}$$

may be a valid well-based Hahn series while not being grid based.

The larger well-based field has cleaner closure properties and greater generality. The grid-based field has more explicit combinatorics. Edgar's expositions carefully track both settings [22, 23].

6.7 Ratio sets as proof certificates

In practical transseries algebra, a ratio set can do more than generate support. It can certify comparisons. To prove $A \prec B$, one may show that every monomial in A/B is a product of positive powers of the small generators. Similarly, an operator Φ is contractive if differences $\Phi(S) - \Phi(T)$ gain at least one small ratio relative to $S - T$.

This finite certificate is useful because direct comparison of complicated nested exponentials can be cumbersome. The ratio set records the handful of scale reductions that the calculation actually uses.

Worked calculation

Suppose

$$\Phi(S) = a\mu + b\nu + c\mu S + dS^2, \quad \mu, \nu \prec 1.$$

Then

$$\Phi(S) - \Phi(T) = c\mu(S - T) + d(S - T)(S + T).$$

If $S, T \prec 1$, both multipliers $c\mu$ and $d(S + T)$ are small. Hence the difference gains a small factor. Iteration therefore pushes errors to smaller and smaller grid monomials.

6.8 Exercises

Exercise 6.1. Let $\mu = x^{-1}$ and $\nu = e^{-x}$. Expand

$$(1 - \mu - \nu)^{-1}$$

through total grid degree $m + n \leq 3$.

Exercise 6.2. Show that the support $\{x^{-m}e^{-nx} : m, n \geq 0\}$ is reverse well ordered under asymptotic dominance.

Exercise 6.3. Explain why the support $\{x^{-1/n} : n \geq 1\}$ cannot be contained in a grid generated by finitely many power monomials $x^{-a_1}, \dots, x^{-a_k}$ with $a_i > 0$.

Chapter summary

A ratio set is a finite list of small monomials. Its integer products form a multidimensional grid that turns support manipulation into lattice bookkeeping. Grid-based transseries are computationally convenient but form a proper subcollection of all well-based transseries.

Building logarithmic-exponential transseries by height and depth

We now describe the inductive construction behind the standard field $\mathbb{T}$ of logarithmic-exponential transseries. The construction is easier to understand if one keeps two counters:

- *exponential height*: how many genuinely large exponential layers occur;
- *logarithmic depth*: how many iterated logarithms of the main variable are needed.

The details differ slightly among authors and between grid-based and well-based variants. The following version captures the common architecture used in Edgar's exposition and in the modern differential-algebraic literature [22, 38, 2].

7.1 Height zero: generalized power series

Begin with the monomial group

$$\mathfrak{G}_0 = \{x^b : b \in \mathbb{R}\}.$$

The height-zero field is the Hahn field

$$\mathbb{T}_0 = \mathbb{R}[[\mathfrak{G}_0]]. \quad (7.1)$$

Its elements are generalized power series

$$T = \sum_{b \in A} a_b x^b,$$

where $A \subseteq \mathbb{R}$ is reverse well ordered in the usual order of exponents.

This already goes beyond ordinary Laurent series. Exponents may be real and need not lie in an arithmetic progression. Examples include

$$x^{\sqrt{2}} + 3x^{1/7} - 5 + x^{-\pi} + \dots$$

and the well-based but nongrid series

$$\sum_{n \geq 1} 2^{-n} x^{1/n}.$$

7.2 Purely large exponents become new monomials

Suppose $\mathbb{T}_n$ has been constructed. Let $L \in \mathbb{T}_n$ be purely large. We introduce a new monomial e^L . To retain ordinary power factors, define

$$\mathfrak{G}_{n+1} = \{x^b e^L : b \in \mathbb{R}, L \in \mathbb{T}_n \text{ purely large}\}. \quad (7.2)$$

We allow $L = 0$, so $\mathfrak{G}_n$ embeds in later levels. The order is determined by the sign of the leading term of the logarithmic quotient:

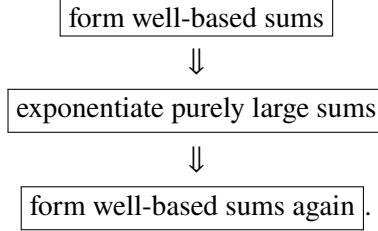
$$\log\left(\frac{x^a e^A}{x^b e^B}\right) = A - B + (a - b) \log x.$$

If this transseries is negative and infinitely large, then $x^a e^A \prec x^b e^B$; if it is positive and infinitely large, the inequality is reversed; if it is zero, the monomials coincide. More plainly, compare A and B first, and if they cancel to logarithmic size, compare the powers of x .

The next field is

$$\mathbb{T}_{n+1} = \mathbb{R}[[\mathfrak{G}_{n+1}]]. \quad (7.3)$$

Thus one alternates:



7.3 Examples of height

Expression	Height	Reason
$x^\pi + x^{-1}$	0	generalized powers only
$e^{x^2} + e^{-x}$	1	exponentials of height-zero large exponents
e^{e^x}	2	exponential of a height-one purely large term
$\exp(e^{e^x} + x)$	3	outer exponential raises the height again
$e^x(1 + e^{-x} + e^{-2x} + \dots)$	1	the parenthesized part is a series on a height-one grid

An expression's visual nesting can overstate its genuine height. For example,

$$\exp(x + e^{-x}) = e^x \exp(e^{-x}) = e^x \left(1 + e^{-x} + \frac{1}{2}e^{-2x} + \dots\right).$$

The small exponential in the exponent is expanded as a power series; it does not create a new large exponential monomial. The resulting transseries has height 1, not 2.

7.4 The log-free union

The union

$$\mathbb{T}_{\text{lf}} = \bigcup_{n \geq 0} \mathbb{T}_n \quad (7.4)$$

is called the *log-free* transseries field. Every one of its elements has finite exponential height, although there is no uniform bound on the heights of all elements.

The phrase “finite height” is crucial. The formal infinite tower

$$\exp(\exp(\exp(\cdots)))$$

is not an element of the standard field. Enlarged fields with superexponentials or infinite-height structures exist, but they require extra construction; Schmeling’s thesis is a foundational source in that direction [35].

7.5 Adding logarithmic depth

The log-free field contains x and exponentials built from x , but not yet every desired iterated logarithm as a basic scale. Let

$$\log_0 x = x, \quad \log_1 x = \log x, \quad \log_2 x = \log \log x, \quad \dots$$

For $m \geq 0$, compose log-free transseries with $\log_m x$:

$$\mathbb{T}_{n,m} = \{T \circ \log_m : T \in \mathbb{T}_n\}.$$

Then form the union over finite height and finite depth:

$$\mathbb{T} = \bigcup_{m \geq 0} \bigcup_{n \geq 0} \mathbb{T}_{n,m}. \quad (7.5)$$

The precise nesting of these unions is handled carefully in formal treatments, but the practical meaning is simple: any individual transseries uses only finitely many iterated logarithms and finitely many large exponential layers.

Examples:

$$\begin{aligned} \log x + \frac{1}{\log x} + \frac{1}{(\log x)^2} + \cdots & \text{ has depth 1,} \\ e^{\sqrt{\log x}} (\log \log x)^3 & \text{ uses depth 2,} \\ \exp(\exp(\sqrt{\log \log x})) & \text{ has finite height and depth 2.} \end{aligned}$$

7.6 A tree view of monomials

A transmonomial can be viewed as a finite syntax tree:

The construction is recursive on these trees. Ordering two outer exponentials requires ordering their exponent subtrees. Differentiation follows the chain rule down the tree. Exponentiating a purely large series wraps it in a new node.

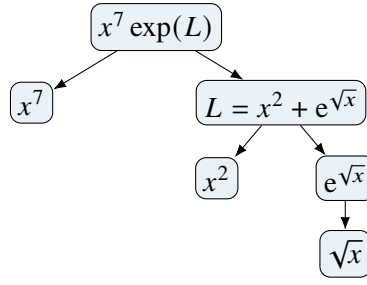


Figure 7.1: A finite-height monomial as a syntax tree. Supports then form well-based forests of such trees.

7.7 Why the exponent must be purely large

Any transseries A has a unique additive decomposition

$$A = L + c + S$$

into purely large, constant, and small parts. Then

$$e^A = e^c e^L e^S.$$

Only e^L needs to be declared a new monomial. The constant e^c is a coefficient, and e^S expands as

$$1 + S + \frac{S^2}{2!} + \cdots.$$

If arbitrary small and constant pieces were hidden inside monomials, the same transseries would have many incompatible monomial representations. The purely large rule gives a canonical normal form.

7.8 Finite complexity versus infinite support

A transseries may have infinitely many terms while each monomial has finite syntactic complexity. This is analogous to an ordinary power series: the series has infinitely many monomials x^n , but each monomial is a finite expression.

The standard field permits:

$$\sum_{n \geq 0} e^{-nx}, \quad \sum_{n \geq 0} x^{-n} e^{-e^x}, \quad \sum_{n \geq 1} x^{1/n},$$

when the chosen well-based variant supports them. It does not automatically permit one term of height n for every n if those terms violate the support architecture. Enlargements such as omega-series and hyperseries address broader closure goals; we return to them in Chapter 29.

7.9 Exercises

Exercise 7.1. Assign the smallest plausible exponential height and logarithmic depth to each expression:

$$e^{x+1/x}, \quad e^{e^x+x}, \quad \log \log x + x^{-1}, \quad \exp(\sqrt{\log x} + e^{-\log x}).$$

Remember to expand small pieces in exponents when appropriate.

Exercise 7.2. Write $\exp(x + 2 + x^{-1} + e^{-x})$ as a constant coefficient times a monomial times a small formal series through terms of dominance at least $e^x x^{-2}$.

Exercise 7.3. Why would allowing $\exp(c + L)$ and $\exp(L)$ as unrelated monomials destroy a canonical coefficient-monomial decomposition?

Chapter summary

The transseries field is built in stages. Height zero consists of generalized powers. Purely large elements at one stage are exponentiated to create monomials at the next. Finite composition with iterated logarithms adds logarithmic depth. Constant and small exponent pieces are kept out of monomials to preserve canonical form.

Arithmetic, inversion, and real powers

The Hahn-field architecture makes addition and multiplication automatic. Division, roots, and general real powers follow from one canonical factorization.

8.1 Canonical multiplicative decomposition

Let $T \neq 0$. Extract its leading term:

$$T = a\mathfrak{m} + \text{smaller terms}, \quad a = \text{lc}(T), \quad \mathfrak{m} = \text{lm}(T).$$

Factor it:

$$T = a\mathfrak{m}(1 + S), \quad S \prec 1. \quad (8.1)$$

This decomposition is unique. It plays the role of writing a nonzero formal power series as

$$a_k t^k (1 + \text{higher powers of } t).$$

8.2 Inversion

Using the formal geometric series,

$$T^{-1} = a^{-1}\mathfrak{m}^{-1} \left(1 - S + S^2 - S^3 + \cdots \right). \quad (8.2)$$

Every nonzero transseries therefore has an inverse.

Worked calculation

Invert

$$T = e^x + x^2 + 1.$$

The leading monomial is e^x , so

$$T = e^x \left(1 + x^2 e^{-x} + e^{-x} \right).$$

Set $S = (x^2 + 1)e^{-x}$. Then

$$T^{-1} = e^{-x} \left(1 - S + S^2 - S^3 + \cdots \right)$$

$$= e^{-x} - (x^2 + 1)e^{-2x} + (x^2 + 1)^2 e^{-3x} - (x^2 + 1)^3 e^{-4x} + \dots$$

Each correction is exponentially smaller than the preceding sector.

8.3 Integer and rational powers

For $r \in \mathbb{Z}$, T^r is defined by multiplication and inversion. If $T > 0$ and $q \geq 1$ is an integer, a q th root can be constructed from Equation (8.1):

$$T^{1/q} = a^{1/q} m^{1/q} (1 + S)^{1/q}.$$

The binomial series is

$$(1 + S)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} S^n, \quad \binom{\alpha}{n} = \frac{\alpha(\alpha - 1) \cdots (\alpha - n + 1)}{n!}. \quad (8.3)$$

It is formally summable whenever $S \prec 1$. For rational α , this gives algebraic roots; after logarithms are defined, the same formula agrees with $\exp(\alpha \log(1 + S))$ for every real α .

Example 8.1. For $x \succ 1$,

$$\begin{aligned} \sqrt{x^2 + x} &= x \sqrt{1 + x^{-1}} \\ &= x \left(1 + \frac{1}{2x} - \frac{1}{8x^2} + \frac{1}{16x^3} - \frac{5}{128x^4} + \cdots \right) \\ &= x + \frac{1}{2} - \frac{1}{8x} + \frac{1}{16x^2} - \frac{5}{128x^3} + \cdots \end{aligned}$$

The familiar binomial asymptotic expansion is an identity in the Hahn field.

8.4 Order and sign

Define $T > 0$ if $\text{lc}(T) > 0$. This order extends the real order and is compatible with field operations. In particular:

$$S < T \iff T - S > 0.$$

An infinitesimal can be positive without being zero:

$$0 < x^{-1} < 1, \quad 0 < e^{-x} < x^{-N} \quad \text{for every fixed } N.$$

Similarly, x is infinitely large because $x > n$ for every real constant n in the formal order.

The field is therefore *non-Archimedean*: the Archimedean principle that every number is bounded by an integer fails.

8.5 Valuation language

The leading monomial defines a valuation-like measure of size. One may assign to T the value corresponding to $\text{lm}(T)$, with the convention that smaller monomials have larger valuation. Then

$$v(ST) = v(S) + v(T), \quad v(S + T) \geq \min\{v(S), v(T)\},$$

with equality unless leading terms cancel.

For beginners, leading-monomial language is usually more concrete. Valuations become valuable when studying differential equations abstractly, because they turn asymptotic size into algebraic inequalities.

8.6 Cancellation and the next term

Suppose

$$S = 3e^x + 2x + \cdots, \quad T = -3e^x + 5x + \cdots.$$

Then

$$S + T = 7x + \cdots.$$

The leading exponential cancels, and the sum drops to a much smaller scale. Algorithms must therefore combine equal monomials before determining the leading term. A sparse dictionary representation naturally enforces this.

A common trap

It is unsafe to infer the leading term of a sum from the leading terms of the summands without checking cancellation. Dominance is not a floating-point approximation; coefficients are exact data.

8.7 Real closedness

A major structural theorem says that the real transseries field is real closed. Informally, this means it has the same algebraic completeness properties as the real numbers.

Theorem 8.1 (Real closedness, stated without proof). *The standard field $\mathbb{T}$ of real logarithmic-exponential transseries is real closed. Consequently:*

1. *every positive transseries has a square root;*
2. *every polynomial of odd degree over $\mathbb{T}$ has a root in $\mathbb{T}$;*
3. *every polynomial over $\mathbb{T}$ factors into linear and irreducible quadratic factors.*

At the Hahn-field level, divisibility of the monomial group and real closedness of the coefficient field are the key ingredients. The inductive exponential-logarithmic construction preserves the required structure. Full proofs are found in the generalized-series and transseries literature [36, 37, 38].

Real closedness is an existence theorem, not an automatic recipe. Dominant balance and fixed-point iteration provide practical expansions of roots; we develop them in Part IV.

8.8 Newton's method as formal algebra

Suppose $P(Y) \in \mathbb{T}[Y]$ and an approximation Y_0 makes $P(Y_0)$ small relative to $P'(Y_0)$. The Newton correction is

$$Y_1 = Y_0 - \frac{P(Y_0)}{P'(Y_0)}.$$

If the correction gains dominance order, iteration often stabilizes coefficient by coefficient. This is not numerical convergence in a metric; it is convergence in the valuation topology.

For instance, to solve $Y^2 = x^2 + x$, start with $Y_0 = x$. Then

$$Y_1 = x - \frac{x^2 - (x^2 + x)}{2x} = x + \frac{1}{2}.$$

A second step yields

$$Y_2 = x + \frac{1}{2} - \frac{(x + 1/2)^2 - (x^2 + x)}{2x + 1} = x + \frac{1}{2} - \frac{1}{8x} + O(x^{-2}),$$

recovering the binomial expansion progressively.

8.9 Exercises

Exercise 8.1. Invert $x^2 - 3x + 1$ through order x^{-6} .

Exercise 8.2. Expand $(1 + 2x^{-1} + e^{-x})^{1/2}$ through all monomials at least as large as $x^{-2}e^{-x}$.

Exercise 8.3. Let $S, T > 0$ with $S \sim T$. Show formally that $S^\alpha \sim T^\alpha$ for any fixed real α , assuming the logarithm and exponential constructions of the next chapter.

Chapter summary

Every nonzero transseries factors uniquely as coefficient times leading monomial times $1 +$ an infinitesimal. Geometric and binomial series then provide inverses and roots. The resulting ordered field is non-Archimedean and, in its standard form, real closed.

Exponentials and logarithms without numerical convergence

The most important formal operations are defined by separating the part that must become a new monomial from the part that can be expanded as an ordinary power series.

9.1 The exponential of a transseries

Write the canonical additive decomposition

$$T = L + c + S,$$

where L is purely large, $c \in \mathbb{R}$, and $S \prec 1$. Define

$$\exp(T) = e^c e^L \sum_{n=0}^{\infty} \frac{S^n}{n!}. \quad (9.1)$$

Each factor has a distinct role:

- e^L is a transmonomial introduced by the height construction;
- e^c is an ordinary real coefficient;
- $\sum S^n/n!$ is a well-based formal series because S is infinitesimal.

No claim is made that substituting a finite numerical x into the series converges. Formal summability follows from support moving to smaller monomials as powers of S increase.

Worked calculation

Expand

$$\exp(x + 2 + x^{-1} + 3e^{-x}).$$

Here

$$L = x, \quad c = 2, \quad S = x^{-1} + 3e^{-x}.$$

Therefore

$$\exp(T) = e^2 e^x \left(1 + S + \frac{S^2}{2} + \frac{S^3}{6} + \cdots \right).$$

The beginning, in dominance order, is

$$e^2 e^x \left(1 + x^{-1} + \frac{1}{2} x^{-2} + \frac{1}{6} x^{-3} + \dots + 3e^{-x} + 3x^{-1}e^{-x} + \frac{3}{2} x^{-2} e^{-x} + \dots + \frac{9}{2} e^{-2x} + \dots \right).$$

After multiplying by e^x , the sector $3e^{-x}$ inside the parentheses contributes the constant term $3e^2$. Formal dominance must be assessed after all monomial factors are combined.

9.2 Why exp respects addition

Given A and B , one expects

$$\exp(A + B) = \exp(A) \exp(B).$$

The proof reduces to the ordinary formal identity for small parts, together with multiplication of monomials for large parts and real exponentials for constants. The support lemmas justify rearranging the finite contribution to each monomial. Thus the exponential is a homomorphism from the additive group of $\mathbb{T}$ to its positive multiplicative group.

9.3 The logarithm of a positive transseries

Let $T > 0$. Its multiplicative decomposition has positive leading coefficient:

$$T = a m(1 + S), \quad a > 0, \quad S \prec 1.$$

Define

$$\log T = \log a + \log m + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n. \quad (9.2)$$

The logarithm of a monomial is already encoded in its construction. If

$$m = x^b e^L,$$

then

$$\log m = b \log x + L. \quad (9.3)$$

The small factor uses the familiar series

$$\log(1 + S) = S - \frac{S^2}{2} + \frac{S^3}{3} - \dots$$

Worked calculation

Find $\log(e^x + x^2)$. Factor the leading monomial:

$$e^x + x^2 = e^x(1 + x^2 e^{-x}).$$

Therefore

$$\begin{aligned}\log(e^x + x^2) &= x + \log(1 + x^2 e^{-x}) \\ &= x + x^2 e^{-x} - \frac{1}{2} x^4 e^{-2x} + \frac{1}{3} x^6 e^{-3x} - \dots.\end{aligned}$$

The correction is exponentially small even though x^2 by itself is large.

9.4 Formal inverse identities

The definitions are designed so that

$$\log(\exp T) = T$$

for all real transseries T , and

$$\exp(\log T) = T$$

for all positive transseries T . On the small part, these are the standard identities of formal power series. The canonical decompositions ensure that large and constant pieces are recovered exactly.

Proposition 9.1. *The exponential is an order-preserving group isomorphism*

$$(\mathbb{T}, +) \longrightarrow (\mathbb{T}_{>0}, \times),$$

with inverse logarithm.

This is one reason transseries are more than a collection of asymptotic notations. They form an ordered exponential field.

9.5 General real powers

For $T > 0$ and $\alpha \in \mathbb{R}$, define

$$T^\alpha = \exp(\alpha \log T). \tag{9.4}$$

Using $T = a\mathfrak{m}(1 + S)$,

$$T^\alpha = a^\alpha \mathfrak{m}^\alpha (1 + S)^\alpha,$$

and the last factor is the binomial series Equation (8.3).

Example 9.1. Let

$$T = x^3 e^{\sqrt{x}} (1 + 2x^{-1} + e^{-x}).$$

Then

$$T^{\sqrt{2}} = x^{3\sqrt{2}} e^{\sqrt{2x}} \left(1 + 2x^{-1} + e^{-x}\right)^{\sqrt{2}}.$$

The small factor begins

$$1 + \sqrt{2}(2x^{-1} + e^{-x}) + \frac{\sqrt{2}(\sqrt{2} - 1)}{2}(2x^{-1} + e^{-x})^2 + \dots.$$

9.6 Nested examples

The recursive construction handles expressions that would be awkward in a single flat formula.

Example 1: a small perturbation of a huge exponential

Let

$$T = \exp\left(e^x + x^{-1}\right).$$

Then

$$T = e^{e^x} \exp(x^{-1}) = e^{e^x} \left(1 + x^{-1} + \frac{1}{2x^2} + \cdots\right).$$

The monomial has height 2, while the power corrections remain in the same sector.

Example 2: a logarithm peels off the top layer

Let

$$T = 3x^7 \exp(e^x - 2\sqrt{x}) \left(1 + x^{-1} + e^{-x}\right).$$

Then

$$\begin{aligned} \log T = & \log 3 + 7 \log x + e^x - 2\sqrt{x} \\ & + \left(x^{-1} + e^{-x}\right) - \frac{1}{2} \left(x^{-1} + e^{-x}\right)^2 + \cdots. \end{aligned}$$

The logarithm converts multiplicative layers into additive layers and lowers the outer exponential height.

9.7 Complex logarithms

The real ordered field has a logarithm only for positive elements. One can adjoin i and work with complex coefficients, but then there is no compatible total order and logarithms become branch dependent:

$$\text{Log } T = \log |T| + i \arg T.$$

Resurgent analysis necessarily uses complex directions and branches. The formal ordered-field chapters avoid this complication until Part V.

9.8 Exercises

Exercise 9.1. Expand $\exp(x + \log x + x^{-1})$ through order $e^x x^{-3}$.

Exercise 9.2. Compute the first four nonzero correction terms in

$$\log(x^5 + e^{-x}).$$

Exercise 9.3. Show formally that

$$\log(AB) = \log A + \log B$$

for positive A, B by reducing to their leading coefficients, monomials, and small factors.

Chapter summary

To exponentiate, split a transseries into purely large, constant, and small parts. The large part creates a monomial, the constant becomes a coefficient, and the small part is expanded. To take a logarithm, factor a positive transseries into coefficient, monomial, and $1 + \text{infinitesimal}$. These definitions make $\exp$ and $\log$ exact formal inverses.

What formal equality does and does not mean

Before differentiating or solving equations, it is worth separating several notions that are often compressed into the same equals sign.

10.1 Four levels of statement

10.1.1 Identity of formal transseries

Two transseries are equal if every monomial has the same coefficient. For example,

$$(1 + x^{-1})^{-1} = 1 - x^{-1} + x^{-2} - x^{-3} + \dots$$

is an exact identity in the formal field. It does not depend on choosing a numerical value of x .

10.1.2 Asymptotic expansion of a function

A function f may satisfy

$$f(x) \sim T(x)$$

in the sense that every truncation of T approximates f to the order of the next omitted monomial. This is an analytic theorem and may hold only in a sector of the complex plane.

10.1.3 Equality of germs

Two functions define the same germ at $+\infty$ if they agree for all sufficiently large x . A Hardy field is a field of such germs. A transseries may map to a germ under an asymptotic expansion map, but constructing an injective map for broad classes is a substantial theorem, not notation.

10.1.4 Equality after summation

A divergent formal transseries may be assigned an analytic function by a summation method. Directional Borel sums can differ across a Stokes ray. Thus one should write

$$S_\theta \widehat{F}$$

rather than silently replacing $\widehat{F}$ by a unique function.

10.2 The danger of plugging in numbers

Take

$$T(x) = \sum_{n \geq 0} \frac{n!}{x^{n+1}}.$$

As a formal transseries it is perfectly valid. At every fixed finite x , its numerical terms eventually grow, so the numerical sum diverges. Writing $T(10)$ without specifying truncation or summation is meaningless.

By contrast,

$$G(x) = \sum_{n \geq 0} x^{-n}$$

converges numerically for $x > 1$ and equals $x/(x - 1)$. It is simultaneously a formal identity and a convergent analytic representation. These are two independent virtues.

10.3 Truncation is the native observation operation

The natural way to inspect a transseries is not to evaluate it but to truncate it by dominance. Given a threshold m , retain all terms larger than m :

$$T_{>m} = \sum_{n > m} a_n n.$$

An algorithm is correct to order m if its output has the same truncation as the exact formal answer above that threshold.

This viewpoint resembles exact arithmetic with lazily generated coefficients. One asks for as much of the ordered support as a later calculation needs.

10.4 Closure properties of the standard field

The standard logarithmic-exponential transseries field is closed under

$$+, -, \times, /, \exp, \log \text{ on positives,}$$

and, as later chapters explain, under differentiation, antiderivatives, and suitable composition. It is real closed and supports a rich differential algebra [38, 2].

Yet it is not literally “all asymptotic expressions.” Important phenomena outside the basic real ordered field include:

- oscillatory monomials such as e^{ix} , which do not possess an eventual real order;
- arbitrary infinitely nested exponentials and logarithms;
- transmonomials requiring superexponentials or hyperserial structure;
- functions with wild zero sets or oscillations incompatible with Hardy-field behavior;
- nonperturbative analytic data, such as Stokes constants, not determined by formal field operations alone.

Enlargements and complex variants handle portions of this list, but no single framework erases all distinctions.

10.5 A useful discipline for calculations

At each step, ask four questions:

1. What is the monomial group and dominance order?
2. Is the support well based or grid based so that the operation is formally defined?
3. Is the statement only formal, or is an actual asymptotic function involved?
4. If a divergent series is being assigned a value, what summation direction or prescription is used?

These questions prevent most conceptual mistakes in introductory transseries work.

Checkpoint

Formal algebra answers “which coefficients must occur?” Asymptotic analysis answers “how well do truncations approximate a function?” Summability answers “which function, if any, is reconstructed from a divergent object?” Resurgence answers “how are the sectors and their discontinuities related?”

10.6 Exercises

Exercise 10.1. For each statement below, classify it as a formal identity, an asymptotic assertion, an equality of functions, or a summation assertion:

1. $(1 - S)^{-1} = \sum_{n \geq 0} S^n$ for $S \prec 1$;
2. $\text{Ei}(x)e^{-x} \sim \sum_{n \geq 0} n!x^{-n-1}$;
3. $\sum_{n \geq 0} x^{-n} = x/(x - 1)$ for $x > 1$;
4. $\mathcal{S}_{0+}\widehat{F} - \mathcal{S}_{0-}\widehat{F} = 2\pi ie^{-x}$.

Exercise 10.2. Give two distinct functions with the same asymptotic expansion in powers of x^{-1} , and then give a transseries scale that distinguishes them.

Chapter summary

Formal identity, asymptotic expansion, equality of germs, and equality after summation are different claims. A transseries is natively inspected by dominance truncation. Numerical substitution is justified only when convergence or an explicit summation rule has been established.

Part III

Calculus, composition, and recursive construction

Differentiation

Formal differentiation is one of the strongest reasons to impose disciplined support conditions. We want to differentiate term by term, preserve the transseries field, and retain the product, quotient, and chain rules.

11.1 Differentiate monomials first

A general log-free transmonomial can be written

$$\mathfrak{m} = x^a e^L,$$

where L is purely large at a lower height. Its derivative is

$$\mathfrak{m}' = \left(\frac{a}{x} + L' \right) \mathfrak{m}. \quad (11.1)$$

This recursively reduces differentiation of a monomial to differentiation of its exponent.

At logarithmic depth, the usual chain rule applies. For example,

$$\begin{aligned} \frac{d}{dx} (\log x)^a &= \frac{a}{x \log x} (\log x)^a, \\ \frac{d}{dx} e^{\sqrt{\log x}} &= \frac{1}{2x \sqrt{\log x}} e^{\sqrt{\log x}}, \\ \frac{d}{dx} e^{-e^x} &= -e^x e^{-e^x}. \end{aligned}$$

The multiplier $\mathfrak{m}'/\mathfrak{m}$ is the logarithmic derivative. It often changes the power or logarithmic prefactor but preserves the essential exponential scale.

11.2 Termwise differentiation

For

$$T = \sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m},$$

define

$$T' = \sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m}'. \quad (11.2)$$

The right side must be a formally summable family. This is not automatic for arbitrary generalized series, but the transseries monomial hierarchy is constructed so that differentiation preserves well-based support.

Theorem 11.1 (Closure under differentiation). *The standard logarithmic-exponential transseries field $\mathbb{T}$ carries a derivation $T \mapsto T'$ satisfying:*

$$\begin{aligned}(S + T)' &= S' + T', \\ (ST)' &= S'T + ST', \\ (T^{-1})' &= -T^{-2}T', \\ (\exp T)' &= T' \exp T, \\ (\log T)' &= T'/T \quad (T > 0).\end{aligned}$$

Its constant field is $\mathbb{R}$: if $T' = 0$, then $T \in \mathbb{R}$.

The support proof is technical because differentiation may replace one monomial by a whole transseries times that monomial. At a conceptual level, the exponent and logarithmic-depth recursion ensures that derivatives move within a controlled collection of lower-height factors. See [38, 22, 2] for rigorous treatments.

11.3 Power and logarithmic series

At height zero, differentiation is familiar:

$$\left(\sum_b a_b x^b \right)' = \sum_b b a_b x^{b-1}.$$

Examples:

$$\begin{aligned}\left(x^{\sqrt{2}} + 3 + x^{-\pi} \right)' &= \sqrt{2}x^{\sqrt{2}-1} - \pi x^{-\pi-1}, \\ \left(\log x + \frac{1}{\log x} + \frac{2}{(\log x)^2} + \cdots \right)' &= \frac{1}{x} - \frac{1}{x(\log x)^2} - \frac{4}{x(\log x)^3} + \cdots.\end{aligned}$$

A constant term disappears, which can reveal a much smaller leading derivative. This is ordinary cancellation, but across transseries scales the drop can be dramatic.

Example 11.1. Let

$$T = x + 100 + e^{-x^2}.$$

Then $T \sim x$, while

$$T' = 1 - 2xe^{-x^2} \sim 1.$$

The leading scale drops from x to 1.

Example 11.2. Let

$$T = 1 + e^{-e^x}.$$

The nonconstant part is extraordinarily small, but

$$T' = -e^x e^{-e^x}.$$

The derivative is larger than e^{-e^x} by a factor e^x , yet it remains smaller than every e^{-Nx} .

11.4 Differentiating an exponential sector expansion

Suppose

$$T = \sum_{k \geq 0} C^k e^{-kAx} T_k(x),$$

where $A > 0$ and each T_k is an inverse-power series. Then

$$T' = \sum_{k \geq 0} C^k e^{-kAx} (T'_k - kAT_k). \quad (11.3)$$

Differentiation preserves the sector index k . Nonlinear multiplication, by contrast, convolves sector indices. This separation is what makes coefficient equations for nonlinear ODEs manageable.

11.5 The chain rule

When composition is defined,

$$(T \circ S)' = (T' \circ S)S'. \quad (11.4)$$

For example,

$$T(x) = e^{x^2+x^{-1}}, \quad S(x) = \log x.$$

Then

$$(T \circ S)(x) = \exp\left((\log x)^2 + (\log x)^{-1}\right),$$

and

$$(T \circ S)' = \left(2 \log x - (\log x)^{-2}\right) \frac{1}{x} \exp\left((\log x)^2 + (\log x)^{-1}\right).$$

11.6 Logarithmic differentiation as a simplifier

For a product of complicated factors, compute T'/T :

$$T = x^a (\log x)^b \exp(L).$$

Then

$$\frac{T'}{T} = \frac{a}{x} + \frac{b}{x \log x} + L'. \quad (11.5)$$

This avoids repeated product rules and immediately reveals the derivative's scale.

Worked calculation

Let

$$T = x^{-3} (\log x)^7 \exp\left(-x^2 + e^{\sqrt{x}}\right).$$

Then

$$\frac{T'}{T} = -\frac{3}{x} + \frac{7}{x \log x} - 2x + \frac{e^{\sqrt{x}}}{2\sqrt{x}}.$$

The last term dominates $2x$, so

$$T' \sim \frac{e^{\sqrt{x}}}{2\sqrt{x}} T.$$

Even though T contains the decaying factor e^{-x^2} , the nested exponential in its exponent controls the logarithmic derivative.

11.7 Differential polynomials

A differential polynomial in an unknown Y is a polynomial expression in

$$Y, Y', Y'', \dots, Y^{(r)}$$

with transseries coefficients. For example,

$$P(Y) = Y'' + xY' - e^x Y^2 + x^{-1}.$$

Substituting a transseries Y and collecting monomials gives another transseries. Solving $P(Y) = 0$ means making every monomial coefficient vanish. Dominant balance identifies the first term; recursive cancellation determines the rest.

The differential-algebraic strength of $\mathbb{T}$ is much deeper than closure under differentiation. The field satisfies powerful existence and model-theoretic properties discussed in Chapter 27.

11.8 Exercises

Exercise 11.1. Differentiate

$$x^4 (\log x)^{-2} \exp(x^{1/3} - e^{-x})$$

using a logarithmic derivative.

Exercise 11.2. Let

$$T = \sum_{n \geq 0} n! x^{-n-1} + C e^{-x}.$$

Verify directly that $T' + T = x^{-1}$ as a formal identity.

Exercise 11.3. Suppose $S \prec 1$. Differentiate the formal identity

$$\log(1 + S) = S - \frac{S^2}{2} + \frac{S^3}{3} - \dots$$

and recover $(\log(1 + S))' = S'/(1 + S)$.

Chapter summary

Differentiate monomials through logarithmic derivatives, then differentiate well-based sums term by term. The transseries support architecture guarantees closure and the usual calculus rules. Sector differentiation preserves exponential sector labels, while nonlinear products

later mix them.

Formal integration and asymptotic antiderivatives

Integration is subtler than differentiation. The derivative of a single monomial is usually a monomial times a nonconstant transseries, so its antiderivative is often an infinite expansion rather than one term.

12.1 Elementary power cases

For $a \neq -1$,

$$\int x^a dx = \frac{x^{a+1}}{a+1}.$$

At the resonant exponent $a = -1$,

$$\int x^{-1} dx = \log x.$$

Then another logarithmic hierarchy appears:

$$\begin{aligned} \int \frac{1}{x \log x} dx &= \log \log x, \\ \int \frac{1}{x \log x \log \log x} dx &= \log_3 x. \end{aligned}$$

These are reminders that logarithmic depth is forced by integration, not added merely for decoration.

12.2 Antidifferentiating a fast monomial

Let m be a positive monomial and set

$$A = \frac{m'}{m} = (\log m)'.$$

We seek an antiderivative in the form

$$F = \frac{m}{A}(1 + U), \quad U \prec 1. \quad (12.1)$$

Define

$$\delta = \frac{A'}{A^2}.$$

Differentiate:

$$\begin{aligned} F' &= m(1+U) + \frac{m}{A}U' - \frac{mA'}{A^2}(1+U) \\ &= m \left[1+U + \frac{U'}{A} - \delta(1+U) \right]. \end{aligned}$$

The condition $F' = m$ becomes

$$U = \delta + \delta U - \frac{U'}{A}. \quad (12.2)$$

When $\delta \prec 1$ and differentiation divided by A makes terms smaller, this is a contractive formal fixed-point equation. Starting with $U_0 = 0$ and iterating the right side produces successively smaller corrections.

Intuition

The first guess m/A comes from treating $A = (\log m)'$ as locally constant: since $m' = Am$, division by A almost reverses differentiation. The infinite correction compensates for the fact that A itself varies.

12.3 Example: $\int e^{x^2} dx$

Here

$$m = e^{x^2}, \quad A = 2x, \quad \delta = \frac{2}{4x^2} = \frac{1}{2x^2}.$$

Seek

$$F = \frac{e^{x^2}}{2x} \left(1 + c_1 x^{-2} + c_2 x^{-4} + \cdots \right).$$

Substitution or repeated use of Equation (12.2) gives

$$c_1 = \frac{1}{2}, \quad c_2 = \frac{3}{4}, \quad c_3 = \frac{15}{8}, \quad c_4 = \frac{105}{16},$$

and in general

$$c_n = \frac{(2n-1)!!}{2^n}.$$

Therefore

$$\int e^{x^2} dx \sim \frac{e^{x^2}}{2x} \left(1 + \frac{1}{2x^2} + \frac{3}{4x^4} + \frac{15}{8x^6} + \frac{105}{16x^8} + \cdots \right). \quad (12.3)$$

The series diverges factorially, but it is a valid formal antiderivative.

Worked calculation

Differentiate the first three corrections:

$$F_2 = \frac{e^{x^2}}{2x} \left(1 + \frac{1}{2x^2} + \frac{3}{4x^4} \right).$$

A direct calculation gives

$$F'_2 = e^{x^2} \left(1 - \frac{15}{8x^6} \right).$$

The first omitted coefficient is exactly the one needed to cancel the residual term.

12.4 Example: $\int e^{e^x} dx$

Let $m = e^{e^x}$. Then

$$A = e^x, \quad \delta = e^{-x}.$$

The fixed-point expansion produces factorial coefficients:

$$\int e^{e^x} dx \sim e^{e^x - x} \left(1 + e^{-x} + 2e^{-2x} + 6e^{-3x} + 24e^{-4x} + \cdots \right). \quad (12.4)$$

Indeed, substituting $t = e^x$ gives

$$\int e^{e^x} dx = \int \frac{e^t}{t} dt = \text{Ei}(t),$$

and the familiar factorial expansion of $\text{Ei}(t)$ reappears with $t = e^x$.

This example is a useful scale elevator: the same divergent pattern occurs one exponential level higher.

12.5 Slow monomials and integration by parts

If m varies slowly, the approximation m/A may not be the right starting point. For example,

$$\int \frac{\log x}{x^2} dx = -\frac{\log x + 1}{x}.$$

One may integrate powers first and treat logarithms by ordinary integration by parts.

A practical integration algorithm classifies the leading monomial according to the size of its logarithmic derivative, chooses a leading antiderivative, subtracts its derivative, and repeats on the smaller residual. Van der Hoeven's algorithmic treatment makes this systematic [38].

12.6 Termwise integration and constants

It is tempting to write

$$\int \sum_m a_m m dx = \sum_m a_m \int m dx.$$

This is often correct, but formal summability of the family of antiderivatives must be checked. The general closure theorem is stronger than a naive termwise rule.

Theorem 12.1 (Antiderivative closure, stated without proof). *For every $T \in \mathbb{T}$ there exists $F \in \mathbb{T}$ such that $F' = T$. Any two antiderivatives differ by a real constant.*

Together with real closedness and closure under exponential integration, this is part of the statement that $\mathbb{T}$ is *Liouville closed*. The differential-algebraic meaning is discussed in Chapter 27.

12.7 An inverse-power recurrence

Suppose we want

$$\int x^\alpha e^{\lambda x} dx, \quad \lambda \neq 0.$$

Set

$$F = \frac{x^\alpha e^{\lambda x}}{\lambda} \sum_{n \geq 0} c_n x^{-n}, \quad c_0 = 1.$$

Differentiation gives the recurrence

$$c_{n+1} = -\frac{\alpha - n}{\lambda} c_n,$$

so

$$\int x^\alpha e^{\lambda x} dx \sim \frac{x^\alpha e^{\lambda x}}{\lambda} \left(1 - \frac{\alpha}{\lambda x} + \frac{\alpha(\alpha-1)}{\lambda^2 x^2} - \dots \right). \quad (12.5)$$

For nonnegative integer α , the series terminates and gives the elementary exact antiderivative. For general α , it is infinite and typically divergent.

12.8 Exercises

Exercise 12.1. Derive the recurrence for Equation (12.3) directly by substituting

$$F = e^{x^2} \sum_{n \geq 0} a_n x^{-2n-1}.$$

Exercise 12.2. Find the first four terms of an asymptotic antiderivative of

$$x^3 e^{-x^2}.$$

In this case the exact integral is elementary; compare the results.

Exercise 12.3. Apply Equation (12.5) to $\alpha = -1$ and $\lambda = 1$. Explain the relationship with the Euler factorial series.

Chapter summary

A monomial's logarithmic derivative suggests its leading antiderivative. Correcting m/A leads to a contractive fixed-point equation and often a divergent but useful transseries. Logarithmic depth appears at resonant power exponents, and the full transseries field is closed under antiderivatives.

Composition and formal Taylor expansion

Composition is where transseries become a language of functions rather than merely an ordered field. It is also one of the technically delicate operations, because substituting an infinite series into another infinite series can destroy support control if done indiscriminately.

13.1 Which inner arguments are allowed?

For the standard real transseries composition, the inner argument S is usually required to be positive and infinitely large:

$$S > 0, \quad S \succ 1.$$

This matches the interpretation that $T(x)$ describes behavior at $x = +\infty$ and we are replacing x by another quantity tending to $+\infty$.

At the elementary level, define composition recursively:

$$\begin{aligned} c \circ S &= c, \\ x \circ S &= S, \\ (A + B) \circ S &= (A \circ S) + (B \circ S), \\ (AB) \circ S &= (A \circ S)(B \circ S), \\ (\exp A) \circ S &= \exp(A \circ S), \\ (\log A) \circ S &= \log(A \circ S), \end{aligned}$$

followed by extension to well-based sums. The last extension is the hard part; Edgar proves appropriate support and convergence properties in [23].

13.2 First examples

Example 13.1. If

$$T(x) = x^2 + e^{-x}, \quad S(x) = \log x + x^{-1},$$

then

$$(T \circ S)(x) = (\log x + x^{-1})^2 + \exp(-\log x - x^{-1}).$$

The exponential term becomes

$$x^{-1} \exp(-x^{-1}) = x^{-1} - x^{-2} + \frac{1}{2}x^{-3} - \dots$$

Example 13.2. For $T(x) = \log x$ and $S = x + x^{-1}$,

$$\begin{aligned} T \circ S &= \log(x + x^{-1}) \\ &= \log x + \log(1 + x^{-2}) \\ &= \log x + x^{-2} - \frac{1}{2}x^{-4} + \frac{1}{3}x^{-6} - \dots \end{aligned}$$

13.3 Composition with a small perturbation

Suppose

$$S = S_0 + U, \quad U \prec S_0,$$

and the derivatives of T are well behaved relative to U . One expects a formal Taylor expansion

$$T(S_0 + U) = \sum_{n=0}^{\infty} \frac{T^{(n)}(S_0)}{n!} U^n. \quad (13.1)$$

For ordinary analytic functions this follows from convergence near a point. In transseries it is a theorem about support and formal summability. Its hypotheses depend on the relative depth of T , S_0 , and U ; careless use in “deep” cases can fail.

For the common case $U \prec 1$ and $S_0 = x$, the formula behaves exactly as intuition suggests.

Worked calculation

Expand $e^{x+x^{-1}}$ around x :

$$T(x) = e^x, \quad U = x^{-1}.$$

Since $T^{(n)}(x) = e^x$,

$$T(x + U) = e^x \sum_{n \geq 0} \frac{x^{-n}}{n!} = e^x \left(1 + x^{-1} + \frac{1}{2x^2} + \dots \right).$$

This agrees with the exponential decomposition of Chapter 9.

13.4 A less trivial Taylor calculation

Let $T(x) = \log x$ and $U = e^{-x}$. Since

$$T^{(n)}(x) = (-1)^{n-1} (n-1)! x^{-n} \quad (n \geq 1),$$

we obtain

$$\begin{aligned} \log(x + e^{-x}) &= \log x + \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} x^{-n} e^{-nx} \\ &= \log x + x^{-1} e^{-x} - \frac{1}{2} x^{-2} e^{-2x} + \frac{1}{3} x^{-3} e^{-3x} - \dots \end{aligned}$$

Again this is the same result obtained by factoring x and expanding $\log(1 + x^{-1}e^{-x})$.

13.5 Associativity and the chain rule

For admissible positive infinitely large arguments,

$$(T \circ S) \circ U = T \circ (S \circ U).$$

Composition has identity x :

$$T \circ x = x \circ T = T.$$

Together with Equation (11.4), these properties make suitable transseries into a group under composition whenever compositional inverses exist.

13.6 Compositional inverses

Suppose T is positive, infinitely large, and strictly increasing in the formal sense $T' > 0$. Under broad conditions there is a transseries S such that

$$T \circ S = S \circ T = x.$$

The leading monomial of S is found by reversing the dominant behavior of T . Corrections can then be obtained by fixed-point or Newton iteration.

Examples:

$$\begin{aligned} T = x^a \quad (a > 0) &\implies T^{-1} = x^{1/a}, \\ T = e^x &\implies T^{-1} = \log x, \\ T = x + \log x &\implies T^{-1} = x - \log x + \frac{\log x}{x} + \cdots. \end{aligned}$$

The inverse of xe^x is Lambert's W function and provides our main worked example in Chapter 14.

13.7 Taylor's theorem in the transseries setting

A modern formal Taylor theorem can be summarized as follows.

Theorem 13.1 (Formal Taylor principle, schematic version). *Let $T, S, U \in \mathbb{T}$, with S positive and infinitely large and U sufficiently small relative to the variation scale of T at S . Then the family*

$$\left(\frac{T^{(n)} \circ S}{n!} U^n \right)_{n \geq 0}$$

is formally summable and its sum equals $T \circ (S + U)$.

The phrase “sufficiently small” cannot be deleted. Edgar distinguishes shallow, moderate, and deep substitutions according to comparisons between U and logarithmic derivatives of monomials [23, 24]. Recent work continues to refine Taylor principles for generalized series and broader transserial settings [5, 31].

13.8 Why composition can fail naively

Consider a formal sum $T = \sum_n m_n$. After substitution, it could happen that infinitely many distinct $m_n \circ S$ collapse onto the same monomial, creating an undefined infinite coefficient sum. Or the substituted support might acquire an infinite increasing chain. The composition theorems rule out such behavior under their hypotheses.

A second danger is using a Taylor expansion when U is not small on the derivative scale. For a very deep monomial m , the quantity

$$\frac{m'}{m}U$$

may be large even when $U \prec x$. Then successive Taylor terms need not decrease.

A common trap

The condition $U \prec S$ is not by itself enough for every Taylor expansion. Check the logarithmic derivative: a useful first test is whether $U(T'/T) \circ S \prec 1$ for the monomials that dominate T .

13.9 Exercises

Exercise 13.1. Compute $\log(x + \log x)$ through terms of order $(\log x)^3/x^3$.

Exercise 13.2. Let $T(x) = x^3$ and $S = x + e^{-x}$. Verify the formal Taylor expansion directly and observe that it terminates.

Exercise 13.3. Find the first three corrections to the compositional inverse of $T = x + \log x$.

Chapter summary

Composition substitutes one positive infinitely large transseries into another, with support conditions doing the work that analytic convergence would do for ordinary functions. Formal Taylor expansion is valid when the perturbation is small on the relevant derivative scale. Composition supports chain rules, inverses, and iteration.

Lambert's W function as a transseries inverse

Lambert's function W is defined by

$$W(x)e^{W(x)} = x. \quad (14.1)$$

It is the compositional inverse of $T \mapsto Te^T$. Its expansion at $+\infty$ is a model calculation: dominant balance finds the logarithmic scales, and a contractive equation generates all coefficients.

14.1 Find the dominant term

Taking logarithms of Equation (14.1) gives

$$W + \log W = \log x. \quad (14.2)$$

Since $\log W \prec W$ for a large W , the first approximation is

$$W \sim \log x.$$

Substituting this back into the smaller term suggests

$$W \approx \log x - \log \log x.$$

Introduce the standard abbreviations

$$L_1 = \log x, \quad L_2 = \log \log x. \quad (14.3)$$

We therefore write

$$W = L_1 - L_2 + u, \quad u \prec L_2. \quad (14.4)$$

14.2 Derive a fixed-point equation

Because

$$W = L_1 \left(1 - \frac{L_2 - u}{L_1} \right),$$

we have

$$\log W = L_2 + \log \left(1 - \frac{L_2 - u}{L_1} \right).$$

Substitute into Equation (14.2). The L_1 and L_2 terms cancel, leaving

$$u = -\log \left(1 - \frac{L_2 - u}{L_1} \right). \quad (14.5)$$

The ratio L_2/L_1 is small. The right side can therefore be expanded and iterated.

A tempting alternative is to set $W = L_1 + Q$ and use

$$Q = -\log(L_1 + Q).$$

That also works after proper normalization, but Equation (14.5) has already removed the large correction $-L_2$ and is more strongly contractive.

14.3 Coefficient extraction

Set

$$r = L_1^{-1}, \quad \ell = L_2, \quad u = \sum_{n \geq 1} p_n(\ell) r^n.$$

Equation (14.5) becomes

$$u = -\log(1 - r(\ell - u)) = \sum_{k \geq 1} \frac{r^k (\ell - u)^k}{k}.$$

Equate powers of r .

At order r ,

$$p_1 = \ell.$$

At order r^2 ,

$$p_2 = -p_1 + \frac{\ell^2}{2} = \frac{\ell(-2 + \ell)}{2}.$$

Continuing gives

$$\begin{aligned} p_3(\ell) &= \frac{\ell(6 - 9\ell + 2\ell^2)}{6}, \\ p_4(\ell) &= \frac{\ell(-12 + 36\ell - 22\ell^2 + 3\ell^3)}{12}, \\ p_5(\ell) &= \frac{\ell(60 - 300\ell + 350\ell^2 - 125\ell^3 + 12\ell^4)}{60}. \end{aligned}$$

Thus

$$\begin{aligned}
 W(x) \sim & L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2 + L_2)}{2L_1^2} \\
 & + \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} \\
 & + \frac{L_2(-12 + 36L_2 - 22L_2^2 + 3L_2^3)}{12L_1^4} \\
 & + \frac{L_2(60 - 300L_2 + 350L_2^2 - 125L_2^3 + 12L_2^4)}{60L_1^5} + \dots .
 \end{aligned} \tag{14.6}$$

This is a logarithmic-power transseries: each power of $1/L_1$ carries a polynomial in L_2 .

14.4 Why the recursion stabilizes

Let

$$\Phi(u) = -\log\left(1 - \frac{L_2 - u}{L_1}\right).$$

Then

$$\Phi'(u) = -\frac{1/L_1}{1 - (L_2 - u)/L_1}.$$

Formally, $\Phi'(u) \sim -L_1^{-1} \prec 1$. Therefore a difference between two approximations gains a factor of order $1/L_1$ after one iteration. The coefficient of L_1^{-N} eventually becomes permanent.

Starting with $u_0 = 0$:

$$\begin{aligned}
 u_1 &= -\log(1 - L_2/L_1), \\
 u_2 &= -\log\left(1 - \frac{L_2 - u_1}{L_1}\right), \\
 &\vdots
 \end{aligned}$$

Each iterate contains a longer correct initial segment of Equation (14.6).

14.5 A quick formal verification

Differentiate Equation (14.1):

$$W'e^W + We^W W' = 1.$$

Since $We^W = x$,

$$W'(x) = \frac{W(x)}{x(1 + W(x))}. \tag{14.7}$$

Substituting the first terms

$$W = L_1 - L_2 + \mathcal{O}(L_2/L_1)$$

gives

$$W' = \frac{1}{x} \left(1 - \frac{1}{L_1} + \frac{1 - L_2}{L_1^2} + \cdots \right),$$

consistent with termwise differentiation of Equation (14.6).

14.6 Accuracy and truncation

Unlike the Euler factorial series, the classical large- x expansion of W in inverse powers of L_1 has a more delicate convergence/asymptotic story depending on the domain and organization of terms. For transseries purposes, the safe statement is formal-asymptotic: for fixed truncation order, the residual in Equation (14.2) moves to a smaller logarithmic scale.

At $x = e^{100}$,

$$L_1 = 100, \quad L_2 = \log 100 \approx 4.60517.$$

Already

$$L_1 - L_2 + L_2/L_1$$

provides a strong approximation because $L_2/L_1 \approx 0.046$.

14.7 The real branch W_{-1} near zero

For $-e^{-1} < x < 0$, Lambert W has two real branches. The branch $W_{-1}(x)$ tends to $-\infty$ as $x \rightarrow 0^-$. Write

$$x = -e^{-\eta}, \quad \eta \rightarrow +\infty, \quad W_{-1}(x) = -v.$$

Then

$$ve^{-v} = e^{-\eta}, \quad v - \log v = \eta.$$

Let $L = \log \eta$. Reversing the same logarithmic calculation gives

$$\begin{aligned} W_{-1}(-e^{-\eta}) \sim & -\eta - L - \frac{L}{\eta} + \frac{L(-2 + L)}{2\eta^2} \\ & - \frac{L(6 - 9L + 2L^2)}{6\eta^3} + \cdots \end{aligned} \quad (14.8)$$

The two real branches thus share the same universal polynomial pattern after the appropriate large logarithm and signs are chosen.

14.8 The branch point uses a different scale

Near $x = -e^{-1}$, the derivative Equation (14.7) blows up because $1 + W = 0$. The correct local variable is a square root, not an inverse logarithm. Set

$$p = \sqrt{2(1 + ex)}.$$

Then the two branches have Puiseux expansions

$$W_{0,-1}(x) = -1 \pm p - \frac{p^2}{3} \pm \frac{11p^3}{72} - \frac{43p^4}{540} + \dots \quad (14.9)$$

This is a useful reminder: the dominant balance determines the correct transseries scale. A logarithmic expansion suited to infinity is not transported unchanged to a branch point.

14.9 A general inversion recipe

For a monotone large transseries F :

- Algorithm 14.1** (Formal compositional inversion). 1. Find a leading inverse G_0 from dominant balance.
2. Write $G = G_0 + U$ with $U \prec G_0$.
 3. Substitute into $F(G) = x$.
 4. Isolate the dominant linear occurrence of U and rewrite as $U = \Phi(U)$.
 5. Iterate until the desired dominance threshold stabilizes.
 6. Verify by composing the truncated answer back into F .

Lambert W is the prototype because the inverse necessarily mixes two logarithmic levels.

14.10 Exercises

Exercise 14.1. Starting from Equation (14.5), derive $p_3(\ell)$ explicitly.

Exercise 14.2. Substitute the expansion through L_1^{-2} into $W + \log W - L_1$ and show that the residual is of order L_2^3/L_1^3 .

Exercise 14.3. Derive the first two corrections in Equation (14.8) directly from $v - \log v = \eta$.

Chapter summary

Lambert W is found by peeling away the dominant logarithm and the next iterated logarithm, then solving a contractive fixed-point equation in $1/\log x$. Its coefficient polynomials illustrate how transseries naturally combine several nested scales. Different singular points require different scale hierarchies.

Formal fixed points and coefficient stabilization

Fixed-point iteration in transseries resembles numerical iteration only superficially. The relevant notion of closeness is dominance: an error becomes smaller when its leading monomial moves downward in the support order.

15.1 A valuation contraction principle

Let Φ act on a class of transseries. Suppose there is a small monomial $\mu \prec 1$ such that

$$\Phi(S) - \Phi(T) \prec \mu(S - T) \quad (15.1)$$

for all S, T in the class. Then each iteration gains at least one factor of μ .

Theorem 15.1 (Formal contraction principle, schematic form). *Under suitable well-based support hypotheses, if Equation (15.1) holds and $T_{n+1} = \Phi(T_n)$, then the differences $T_{n+1} - T_n$ move to successively smaller monomials. The sequence converges coefficientwise to a unique fixed point $T = \Phi(T)$ in the class.*

The rigorous versions use ratio sets, valuations, or complete ultrametric spaces. For calculation, the operational test is simple: every pass must push the unresolved residual beyond the current truncation.

15.2 The Catalan fixed point

Consider

$$S = A + S^2, \quad A \prec 1. \quad (15.2)$$

Start with $S_0 = 0$:

$$\begin{aligned} S_1 &= A, \\ S_2 &= A + A^2, \\ S_3 &= A + A^2 + 2A^3 + A^4, \\ S_4 &= A + A^2 + 2A^3 + 5A^4 + 6A^5 + \cdots \end{aligned}$$

The stable coefficients are the Catalan numbers:

$$S = A + A^2 + 2A^3 + 5A^4 + 14A^5 + 42A^6 + \dots \quad (15.3)$$

Algebraically,

$$S = \frac{1 - \sqrt{1 - 4A}}{2},$$

and the branch is selected by $S \prec 1$.

For two small candidates S, T ,

$$\Phi(S) - \Phi(T) = S^2 - T^2 = (S + T)(S - T),$$

and $S + T \prec 1$, so the map is contractive.

15.3 Residual correction

A flexible method is to work with the residual. Suppose $P(Y) = 0$ and Y_0 is an approximation. Write

$$Y = Y_0 + U.$$

Expand

$$P(Y_0 + U) = P(Y_0) + P'(Y_0)U + \text{higher powers of } U.$$

If $P'(Y_0)$ has an inverse and the nonlinear remainder is smaller, isolate

$$U = -\frac{P(Y_0)}{P'(Y_0)} + \text{smaller terms involving } U. \quad (15.4)$$

This is Newton's correction reorganized as a fixed-point equation.

15.4 A square-root example

Solve

$$Y^2 = x^2 + x + e^{-x}$$

with $Y > 0$. Dominant balance gives $Y_0 = x$. Write

$$Y = x(1 + S).$$

Then

$$(1 + S)^2 = 1 + x^{-1} + x^{-2}e^{-x},$$

so

$$2S + S^2 = x^{-1} + x^{-2}e^{-x}$$

and

$$S = \frac{1}{2}x^{-1} + \frac{1}{2}x^{-2}e^{-x} - \frac{1}{2}S^2. \quad (15.5)$$

Iteration gives

$$S = \frac{1}{2x} - \frac{1}{8x^2} + \frac{1}{16x^3} - \frac{5}{128x^4} + \cdots \\ + \frac{1}{2}x^{-2}e^{-x} - \frac{1}{4}x^{-3}e^{-x} + \frac{3}{16}x^{-4}e^{-x} + \cdots + O(e^{-2x}).$$

The power sector is the binomial expansion of $\sqrt{1+x^{-1}}$; the exponential forcing generates its own correction sector.

15.5 Fixed points for differential operators

The integration equation Equation (12.2) has the form

$$U = \delta + \mathcal{A}(U), \quad \mathcal{A}(U) = \delta U - U'/A.$$

If both multiplication by δ and the operator $A^{-1}\partial$ make monomials smaller, then $\mathcal{A}$ is contractive. The solution is a Neumann series in the operator:

$$U = (I - \mathcal{A})^{-1}\delta = \delta + \mathcal{A}\delta + \mathcal{A}^2\delta + \cdots.$$

This is the operator analogue of $1/(1-S) = 1 + S + S^2 + \cdots$.

15.6 Newton iteration versus plain iteration

A fixed-point rearrangement can converge slowly in dominance order if it gains only one small ratio per step. Newton iteration often doubles the amount of correct support, because the new residual is quadratic in the old error:

$$Y_{n+1} = Y_n - \frac{P(Y_n)}{P'(Y_n)}.$$

For formal power series, this underlies fast algorithms for reciprocal, square root, exponential, and reversion. The same idea extends to finite truncations of grid-based transseries, although support comparison and composition are more expensive.

15.7 How an iteration fails

An equation can be algebraically equivalent yet computationally useless. In the Lambert problem, writing

$$Q = e^{-Q} - \log x$$

and starting from $Q_0 = 0$ gives

$$Q_1 = 1 - \log x, \quad Q_2 = x/e - \log x,$$

which moves to a larger scale instead of a smaller one. The map is not contractive in the intended region.

A good rearrangement isolates the variable through an operator whose derivative is small. This is why

$$u = -\log\left(1 - \frac{L_2 - u}{L_1}\right)$$

works: differentiation with respect to u introduces $1/L_1 \prec 1$.

15.8 A practical stopping rule

Suppose the goal is an answer through threshold m_* . Stop when

$$T_{n+1} - T_n \prec m_*.$$

Then all terms larger than or equal to m_* have stabilized, assuming the contraction estimate persists. This is the formal analogue of stopping a numerical iteration when the error estimate falls below tolerance.

Listing 15.1: Dominance-driven fixed-point iteration

```
input: map Phi, initial approximation T, monomial threshold cutoff
repeat:
  U := truncate(Phi(T), cutoff)
  if truncate(U - T, cutoff) = 0:
    return U
  T := U
```

The operation `truncate` is essential. Exact untruncated intermediates may be infinite and need not be materialized.

15.9 Exercises

Exercise 15.1. Use fixed-point iteration to solve $S = A(1 + S)^2$ through A^5 .

Exercise 15.2. Derive a contractive equation for the positive solution of $Y^3 = x^3 + x$ and compute the first four corrections.

Exercise 15.3. For Equation (15.2), prove uniqueness among infinitesimal solutions by subtracting two fixed-point equations.

Chapter summary

Formal fixed-point iteration converges when differences gain small monomial factors. Coefficients stabilize one dominance layer at a time. The art lies in choosing a rearrangement whose linearized operator is small; algebraic equivalence alone does not guarantee a useful iteration.

Fractional iteration and Abel's equation

Composition permits integer iterates

$$T^{[0]} = x, \quad T^{[1]} = T, \quad T^{[2]} = T \circ T, \quad \dots$$

A natural question is whether the discrete index can be replaced by a real parameter s so that

$$T^{[s+t]} = T^{[s]} \circ T^{[t]}, \quad T^{[1]} = T. \quad (16.1)$$

Such a family is an iteration group.

16.1 Abel's linearizing coordinate

Suppose there is a transseries V satisfying Abel's equation

$$V \circ T = V + 1. \quad (16.2)$$

Then define

$$T^{[s]} = V^{-1} \circ (V + s). \quad (16.3)$$

Indeed,

$$\begin{aligned} T^{[s]} \circ T^{[t]} &= V^{-1} \circ (V + s) \circ V^{-1} \circ (V + t) \\ &= V^{-1} \circ (V + s + t) = T^{[s+t]}. \end{aligned}$$

Abel's coordinate turns composition by T into ordinary translation by 1.

16.2 Translation

For

$$T(x) = x + 1,$$

take $V(x) = x$. Then

$$T^{[s]}(x) = x + s.$$

This is the simplest possible iteration group.

16.3 Scaling

Let

$$T(x) = ax, \quad a > 0, \quad a \neq 1.$$

Take

$$V(x) = \frac{\log x}{\log a}.$$

Then

$$V(ax) = V(x) + 1,$$

so

$$T^{[s]}(x) = a^s x. \quad (16.4)$$

The familiar real power a^s is exactly a fractional iterate of multiplication by a .

16.4 Power maps

Let

$$T(x) = x^a, \quad a > 0, \quad a \neq 1.$$

Now

$$V(x) = \frac{\log \log x}{\log a}$$

satisfies

$$V(x^a) = \frac{\log(a \log x)}{\log a} = V(x) + 1.$$

Therefore

$$T^{[s]}(x) = x^{a^s}. \quad (16.5)$$

At $s = 1/2$, the compositional square root of x^a is $x^{\sqrt{a}}$:

$$(x^{\sqrt{a}}) \circ (x^{\sqrt{a}}) = x^a.$$

16.5 Near-identity maps and iterative logarithms

If

$$T = x + \text{small correction},$$

one may seek a formal vector field

$$\mathcal{V} = v(x) \frac{d}{dx}$$

whose time-one flow is T . Formally,

$$T = \exp(\mathcal{V})x = x + \mathcal{V}x + \frac{\mathcal{V}^2 x}{2!} + \cdots. \quad (16.6)$$

The operator $\mathcal{V}$ is called an iterative logarithm, and

$$T^{[s]} = \exp(s\mathcal{V})x.$$

This is composition's analogue of writing $a^s = \exp(s \log a)$.

For example, if

$$T = x + ax^{-1} + O(x^{-2}),$$

then the leading vector field is

$$\mathcal{V} \sim ax^{-1} \partial_x,$$

and the fractional iterate begins

$$T^{[s]} = x + sax^{-1} + \dots$$

Higher coefficients must correct the difference between a single displacement and a flow.

16.6 Shallow, moderate, and deep iteration

The Taylor expansion used to compose T with a perturbation depends on how the perturbation compares with logarithmic derivatives of the monomials. Edgar classifies cases as shallow, moderate, and deep [24]. Roughly:

- In a shallow case, each composition correction is clearly smaller and a direct Taylor-recursive construction works.
- In a moderate case, leading derivative effects are comparable and must be resummed or normalized.
- In a deep case, a nominally small displacement can produce large relative changes in some monomials, so the naive Taylor iteration fails.

The classification is not a judgment of difficulty alone; it is a comparison of transseries scales.

16.7 Why the exponential map is special

For $T(x) = e^x$, integer iteration raises exponential height:

$$x, \quad e^x, \quad e^{e^x}, \quad \dots$$

A fractional iterate would be a form of tetration. It cannot be represented by simply interpolating coefficients inside one fixed finite-height sector. Standard logarithmic-exponential transseries are not designed to contain a globally satisfactory half-iterate of the exponential. Larger structures involving superexponentials, omega-series, surreal analysis, or other extensions are needed for such questions [35, 10].

16.8 Conjugacy is the unifying idea

Abel's equation is one instance of conjugating a complicated map to a simple normal form:

$$V \circ T \circ V^{-1} = x + 1.$$

Near a fixed point with multiplier $\lambda \neq 0, 1$, Schröder's equation instead seeks

$$V \circ T = \lambda V,$$

leading to

$$T^{[s]} = V^{-1}(\lambda^s V).$$

Transseries provide the asymptotic language for constructing such conjugacies when ordinary Taylor series are insufficient.

16.9 Exercises

Exercise 16.1. Verify Equation (16.5) directly from the group law.

Exercise 16.2. Find an Abel coordinate and fractional iterates for $T(x) = x + c$ with nonzero real c .

Exercise 16.3. Suppose $T = V^{-1} \circ (V + 1)$. Show that replacing V by $V + c$ does not change the resulting iteration group.

Chapter summary

Fractional iteration is obtained by conjugating composition to translation or scaling. Abel's equation gives $T^{[s]} = V^{-1}(V + s)$. Simple maps yield familiar formulas, while the exponential map points beyond finite-height logarithmic-exponential transseries toward tetration and larger transserial fields.

Part IV

Solving equations with transseries

Dominant balance and transseries

Newton methods

A zero sum cannot have exactly one uncanceled largest monomial. This elementary observation is the transseries version of dominant balance and the starting point for solving algebraic and differential equations.

17.1 The largest terms must cancel

Suppose

$$T_1 + T_2 + \cdots + T_r = 0.$$

If one T_j were strictly larger than all the others, its leading monomial could not be canceled. Therefore at least two summands must share the maximal dominance scale, and their leading coefficients must add to zero.

For a polynomial equation

$$P(Y) = a_0 + a_1Y + \cdots + a_dY^d = 0,$$

we guess a leading monomial $Y \sim c\mathfrak{m}$. Each term has leading monomial

$$\text{lm}(a_j)\mathfrak{m}^j.$$

A viable guess makes the maximum among these monomials occur at least twice. This is the valuation-theoretic form of the Newton polygon rule.

Algorithm 17.1 (Dominant balance). 1. List the monomial scale of every term under a trial $Y \sim c\mathfrak{m}$.

2. Equate pairs of candidate dominant scales to determine $\mathfrak{m}$.
3. Reject candidates for which another term becomes strictly larger.
4. Solve the leading coefficient equation for c .
5. Write $Y = c\mathfrak{m}(1 + S)$ with $S \prec 1$ and derive a fixed-point equation for S .

17.2 A cubic with an exponential coefficient

Consider

$$Y^3 - xY - e^x = 0. \quad (17.1)$$

The dominant exponential suggests balancing Y^3 with e^x :

$$Y \sim e^{x/3}.$$

Then

$$xY \sim xe^{x/3} \prec e^x,$$

so the third term is indeed smaller. The real leading coefficient satisfies $c^3 = 1$, giving $c = 1$.

Set

$$Y = e^{x/3}(1 + u), \quad \varepsilon = xe^{-2x/3} \prec 1.$$

After division by e^x , Equation (17.1) becomes

$$(1 + u)^3 - 1 = \varepsilon(1 + u). \quad (17.2)$$

Expand:

$$3u + 3u^2 + u^3 = \varepsilon + \varepsilon u.$$

Solving recursively gives

$$u = \frac{\varepsilon}{3} - \frac{\varepsilon^3}{81} + \frac{\varepsilon^4}{243} - \frac{4\varepsilon^6}{6561} + \frac{5\varepsilon^7}{19683} + \dots \quad (17.3)$$

Therefore the real root has transseries

$$\begin{aligned} Y = e^{x/3} + \frac{x}{3}e^{-x/3} - \frac{x^3}{81}e^{-5x/3} + \frac{x^4}{243}e^{-7x/3} \\ - \frac{4x^6}{6561}e^{-11x/3} + \frac{5x^7}{19683}e^{-13x/3} + \dots \end{aligned} \quad (17.4)$$

The other two roots over $\mathbb{C}$ begin with $\omega e^{x/3}$, where $\omega^3 = 1$ and $\omega \neq 1$; their correction equations use the corresponding coefficient.

17.3 How to obtain the coefficients

Write

$$u = \sum_{n \geq 1} a_n \varepsilon^n.$$

Substitute into Equation (17.2) and compare powers. At order ε ,

$$3a_1 = 1,$$

so $a_1 = 1/3$. At order ε^2 ,

$$3a_2 + 3a_1^2 = a_1,$$

so $a_2 = 0$. At order ε^3 ,

$$3a_3 + 6a_1a_2 + a_1^3 = a_2,$$

so $a_3 = -1/81$. The zero coefficients are genuine structural cancellations, not missing work.

17.4 A richer fifth-degree example

Edgar's introductory paper studies

$$Y^5 + e^x Y^2 - xY - 9 = 0 \quad (17.5)$$

by exactly this method [22]. Let $Y \sim ax^b e^L$. The exponential parts of the four terms are

$$5L, \quad x + 2L, \quad L, \quad 0.$$

A dominant cancellation is possible only at

$$L = \frac{x}{3} \quad \text{or} \quad L = -\frac{x}{2}.$$

For $L = x/3$, balancing Y^5 with $e^x Y^2$ gives

$$a^5 + a^2 = 0, \quad a^3 = -1.$$

There are three complex large roots; one is real with $a = -1$. Its beginning is

$$Y = -e^{x/3} - \frac{x}{3}e^{-x} + 3e^{-4x/3} + \frac{2x^2}{9}e^{-7x/3} + \dots \quad (17.6)$$

For $L = -x/2$, balancing $e^x Y^2$ with 9 gives $a = \pm 3$, producing two small real roots:

$$Y = \pm 3e^{-x/2} + \frac{x}{2}e^{-x} \pm \frac{x^2}{24}e^{-3x/2} + \dots \quad (17.7)$$

Together these balances account for all five roots. A seemingly difficult quintic becomes a collection of recursive small-correction problems.

17.5 The Newton polygon picture

Suppose the coefficient a_j has a valuation or exponential order v_j . Plot the points

$$(j, v_j).$$

A candidate root valuation corresponds to a slope of a supporting edge of the lower convex hull, with sign conventions depending on how the valuation is oriented. Points on the same edge represent terms that tie for dominance. Their coefficients form the leading polynomial for c .

For transseries coefficients, a single real-valued valuation may not capture the full nested hierarchy, but the same “maximum attained twice” principle survives in the ordered monomial group.

17.6 Newton correction after dominant balance

Once Y_0 contains several terms, compute

$$R = P(Y_0).$$

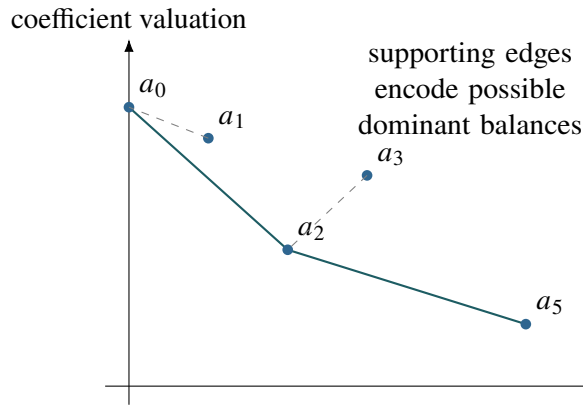


Figure 17.1: A schematic Newton polygon. Only points on a supporting edge participate in a leading cancellation.

If $P'(Y_0)$ is nonzero, set

$$\Delta = -\frac{R}{P'(Y_0)}.$$

Then

$$P(Y_0 + \Delta) = O(\Delta^2 P''(Y_0)),$$

so the residual often drops rapidly in dominance. This is an efficient way to extend a long expansion without re-deriving a fixed-point equation at every order.

17.7 Degenerate balances

If the leading coefficient polynomial has a multiple root, then $P'(Y_0)$ may be unusually small and ordinary Newton correction loses strength. New scales, including fractional powers or logarithms, can appear. Near a turning point or algebraic branch point, Puiseux series are the simplest example.

For differential equations, resonance can play the same role: the linearized operator fails to invert on a particular monomial, forcing logarithmic factors or additional free parameters.

17.8 Exercises

Exercise 17.1. Find all leading balances for

$$Y^4 - e^x Y + x^2 = 0.$$

How many real leading coefficients arise from each balance?

Exercise 17.2. Derive the coefficient of ε^4 in Equation (17.3).

Exercise 17.3. For Equation (17.5), verify that no value of L other than $x/3$ or $-x/2$ can make the maximal exponential order occur twice.

Chapter summary

A valid leading balance makes at least two maximal monomials cancel. After the leading monomial and coefficient are fixed, the equation becomes a small-correction fixed point or Newton iteration. The method is a transseries generalization of Newton polygons and Puiseux expansions.

Linear differential equations and WKB transseries

Linear equations already exhibit divergent sectors, exponentially small complementary solutions, turning points, and Stokes phenomena. Their structure is simpler than that of nonlinear equations because sectors do not multiply into new sectors.

18.1 First-order equations

Consider

$$y' + a(x)y = b(x). \quad (18.1)$$

Let

$$A(x) = \int a(x) \, dx.$$

Formally,

$$(e^A y)' = e^A b,$$

so

$$y = e^{-A} \left(C + \int e^A b \, dx \right). \quad (18.2)$$

The homogeneous monomial e^{-A} is the scale on which the integration constant lives. A power-series particular solution may be blind to it, exactly as in Chapter 2.

Example 18.1. For

$$y' + xy = 1,$$

we have $A = x^2/2$ and

$$y = e^{-x^2/2} \left(C + \int e^{x^2/2} \, dx \right).$$

Formal integration gives

$$y \sim \frac{1}{x} + \frac{1}{x^3} + \frac{3}{x^5} + \frac{15}{x^7} + \cdots + C e^{-x^2/2}.$$

The coefficients are odd double factorials, and the missing constant lies on the scale $e^{-x^2/2}$.

18.2 Second-order normal form

A broad class of equations can be transformed into

$$y'' = Q(x)y. \quad (18.3)$$

Set

$$y = \exp \left(\int p(x) dx \right).$$

Then

$$y' = py, \quad y'' = (p' + p^2)y,$$

so p satisfies the Riccati equation

$$p' + p^2 = Q. \quad (18.4)$$

When $Q \succ 1$ is positive, dominant balance gives

$$p \sim \pm \sqrt{Q}.$$

The two signs generate growing and decaying WKB sectors.

18.3 The first WKB corrections

Let $q = \sqrt{Q}$ and choose $s = \pm 1$. Write

$$p = sq + r_0 + r_1 + \cdots,$$

with descending corrections. Substitution into Equation (18.4) gives at the first correction

$$2sq r_0 + sq' = 0,$$

so

$$r_0 = -\frac{q'}{2q} = -\frac{Q'}{4Q}. \quad (18.5)$$

At the next order,

$$2sq r_1 + r_0' + r_0^2 = 0,$$

which yields

$$r_1 = s \left(\frac{Q''}{8Q^{3/2}} - \frac{5(Q')^2}{32Q^{5/2}} \right). \quad (18.6)$$

Integrating r_0 produces the universal amplitude $Q^{-1/4}$. Thus

$$y_{\pm}(x) \sim Q(x)^{-1/4} \exp \left(\pm \int^x \sqrt{Q(t)} dt \right) (1 + \text{smaller corrections}). \quad (18.7)$$

18.4 The Airy equation

For

$$y'' = xy, \quad (18.8)$$

we have $Q = x$ and

$$\int \sqrt{x} dx = \frac{2}{3}x^{3/2}.$$

The formal solutions are

$$y_{\pm}(x) \sim x^{-1/4} \exp\left(\pm \frac{2}{3}x^{3/2}\right) \times \left(1 \pm \frac{5}{48}x^{-3/2} + \frac{385}{4608}x^{-3} \pm \frac{85085}{663552}x^{-9/2} + \dots\right). \quad (18.9)$$

Up to normalization, the decaying sign corresponds to $\text{Ai}(x)$ and the growing sign to $\text{Bi}(x)$ on the positive real axis.

The coefficient series diverge factorially. In the complex plane, which exponential is dominant depends on the argument of x , and the actual solution's decomposition changes across Stokes rays. Chapter 23 returns to this example.

18.5 Deriving the Airy recurrence

Take

$$y = x^{-1/4} e^{s\zeta} \sum_{n \geq 0} a_n x^{-3n/2}, \quad \zeta = \frac{2}{3}x^{3/2}, \quad s = \pm 1, \quad a_0 = 1.$$

Substitute into $y'' = xy$. Collecting powers gives a recurrence equivalent to

$$a_{n+1} = s \frac{(6n+1)(6n+5)}{48(n+1)} a_n. \quad (18.10)$$

For the decaying solution $s = -1$, signs alternate; for the growing solution $s = +1$, they do not. The magnitude behaves roughly like a constant times $n!$, explaining divergence.

18.6 Factorization of the differential operator

If p solves Equation (18.4), then noncommutative operator multiplication gives

$$(\partial + p)(\partial - p) = \partial^2 - p' - p^2 = \partial^2 - Q, \quad (18.11)$$

where $\partial = d/dx$. Thus a transseries solution of the Riccati equation formally factors the second-order operator.

Conversely, constructing an asymptotic factorization is equivalent to constructing WKB logarithmic derivatives. This bridge is valuable in symbolic differential algebra.

18.7 Turning points

The WKB scale assumes Q stays away from zero. If $Q(x_0) = 0$, then $Q^{-1/4}$ blows up and the correction hierarchy collapses. Near a simple turning point, rescaling reduces the equation to Airy's equation. A local Airy approximation is then matched to outer WKB expansions.

Turning points are places where the correct transseries changes its monomial architecture. They also emit Stokes curves in the complex plane.

18.8 Inhomogeneous equations

For

$$y'' - Qy = f,$$

a particular transseries can be found by variation of parameters or by formally inverting the differential operator. If y_+, y_- are homogeneous WKB solutions with Wronskian W , then

$$y_p = -y_+ \int \frac{y_- f}{W} dx + y_- \int \frac{y_+ f}{W} dx.$$

Each integral may require its own asymptotic transseries. The full solution includes two independent constants multiplying y_+ and y_- .

18.9 Exercises

Exercise 18.1. Find the formal particular solution and homogeneous scale for

$$y' + x^2 y = x.$$

Exercise 18.2. Derive Equation (18.6) from the Riccati equation.

Exercise 18.3. Use Equation (18.10) to reproduce the first three correction coefficients in Equation (18.9).

Chapter summary

Linear first-order equations attach the integration constant to a homogeneous exponential scale. Second-order equations lead through a Riccati substitution to WKB transseries with two exponential sectors. Their divergent coefficients and changing dominance in the complex plane prepare the Stokes phenomenon.

Nonlinear equations and the tower of sectors

A nonlinear equation does more than attach a single homogeneous exponential to a perturbative solution. Products of the first exponential correction generate its powers, and each power carries a new asymptotic series.

19.1 A Riccati model

Consider

$$y' + y = y^2 + \frac{1}{x}. \quad (19.1)$$

We first seek a perturbative inverse-power sector

$$\phi_0(x) = \sum_{n=1}^{\infty} a_n x^{-n}. \quad (19.2)$$

Substitution gives

$$\phi_0' + \phi_0 = \phi_0^2 + x^{-1}.$$

The coefficient of x^{-1} gives $a_1 = 1$. For $m \geq 2$,

$$a_m = (m-1)a_{m-1} + \sum_{i=1}^{m-1} a_i a_{m-i}. \quad (19.3)$$

Hence

$$\phi_0 = x^{-1} + 2x^{-2} + 8x^{-3} + 44x^{-4} + 296x^{-5} + 2312x^{-6} + \cdots. \quad (19.4)$$

The rapid coefficient growth already suggests divergence.

19.2 Linearize to discover the first exponential

Write

$$y = \phi_0 + \eta.$$

Subtracting the equation for ϕ_0 gives

$$\eta' + (1 - 2\phi_0)\eta = \eta^2. \quad (19.5)$$

Ignoring the quadratic term for an infinitesimal η , the linearized equation has solution

$$\eta \sim C \exp\left(-\int (1 - 2\phi_0) dx\right).$$

Since $\phi_0 \sim x^{-1}$,

$$-\int (1 - 2\phi_0) dx = -x + 2 \log x + \text{smaller terms}.$$

Therefore the first beyond-all-orders scale is

$$C e^{-x} x^2 (1 + \text{inverse-power corrections}). \quad (19.6)$$

The power x^2 is not guessed arbitrarily; it comes from integrating the $2/x$ correction in the linearized coefficient.

19.3 The full one-parameter transseries

Because the equation contains y^2 , powers of the first correction interact. Use the ansatz

$$\widehat{y}(x; C) = \sum_{k=0}^{\infty} C^k e^{-kx} \phi_k(x), \quad (19.7)$$

where each ϕ_k is an inverse-power or Laurent-type formal series. Substitute into Equation (19.1). Differentiation gives

$$\frac{d}{dx} \left(e^{-kx} \phi_k \right) = e^{-kx} (\phi'_k - k\phi_k),$$

and squaring convolves sector labels:

$$y^2 = \sum_{k \geq 0} C^k e^{-kx} \sum_{i+j=k} \phi_i \phi_j.$$

Thus every sector satisfies

$$\phi'_k + (1 - k)\phi_k = \sum_{i+j=k} \phi_i \phi_j + \frac{\delta_{k0}}{x}. \quad (19.8)$$

This single formula is the bookkeeping payoff of the transseries ansatz.

19.4 The one-instanton sector

For $k = 1$, Equation (19.8) becomes

$$\phi'_1 = 2\phi_0\phi_1.$$

Therefore

$$\phi_1 = K \exp \left(2 \int \phi_0 dx \right).$$

Absorb the normalization K into C . Termwise integration of Equation (19.4) gives

$$\phi_1 = x^2 \left(1 - \frac{4}{x} - \frac{8}{x^3} - \frac{52}{x^4} - \frac{384}{x^5} - \frac{3200}{x^6} + \cdots \right). \quad (19.9)$$

The coefficient of x^{-2} inside the parentheses happens to vanish.

19.5 The two-instanton sector

For $k = 2$,

$$\phi_2' - \phi_2 = 2\phi_0\phi_2 + \phi_1^2.$$

The dominant balance is

$$-\phi_2 \sim \phi_1^2 \sim x^4,$$

so $\phi_2 \sim -x^4$. Continuing gives

$$\phi_2 = x^4 \left(-1 + \frac{6}{x} - \frac{6}{x^2} + \frac{8}{x^3} + \frac{48}{x^4} + \frac{384}{x^5} + \cdots \right). \quad (19.10)$$

The sector $C^2 e^{-2x} \phi_2$ was not present in the linearized equation. It is created by the square of the first sector.

19.6 What the parameter means

The first-order ODE has a one-dimensional family of local solutions. The transseries parameter C is the formal coordinate on that family near the chosen asymptotic regime. It is determined by an initial condition or boundary condition after analytic reconstruction.

A Stokes constant is different. It is a property of the equation and normalization describing how the parameter C changes when the summation direction crosses a Stokes ray. Schematically,

$$C \mapsto C + S_1$$

for a simple one-parameter problem. Confusing C with S_1 obscures which data are solution-dependent and which are equation-dependent.

19.7 A transasymptotic glimpse

The leading behavior of Equation (19.7) is

$$y \approx \phi_0 + z - z^2 + z^3 - \cdots, \quad z = Cx^2 e^{-x},$$

where the signs follow from the leading terms of ϕ_k . The geometric pattern suggests

$$y \approx \phi_0 + \frac{z}{1+z} \quad (19.11)$$

when z is not extremely small even though e^{-x} is. A denominator zero at $z = -1$ predicts the approximate location of movable poles:

$$Cx^2e^{-x} \approx -1.$$

This reorganization, which sums over sectors before expanding deeply within each sector, is called a *transasymptotic* expansion. Chapter 25 develops the idea further.

19.8 General sector algebra

For an equation with polynomial nonlinearities, a product of a k -sector and an ℓ -sector lies in sector $k + \ell$. Thus the nonnegative integers form the sector semigroup. For several independent exponential actions, the labels become multi-indices in $\mathbb{N}^r$.

If a cubic nonlinearity and a symmetry suppress even powers, one may obtain only odd sectors. If the linearized problem has several eigenvalues, each produces its own fundamental action. Resonance occurs when different combinations land on the same action.

19.9 Divergence within every sector

The outer sum over k is only one infinite direction. Each ϕ_k usually has its own divergent large-order expansion:

$$\phi_k(x) = x^{\beta_k} \sum_{n \geq 0} a_{k,n} x^{-n}.$$

A resurgent transseries is therefore often doubly infinite. The large-order behavior of $a_{k,n}$ is controlled by neighboring sectors, and the Borel singularities of ϕ_k encode transitions to them.

19.10 Exercises

Exercise 19.1. Derive recurrence Equation (19.3) and compute a_5 .

Exercise 19.2. Starting from $\phi'_1 = 2\phi_0\phi_1$, reproduce the coefficients $-4, 0, -8$ in Equation (19.9).

Exercise 19.3. Use dominant balance in the k th sector equation to conjecture the leading power and sign of ϕ_k .

Chapter summary

Linearization finds the first exponential correction; nonlinearity generates all of its powers. Sector equations are obtained by convolution of sector labels. The free transseries parameter records the solution, while Stokes constants later describe direction-dependent changes of that parameter.

Singular perturbations and boundary layers

Transseries also arise when a small parameter multiplies the highest derivative. An ordinary power series in that parameter often solves the differential equation in the bulk but cannot satisfy all boundary conditions. Exponentials such as $e^{-x/\varepsilon}$ repair the mismatch.

20.1 A first boundary-layer equation

Let $0 < \varepsilon \ll 1$ and consider

$$\varepsilon y'(x) + y(x) = x, \quad y(0) = 0. \quad (20.1)$$

If one simply sets $\varepsilon = 0$, the reduced equation gives

$$y_0 = x.$$

This leading term happens to satisfy the boundary value at $x = 0$, but that coincidence does not persist at the next algebraic order. To see the nonuniform correction, compute the regular expansion carefully. Write

$$y \sim y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \cdots.$$

Substitution gives

$$y_0 = x, \quad y_1 = -y_0' = -1, \quad y_2 = -y_1' = 0,$$

so the outer solution is

$$y_{\text{out}} = x - \varepsilon.$$

This fails the boundary condition by $-\varepsilon$.

The exact solution is

$$y = x - \varepsilon + \varepsilon e^{-x/\varepsilon}. \quad (20.2)$$

The exponentially small term for fixed $x > 0$ is invisible to every power of ε , yet at $x = 0$ it is exactly large enough to restore the boundary condition.

A common trap

A term can be exponentially small in the outer region and order one in a rescaled boundary layer. “Small” always refers to a specified asymptotic regime.

20.2 The inner variable

Near $x = 0$, introduce

$$X = \frac{x}{\varepsilon}.$$

Then $y'(x) = \varepsilon^{-1}y_X$, and Equation (20.1) becomes

$$y_X + y = \varepsilon X.$$

At leading order in the layer,

$$y_X + y = 0,$$

whose solution is $Ae^{-X} = Ae^{-x/\varepsilon}$. Matching to the outer discrepancy fixes $A = \varepsilon$.

The inner and outer descriptions are two projections of the same transseries

$$y = x - \varepsilon + \varepsilon e^{-x/\varepsilon}.$$

20.3 General forcing

For

$$\varepsilon y' + y = f(x), \tag{20.3}$$

formal inversion of $1 + \varepsilon \partial$ gives

$$y_{\text{out}} = (1 + \varepsilon \partial)^{-1} f = f - \varepsilon f' + \varepsilon^2 f'' - \varepsilon^3 f''' + \dots. \tag{20.4}$$

The homogeneous correction is

$$Ce^{-x/\varepsilon}.$$

Hence

$$\widehat{y} = f - \varepsilon f' + \varepsilon^2 f'' - \dots + Ce^{-x/\varepsilon}. \tag{20.5}$$

For analytic f , the derivative coefficients often grow factorially, so even the outer series may diverge. The boundary layer and Borel divergence then coexist.

20.4 Two boundary layers

For a second-order problem such as

$$\varepsilon y'' + a(x)y' + b(x)y = f(x)$$

on a finite interval, different characteristic roots may generate exponentials attached to the left and right boundaries. A term small from one end can grow toward the other. Boundary conditions select the allowed linear combination.

The sign of the action matters. For $e^{-A(x)/\varepsilon}$, a layer decays where $\Re A(x) > 0$. Curves where $\Re A = 0$ separate regions of growth and decay and are closely related to Stokes geometry.

20.5 A nonlinear layer

Consider

$$\varepsilon y' = y - y^2, \quad 0 < y < 1. \quad (20.6)$$

The exact family is

$$y(x) = \frac{1}{1 + Ce^{-x/\varepsilon}}.$$

For $x/\varepsilon \rightarrow +\infty$,

$$y = 1 - Ce^{-x/\varepsilon} + C^2 e^{-2x/\varepsilon} - C^3 e^{-3x/\varepsilon} + \dots \quad (20.7)$$

This is the same nonlinear sector-generation mechanism seen in Chapter 19, now with action x/ε .

Near a point where $Ce^{-x/\varepsilon}$ is order one, the sector expansion should be resummed into the closed denominator. That is a simple transasymptotic transition.

20.6 Matched asymptotics versus transseries

Classical matched asymptotic expansions construct separate outer and inner series and compare their overlap. A transseries aims to retain all relevant scales in one formal object. The approaches are complementary:

- matching is geometrically intuitive and often easiest for a small number of regions;
- transseries give systematic algebra for multiple exponentially small effects, nonlinear sector interactions, and Stokes switching;
- exponential asymptotics uses late terms of the outer expansion to determine the beyond-all-orders contributions needed for uniformity.

20.7 Turning points and action integrals

For WKB-type singular perturbations,

$$\varepsilon^2 y'' = Q(x)y,$$

the exponential actions are

$$A_{\pm}(x) = \pm \int^x \sqrt{Q(t)} dt,$$

and the sectors contain $e^{A_{\pm}/\varepsilon}$. At a turning point $Q = 0$, the two actions coalesce, ordinary WKB breaks down, and an Airy scaling resolves the local structure.

The action is the analogue of the monomial exponent x in e^{-x} . Its zeros, equalities, and complex phases organize the global asymptotic geometry.

20.8 Exercises

Exercise 20.1. Solve $\varepsilon y' + y = 1$ with $y(0) = 0$ exactly and identify its outer and inner expansions.

Exercise 20.2. For Equation (20.3), determine C in terms of a boundary value $y(0) = y_b$, using a truncation of the outer series.

Exercise 20.3. Expand the exact nonlinear solution Equation (20.7) about the other equilibrium $y = 0$ in a region where $Ce^{-x/\varepsilon}$ is large.

Chapter summary

Singular perturbations force exponentials such as $e^{-x/\varepsilon}$ that are invisible to power series in ε but essential near boundaries. Inner scaling, matching, and transseries are compatible descriptions. Nonlinearity again generates towers of exponential sectors.

Several actions, resonance, and logarithmic sectors

Systems and higher-order equations usually carry more than one independent exponentially small mode. Their transseries parameters and actions become vectors, and arithmetic relations among actions create resonance.

21.1 A multi-parameter template

A common formal ansatz is

$$\widehat{F}(x; \sigma) = \sum_{\mathbf{k} \in \mathbb{N}^r} \sigma^{\mathbf{k}} e^{-\mathbf{k} \cdot \mathbf{A} x} x^{\mathbf{k} \cdot \beta} \Phi_{\mathbf{k}}(x), \quad (21.1)$$

where

$$\sigma^{\mathbf{k}} = \sigma_1^{k_1} \cdots \sigma_r^{k_r}, \quad \mathbf{k} \cdot \mathbf{A} = k_1 A_1 + \cdots + k_r A_r.$$

Each fundamental linearized mode contributes an action A_j and a parameter σ_j . Polynomial nonlinearities add multi-indices:

$$\mathbf{k} + \ell.$$

Thus the sector labels form the additive semigroup $\mathbb{N}^r$.

21.2 Linear systems

For

$$\mathbf{y}' = M(x)\mathbf{y} + \mathbf{f}(x),$$

the eigenvalues of the leading part of M determine exponential factors

$$\exp \left(\int \lambda_j(x) dx \right).$$

If M is asymptotically diagonalizable, each eigenmode carries its own formal power corrections. Off-diagonal terms couple the recurrences but do not by themselves create sums of actions; nonlinear terms do.

If two eigenvalues coalesce, diagonalization may fail and polynomial or logarithmic prefactors arise, just as Jordan blocks produce factors of x multiplying $e^{\lambda x}$ in constant-coefficient systems.

21.3 A simple resonance that forces a logarithm

Consider

$$y' + y = \frac{e^{-x}}{x}. \quad (21.2)$$

The homogeneous mode is e^{-x} . The forcing lies in the same exponential sector. Multiplying by e^x gives

$$(e^x y)' = \frac{1}{x},$$

so

$$y = e^{-x}(C + \log x). \quad (21.3)$$

The logarithm appears because the forcing resonates with the kernel of the linear operator $\partial + 1$. Trying an ansatz $a(x)e^{-x}$ reduces the problem to $a' = 1/x$, whose antiderivative raises logarithmic depth.

Intuition

Resonance means that the operator expected to determine a coefficient cannot be inverted within the originally guessed monomial class. The remedy is usually a generalized prefactor: a power of x , a logarithm, or an additional polynomial in a transseries parameter.

21.4 Commensurate actions

Suppose a system has fundamental actions $A_1 = A$ and $A_2 = 2A$. The sector e^{-2Ax} can arise in two ways:

$$\sigma_2 e^{-2Ax} \quad \text{and} \quad \sigma_1^2 e^{-2Ax}.$$

Nonlinear equations mix these contributions. A clean normal form may require replacing the raw sector coefficient by a polynomial in parameters,

$$\sigma_2 + c\sigma_1^2,$$

or, in stronger resonance, by terms involving x or $\log x$.

When actions satisfy no nontrivial integer relation, sector labels remain easier to separate. This is the nonresonant setting assumed in many rigorous Borel-summability theorems.

21.5 Logarithmic transmonomials in resurgent ansatzes

A resonant transseries may take the form

$$\sum_{\mathbf{k}} \sigma^{\mathbf{k}} e^{-\mathbf{k} \cdot \mathbf{A} / z} z^{\beta_{\mathbf{k}}} \sum_{m=0}^{M_{\mathbf{k}}} (\log z)^m \sum_{n \geq 0} a_{\mathbf{k}, m, n} z^n. \quad (21.4)$$

The logarithmic degree can grow with the sector index. Bookkeeping must then include three coordinates: action, logarithmic power, and perturbative order.

21.6 Parameter counting

An n th-order ODE has an n -dimensional local solution space under generic conditions. A transseries representation should contain the same number of independent free parameters, although asymptotic restrictions may set some to zero in a given sector. Nonlinear generation of infinitely many sectors does not create infinitely many independent constants; most sector coefficients are recursively determined by the finite parameter set.

This is a useful consistency check. If a first-order generic ODE appears to require two unrelated transseries parameters, one has probably double-counted a normalization or overlooked a relation.

21.7 Bridge to Stokes transformations

In a multi-parameter problem, crossing a Stokes ray acts on the parameter vector:

$$\sigma \mapsto \mathfrak{S}_\theta(\sigma).$$

The map can be triangular and nonlinear because sectors generate one another. Resonance can introduce polynomial and logarithmic terms in this parameter transformation. Alien calculus packages these changes as differential operators on transseries parameters; Chapter 24 gives a gentle introduction.

21.8 Exercises

Exercise 21.1. Solve

$$y' - 2y = e^{2x}x^{-1}$$

and identify the resonant logarithmic factor.

Exercise 21.2. In a two-action transseries with actions A and $\sqrt{2}A$, explain why distinct multi-indices cannot produce the same total action.

Exercise 21.3. A second-order nonlinear ODE has two independent linearized decaying modes. Write a schematic two-parameter sector ansatz through total sector degree 2.

Chapter summary

Several linearized modes lead to multi-parameter transseries indexed by $\mathbb{N}^r$. Nonlinear products add sector indices. Integer relations among actions create resonance, mixing sectors and often forcing powers or logarithms. The number of independent parameters still matches the differential equation's solution-space dimension.

Part V

From formal series to analytic functions

Borel transforms and Laplace reconstruction

A formal transseries can be manipulated without assigning a numerical value to it. Analysis begins when we ask a harder question:

Can a divergent formal sector be turned into an actual analytic function, in a way that respects the differential equation and recovers the formal expansion?

For factorially divergent series, the basic answer is *Borel–Laplace summation*. The idea is beautifully simple. Divide away the factorial growth, analytically continue the resulting power series, and then use a Laplace integral to put the factorials back.

This chapter develops that procedure slowly and fixes conventions carefully. Different books shift indices in slightly different ways; no mathematics depends on the choice, but every sign and factor does.

22.1 Why ordinary convergence fails

Consider a formal inverse-power series

$$\widehat{f}(x) = \sum_{n=0}^{\infty} a_n x^{-n-1}, \quad x \longrightarrow \infty. \quad (22.1)$$

If a_n grows approximately like $n!$, then for every fixed x the terms eventually increase:

$$\left| \frac{a_{n+1} x^{-n-2}}{a_n x^{-n-1}} \right| \approx \frac{n+1}{|x|}.$$

The smallest term occurs near $n \approx |x|$. Truncating there can give extraordinary accuracy, but the infinite series itself does not converge.

Factorial divergence is not random noise. It says that the coefficients have exactly the growth produced by repeated Laplace integration:

$$\int_0^{\infty} e^{-x\xi} \xi^n d\xi = \frac{n!}{x^{n+1}}, \quad \Re x > 0. \quad (22.2)$$

Equation (22.2) is the entire motivation for the Borel transform.

22.2 The formal Borel transform

Definition 22.1 (Borel transform). For a formal series of the form (22.1), its order-one formal Borel transform is

$$\mathcal{B}\widehat{f}(\xi) := \sum_{n=0}^{\infty} \frac{a_n}{n!} \xi^n. \quad (22.3)$$

If a_n has at most factorial growth, the new coefficients may have merely geometric growth, so (22.3) has a positive radius of convergence. A divergent series in x^{-1} has become an ordinary analytic germ at $\xi = 0$.

Example 22.1. For

$$\widehat{f}(x) = \sum_{n \geq 0} n! x^{-n-1},$$

we obtain

$$\mathcal{B}\widehat{f}(\xi) = \sum_{n \geq 0} \xi^n = \frac{1}{1-\xi}$$

near $\xi = 0$. The factorial explosion has become a single pole at $\xi = 1$.

The new variable ξ is called the *Borel variable*. The ξ -plane is often more informative than the original x -plane: singularities in the Borel plane reveal exponentially small scales in the original problem.

22.3 The Laplace transform puts the factorials back

Let θ be a direction in the Borel plane. Write

$$e^{i\theta}\mathbb{R}_+ = \{re^{i\theta} : r \geq 0\}.$$

If a function $g(\xi)$ can be continued along this ray and grows no faster than exponentially, define its directional Laplace transform by

$$\mathcal{L}_\theta g(x) := \int_0^{e^{i\theta}\infty} e^{-x\xi} g(\xi) d\xi. \quad (22.4)$$

The integral converges when $\Re(xe^{i\theta})$ is sufficiently positive.

Expanding $g(\xi) = \sum a_n \xi^n / n!$ near the origin and using (22.2) formally gives

$$\mathcal{L}_\theta g(x) \sim \sum_{n \geq 0} a_n x^{-n-1}.$$

This motivates the central definition.

Definition 22.2 (Directional Borel sum). Suppose that $\mathcal{B}\widehat{f}$ converges near 0, admits analytic continuation along the ray of angle θ , and has suitable exponential growth there. The directional Borel sum of $\widehat{f}$ is

$$\mathcal{S}_\theta \widehat{f} := \mathcal{L}_\theta \mathcal{B}\widehat{f}. \quad (22.5)$$

Three logically separate steps are hidden in this notation:

1. the formal Borel transform must converge near 0;
2. that germ must be analytically continued far enough along the chosen ray;
3. the continued function must satisfy a growth bound making the Laplace integral converge.

A formal series can fail at any one of these stages.

Intuition

The Borel transform converts *large-order growth* into *finite-distance singularities*. The Laplace transform converts those singularities back into exponentially small contributions. In this sense, the Borel plane is a microscope for effects that ordinary power asymptotics cannot see.

22.4 Gevrey order: measuring factorial growth

Definition 22.3 (Gevrey–1 series). The formal series (22.1) is Gevrey–1 if there are constants $C, A > 0$ such that

$$|a_n| \leq CA^n n! \quad (n \geq 0). \quad (22.6)$$

Condition (22.6) guarantees that the Borel transform converges for $|\xi| < A^{-1}$. More generally, Gevrey order s corresponds roughly to growth like $\Gamma(1 + sn)$. Ordinary analytic Taylor series have Gevrey order 0; the most common divergent expansions from rank-one irregular differential equations have Gevrey order 1.

A function f on a sector has a Gevrey–1 asymptotic expansion $\sum a_n x^{-n-1}$ if its remainders satisfy a uniform factorial estimate of the form

$$\left| f(x) - \sum_{n=0}^{N-1} a_n x^{-n-1} \right| \leq CA^N N! |x|^{-N-1} \quad (22.7)$$

through every slightly smaller subsector. This estimate is much stronger than the ordinary Poincaré condition for each fixed N .

22.5 The Euler series from beginning to end

Return to the equation from Chapter 2:

$$y' + y = \frac{1}{x}. \quad (22.8)$$

Its formal perturbative solution is

$$\widehat{y}_0(x) = \sum_{n=0}^{\infty} \frac{n!}{x^{n+1}}. \quad (22.9)$$

The Borel transform is

$$\mathcal{B}\widehat{y}_0(\xi) = \frac{1}{1 - \xi}. \quad (22.10)$$

Away from the positive real ray, there is no obstruction to a directional Laplace integral. Along the positive ray, however, the pole at $\xi = 1$ lies directly on the path.

For $x > 0$, define upper and lower lateral sums by paths passing just above or below the pole:

$$\mathcal{S}_{0\pm}\widehat{y}_0(x) := \int_0^{\infty e^{\pm i0}} \frac{e^{-x\xi}}{1-\xi} d\xi. \quad (22.11)$$

Their average is the Cauchy principal value,

$$\frac{1}{2} (\mathcal{S}_{0+} + \mathcal{S}_{0-}) \widehat{y}_0 = e^{-x} \text{Ei}(x), \quad (22.12)$$

with the standard real branch of Ei for $x > 0$. The two lateral paths differ by a small loop around the pole, giving

$$\mathcal{S}_{0+}\widehat{y}_0 - \mathcal{S}_{0-}\widehat{y}_0 = 2\pi i e^{-x}. \quad (22.13)$$

The ambiguity is not arbitrary. It is precisely a multiple of the missing homogeneous solution of (22.8).

Worked calculation

The exact family of solutions is

$$y(x) = e^{-x} \text{Ei}(x) + C e^{-x}.$$

Represent the same analytic solution using either lateral sum:

$$y = \mathcal{S}_{0+}\widehat{y}_0 + C_+ e^{-x} = \mathcal{S}_{0-}\widehat{y}_0 + C_- e^{-x}.$$

By (22.13),

$$C_+ = C_- - 2\pi i.$$

Thus crossing the singular direction changes the transseries coordinate, not the underlying solution.

This example contains the whole elementary Stokes mechanism:

- coefficient growth $n!$;
- a Borel singularity at the action $A = 1$;
- an exponentially small scale $e^{-Ax} = e^{-x}$;
- two lateral reconstructions;
- a jump proportional to the homogeneous exponential;
- a compensating change of the free transseries parameter.

22.6 The operational dictionary

Borel transformation converts several operations into simple operations in the ξ -plane. Assume for the moment that all series have zero constant term in x and use convention (22.3).

22.6.1 Differentiation becomes multiplication

From

$$\frac{d}{dx} \sum_{n \geq 0} a_n x^{-n-1} = - \sum_{n \geq 0} (n+1) a_n x^{-n-2},$$

one finds

$$\mathcal{B}(\widehat{f'})} = -\xi \mathcal{B}\widehat{f}. \quad (22.14)$$

This is why differential equations often become algebraic or convolution equations after Borel transformation.

22.6.2 Products become convolution

Define

$$(g * h)(\xi) := \int_0^\xi g(\eta) h(\xi - \eta) d\eta. \quad (22.15)$$

A beta-integral calculation gives

$$\mathcal{B}(\widehat{fg}) = (\mathcal{B}\widehat{f}) * (\mathcal{B}\widehat{g}). \quad (22.16)$$

Indeed,

$$\int_0^\xi \frac{\eta^m}{m!} \frac{(\xi - \eta)^n}{n!} d\eta = \frac{\xi^{m+n+1}}{(m+n+1)!}.$$

The shift by one matches multiplication of x^{-m-1} and x^{-n-1} .

22.6.3 Multiplication by x^{-1} becomes integration

One similarly obtains

$$\mathcal{B}(x^{-1}\widehat{f})(\xi) = \int_0^\xi \mathcal{B}\widehat{f}(\eta) d\eta. \quad (22.17)$$

These rules let us translate many formal ODE recurrences into analytic integral equations. Convolution also explains why Borel singularities add: if g is singular near A and h near B , then $g * h$ naturally develops singular behavior near $A + B$.

22.7 Why the Laplace sum has the advertised asymptotic series

Split the Laplace integral at a small positive r :

$$\int_0^\infty e^{i\theta} = \int_0^r e^{i\theta} + \int_r^\infty e^{i\theta}.$$

On the short segment, expand

$$g(\xi) = \sum_{n=0}^{N-1} \frac{a_n}{n!} \xi^n + R_N(\xi).$$

Integrating the polynomial gives the first N inverse powers. Analyticity bounds $R_N(\xi)$ by a constant times $|\xi|^N$, producing a remainder of order $|x|^{-N-1}$. The far segment is exponentially small because of $e^{-x\xi}$. This is the basic proof behind Watson-type lemmas.

The exact hypotheses matter when one wants uniform Gevrey bounds, continuation over wide sectors, or uniqueness. For a first encounter, the important direction is:

analytic continuation and an exponential growth bound
 $\Downarrow$
a function with the stated formal asymptotic series

22.8 Uniqueness and exponentially flat functions

An asymptotic expansion does not usually determine a function uniquely. The function e^{-x} has the zero inverse-power expansion as $x \rightarrow +\infty$. More generally, exponentially small functions can be added without changing any coefficient.

On sufficiently wide complex sectors, analyticity and growth restrictions can restore uniqueness. A typical Watson uniqueness theorem says, roughly, that a Gevrey–1 expansion on a sector wider than π determines the function uniquely within the appropriate growth class. The width π is natural because $e^{-1/z}$ can decay on sectors of opening up to π around the positive real z -axis.

This does not contradict Stokes phenomena. Different sectorial sums live on overlapping sectors separated by singular directions; each is unique in its own admissible summation problem, while their analytic continuations may differ by exponentially flat terms on the overlap.

22.9 Other Gevrey orders and acceleration

Not every problem has simple $n!$ growth. Suppose coefficients behave like $\Gamma(1 + n/k)$. One can use a Borel transform of order k , followed by a matching Laplace transform. If several factorial scales occur in succession, *multisummability* performs several transformations at different levels.

Still faster growth may require *acceleration*: a nonlinear change of the Borel variable that turns excessive growth into manageable exponential growth. Écalle’s theory organizes these procedures into a flexible summation machine. Costin’s generalized Borel summation gives a closely related analytic framework for broad classes of nonlinear differential systems [12, 13, 15].

22.10 Borel summation is not automatic

The phrase “take the Borel sum” can conceal serious obstacles:

1. the series may not be Gevrey of the expected order;
2. the Borel transform may have a natural boundary preventing continuation;
3. continuation may exist but grow too rapidly for an ordinary Laplace transform;
4. singularities may lie on the desired direction, forcing lateral or median prescriptions;
5. a complete transseries may require summing infinitely many sectors in a controlled topology.

For particular differential equations, these are theorems to be proved, not formal consequences of writing down an ansatz.

22.11 Exercises

Exercise 22.1. Compute the Borel transforms of

$$\sum_{n \geq 0} (-1)^n n! x^{-n-1} \quad \text{and} \quad \sum_{n \geq 0} A^{-n-1} n! x^{-n-1}.$$

Locate their nearest Borel singularities.

Exercise 22.2. Use the beta integral to prove (22.16) directly for two monomials and then extend it bilinearly.

Exercise 22.3. Apply the Borel transform to $y' + y = x^{-1}$ and solve the resulting algebraic equation for $\widehat{\mathcal{B}y}$.

Exercise 22.4. Suppose g is analytic near the positive ray and satisfies $|g(\xi)| \leq C e^{K|\xi|}$. Show that (22.4) converges for real $x > K$ when $\theta = 0$.

Chapter summary

Borel transformation divides out factorial growth; analytic continuation exposes finite Borel singularities; Laplace transformation reconstructs an analytic function with the original asymptotic expansion. In the Euler example, a pole at $\xi = 1$ produces a jump proportional to e^{-x} . Summability requires analytic hypotheses beyond formal transseries algebra.

Stokes phenomena: when exponentially small terms switch

An asymptotic formula can change its visible form as the argument of a complex variable rotates. A term that is tiny in one direction may become oscillatory on another and dominant on a third. The exact analytic function remains smooth; what changes is the efficient asymptotic description.

This direction-dependent rearrangement is the *Stokes phenomenon*. Transseries are its natural bookkeeping language.

23.1 A geometric picture of exponentials

Consider

$$e^{-A/z}, \quad z = r e^{i\varphi}, \quad A = |A| e^{i\alpha}. \quad (23.1)$$

Its magnitude is

$$\left| e^{-A/z} \right| = \exp \left[-\frac{|A|}{r} \cos(\alpha - \varphi) \right]. \quad (23.2)$$

Therefore:

- it is exponentially small where $\cos(\alpha - \varphi) > 0$;
- it is exponentially large where $\cos(\alpha - \varphi) < 0$;
- it is oscillatory with unit magnitude where $\cos(\alpha - \varphi) = 0$.

The rays $\varphi = \alpha \pm \pi/2$ separate decay from growth.

Meanwhile, the Borel–Laplace integral

$$\int e^{-\xi/z} \mathcal{B}\widehat{f}(\xi) \, d\xi$$

meets a singularity at $\xi = A$ when its integration ray has angle α . Thus the direction $\arg z = \arg A$ is where lateral Borel sums differ.

A common trap

Authors do not all use the words *Stokes line* and *anti-Stokes line* in the same way. Some call the Borel-singular direction a Stokes line; others reserve “Stokes line” for equal phase and call equal magnitude anti-Stokes. The invariant content is the pair of equations

$$\Im(A/z) = 0 \quad \text{and} \quad \Re(A/z) = 0.$$

Always check which one an author is naming.

23.2 Lateral sums and discontinuous asymptotic coordinates

Suppose $\mathcal{B}\hat{f}$ has singularities on the ray of angle θ . Deform the Laplace path slightly above or below that ray:

$$\mathcal{S}_{\theta^\pm} \hat{f} := \int_0^{e^{i(\theta \pm 0)} \infty} e^{-\xi/z} \mathcal{B}\hat{f}(\xi) \frac{d\xi}{z}. \quad (23.3)$$

Here we use the small- z normalization, with $d\xi/z$, so that ξ^n reconstructs $n!z^n$. This is equivalent to the large- x convention of Chapter 22 after $z = 1/x$.

The difference

$$\text{Disc}_\theta \hat{f} := \mathcal{S}_{\theta^+} \hat{f} - \mathcal{S}_{\theta^-} \hat{f} \quad (23.4)$$

is exponentially small in a region where the original perturbative series is useful. The two sums have the same ordinary power expansion, so no finite algebraic-order calculation can detect the jump.

A full one-parameter transseries has the form

$$\hat{\Phi}(z, \sigma) = \sum_{k \geq 0} \sigma^k e^{-kA/z} z^{k\beta} \hat{\phi}_k(z). \quad (23.5)$$

The same analytic solution can be represented on the two sides of a singular direction by different parameter values:

$$\mathcal{S}_{\theta^+} \hat{\Phi}(z, \sigma_+) = \mathcal{S}_{\theta^-} \hat{\Phi}(z, \sigma_-). \quad (23.6)$$

In the simplest normalization,

$$\sigma_+ = \sigma_- + S_\theta, \quad (23.7)$$

where S_θ is a Stokes constant. Depending on the convention for the lateral discontinuity and the normalization of ϕ_1 , the sign may be reversed.

23.3 The Euler jump revisited

For the Euler series, the Borel singularity is the simple pole

$$\frac{1}{1 - \xi}$$

at $A = 1$. The residue theorem gives

$$\mathcal{S}_{0^+} \hat{y}_0 - \mathcal{S}_{0^-} \hat{y}_0 = 2\pi i e^{-x}.$$

The Stokes multiplier is therefore $2\pi i$ in the normalization where the first nonperturbative sector is exactly e^{-x} . If that sector were normalized as $7e^{-x}$, the reported constant would be $2\pi i/7$. Stokes constants are meaningful only together with a stated sector normalization.

23.4 A local singularity produces a whole asymptotic sector

A simple pole produces a pure exponential. More complicated Borel singularities produce exponentials times powers and series.

Suppose near $\xi = A$ the analytic continuation has a branch behavior

$$\mathcal{B}\widehat{\phi}_0(\xi) \sim c(\xi - A)^{-\beta} \left(1 + b_1(\xi - A) + b_2(\xi - A)^2 + \cdots \right). \quad (23.8)$$

The discontinuity across a cut from A is supported near that point. Shift $\xi = A + u$ in the Laplace integral. The factor splits as

$$e^{-(A+u)/z} = e^{-A/z} e^{-u/z}.$$

The resulting jump has the form

$$e^{-A/z} z^{1-\beta} \left(c_0 + c_1 z + c_2 z^2 + \cdots \right). \quad (23.9)$$

Thus the local expansion around a Borel singularity is not merely an obstruction. It contains the fluctuation series of another transseries sector.

23.5 Airy's equation as a map of Stokes geometry

Airy's equation

$$y'' - zy = 0 \quad (23.10)$$

has WKB exponentials

$$\exp \left(\pm \frac{2}{3} z^{3/2} \right). \quad (23.11)$$

Their magnitude ratio is controlled by

$$\Re \left(\frac{4}{3} z^{3/2} \right).$$

The rays of equal magnitude satisfy

$$\Re(z^{3/2}) = 0, \quad \arg z = \frac{(2k+1)\pi}{3}. \quad (23.12)$$

The rays of equal phase satisfy

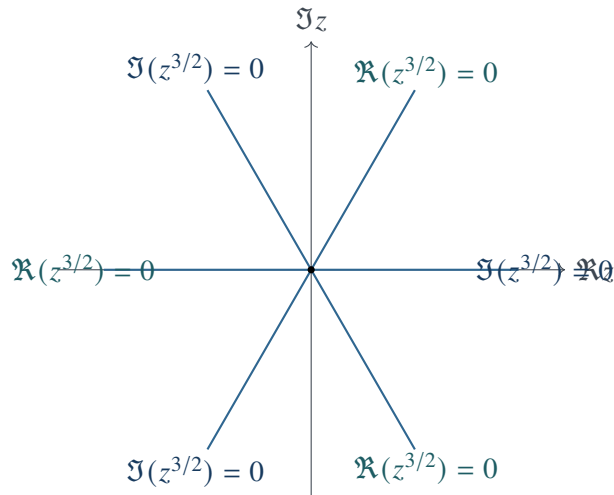
$$\Im(z^{3/2}) = 0, \quad \arg z = \frac{2k\pi}{3}. \quad (23.13)$$

These six rays divide the plane into sectors in which one WKB exponential dominates the other.

For $|\arg z| < \pi$, one standard asymptotic form is

$$\text{Ai}(z) \sim \frac{e^{-\frac{2}{3}z^{3/2}}}{2\sqrt{\pi} z^{1/4}} \sum_{n \geq 0} (-1)^n \frac{u_n}{(\frac{2}{3}z^{3/2})^n}, \quad (23.14)$$

inside suitable closed subsectors. After analytic continuation, a multiple of the opposite exponential appears in the efficient representation. The connection formulas among $\text{Ai}(z)$ and its rotations are exact; the Stokes phenomenon describes how those exact formulas are reflected in asymptotic decompositions.



The diagram is schematic: because $z^{3/2}$ is multivalued, one must specify branches when following a solution around the origin.

23.6 Dominant and subdominant exponentials

Suppose an exact solution is written asymptotically as

$$F(z) \sim F_{\text{dom}}(z) + s(\varphi)F_{\text{sub}}(z).$$

When $F_{\text{sub}}/F_{\text{dom}}$ is exponentially tiny, changing s by an order-one amount changes F by less than every power of z . This is where an apparently discontinuous Stokes multiplier can hide.

After crossing an equal-magnitude ray, the former subdominant term may become dominant. Then its coefficient is no longer negligible and must have been chosen correctly. Stokes analysis supplies exactly that previously invisible information.

23.7 Optimal truncation and the smooth reality behind the jump

The classical asymptotic rule often states that a subdominant exponential “switches on” discontinuously at a Stokes ray. Exact analytic functions do not jump there. The discontinuity belongs to an idealized decomposition into an optimally truncated dominant series plus a separately named exponential remainder.

At finite but large $|z|^{-1}$, the effective multiplier changes rapidly but smoothly across a narrow angular transition region. Under broad conditions, the leading profile is an error function. This is *Berry smoothing*. The angular width typically shrinks like the inverse square root of the large asymptotic parameter. Costin and Kruskal proved precise optimal-truncation and smoothing results for broad differential systems [16].

Intuition

A Stokes jump is like the sharp limit of a rapidly changing coordinate chart. The analytic solution is the landscape; the transseries parameter is a coordinate. Crossing a special ray requires changing coordinates to keep describing the same point.

23.8 Median summation and real solutions

If a formal series has real coefficients but a Borel singularity lies on the positive real ray, the two lateral sums are often complex conjugates for real positive z :

$$\overline{S_{0+}\hat{f}} = S_{0-}\hat{f}.$$

Their average is real but may not by itself respect nonlinear multiplication. Écalle's *median summation* supplements lateral averaging by a half Stokes transformation. Schematically,

$$S_{\text{med}} = S_{0-} \circ \mathfrak{S}_0^{-1/2} = S_{0+} \circ \mathfrak{S}_0^{1/2}. \quad (23.15)$$

The square root is understood as the exponential of half the infinitesimal Stokes operator. In favorable resurgent problems, median summation preserves reality and the algebraic operations needed by the equation.

For the elementary Euler pole, ordinary principal-value averaging already illustrates the real answer. For nonlinear problems, the transseries parameter and the full Stokes automorphism must be included.

23.9 Global monodromy from local jumps

Rotate z through a full turn. One may cross several Stokes rays, applying a Stokes transformation at each. Their ordered product combines with ordinary monodromy from powers z^β and logarithms. Exact global connection formulas impose relations among the Stokes constants.

For a linear equation, these transformations are represented by Stokes matrices. Their product, together with formal monodromy, gives the actual monodromy around the irregular singularity. For nonlinear equations, the transformations act nonlinearly on transseries parameters rather than linearly on a vector space of solutions.

23.10 A practical procedure

For a differential equation near an irregular limit:

1. construct the perturbative formal series;

2. linearize about it and identify exponential actions A_j ;
3. draw rays from the equations $\Re(A_j/z) = 0$ and $\Im(A_j/z) = 0$;
4. Borel transform each formal sector;
5. locate singularities at actions and their combinations;
6. define directional sums away from singular rays;
7. compare lateral sums and infer parameter transformations;
8. use boundary, reality, or global monodromy conditions to fix the parameters.

The hard steps are analytic continuation and singularity control. The geometric map tells us where to look.

23.11 Exercises

Exercise 23.1. For $A = 2 + 2i$, find the rays on which $e^{-A/z}$ has unit magnitude and the ray on which a Borel singularity at $\xi = A$ obstructs radial Laplace summation.

Exercise 23.2. Assume $\widehat{\mathcal{B}f}$ has a simple pole $r/(\xi - A)$ on a positive ray. Compute the difference of upper and lower lateral Laplace integrals, keeping track of the exponential $e^{-A/z}$.

Exercise 23.3. Derive the six rays (23.12)–(23.13) directly from $z = re^{i\varphi}$.

Exercise 23.4. Explain why changing the normalization of a one-instanton sector changes the numerical Stokes constant but not the analytic jump.

Chapter summary

Borel singular directions force upper and lower lateral sums. Their difference is exponentially small and is absorbed by a transformation of transseries parameters. Equal-phase and equal-magnitude rays organize the geometry, though their names vary by convention. Optimal truncation reveals a smooth finite-parameter transition whose sharp limit is the classical Stokes jump.

Resurgence: how sectors remember one another

The Borel transform of the Euler series has one pole, and that pole predicts the missing exponential. In nonlinear problems the picture repeats recursively: after continuing around one singularity, one encounters more singularities, and their local expansions reproduce other sectors of the same transseries.

This self-referential analytic structure is called *resurgence*. The word suggests that information from one sector “rises again” inside the large-order behavior of another.

24.1 From Borel summability to resurgence

Borel summability along one ray asks for analytic continuation only along that ray. Resurgence asks for much more controlled continuation throughout the Borel plane.

A useful informal definition is:

Definition 24.1 (Resurgent germ, informal). A convergent Borel germ at $\xi = 0$ is resurgent if it can be analytically continued along every finite path that avoids a locally finite set of singular points, and if the singular behavior reached in this way remains of a controlled type.

The precise category varies. Écalle’s endlessly continuable germs allow the set of forbidden points to depend on the length of the path but require only finitely many obstructions below every finite path length. More restrictive “simple resurgent” classes permit singularities built from poles and logarithms. For elementary reasoning, the essential properties are:

1. singularities are isolated enough that one can travel around them;
2. their locations have asymptotic meaning;
3. their local expansions can themselves be continued;
4. convolution preserves the class under suitable hypotheses.

24.2 Actions become Borel locations

Suppose a transseries contains the exponential scale

$$e^{-A/z}.$$

Then one expects a Borel singularity displaced by A . If several actions $A_1, \dots, A_r$ occur, nonlinear convolution generates sums

$$\mathbf{k} \cdot \mathbf{A} = k_1 A_1 + \dots + k_r A_r, \quad \mathbf{k} \in \mathbb{N}^r. \quad (24.1)$$

These are exactly the sector actions in [Chapter 21](#). The additive geometry of the Borel plane mirrors multiplication of transmonomials:

$$e^{-A/z} e^{-B/z} = e^{-(A+B)/z}.$$

The singularity pattern need not be only the positive semigroup. Analytic continuation of different sectors may reveal differences of actions, conjugate actions, or repeated sheets of a ramified surface. Nevertheless, (24.1) is the first map to draw.

24.3 Darboux's principle in the Borel plane

Large Taylor coefficients are controlled by nearby singularities. This elementary fact is the bridge from finite-distance Borel geometry to factorial divergence.

Let

$$g(\xi) = \sum_{n \geq 0} b_n \xi^n$$

be analytic at 0. If its nearest singularity is a simple pole

$$g(\xi) \sim \frac{c}{A - \xi}, \quad (24.2)$$

then

$$\frac{c}{A - \xi} = \frac{c}{A} \sum_{n \geq 0} \left(\frac{\xi}{A} \right)^n,$$

so

$$b_n \sim c A^{-n-1}. \quad (24.3)$$

If $g = \mathcal{B}\hat{f}$ and $a_n = n!b_n$, this becomes

$$a_n \sim c \frac{n!}{A^{n+1}}. \quad (24.4)$$

The action A controls the geometric factor and the singularity type controls the power of n multiplying $n!$.

More generally,

$$g(\xi) \sim c \left(1 - \frac{\xi}{A} \right)^{-\beta} \quad (24.5)$$

implies

$$b_n \sim \frac{c}{\Gamma(\beta)} A^{-n} n^{\beta-1}, \quad a_n \sim \frac{c}{\Gamma(\beta)} A^{-n} \Gamma(n + \beta). \quad (24.6)$$

This is a version of Darboux's principle or singularity analysis.

Intuition

The late coefficients of a divergent series are a blurred photograph of the nearest Borel singularities. The exponential action gives the distance and direction; the factorial-over-power correction gives the singularity's local shape.

24.4 Large-order relations between sectors

Write a one-action transseries as

$$\widehat{\Phi}(z, \sigma) = \sum_{k \geq 0} \sigma^k e^{-kA/z} z^{k\beta} \widehat{\phi}_k(z), \quad \widehat{\phi}_k(z) = \sum_{n \geq 0} a_{k,n} z^n. \quad (24.7)$$

A singularity of the Borel transform of ϕ_0 at A has local data related to ϕ_1 . Consequently, the late perturbative coefficients often satisfy an expansion of the schematic form

$$a_{0,n} \sim \frac{S_{0 \rightarrow 1}}{2\pi i} \sum_{m \geq 0} a_{1,m} \frac{\Gamma(n + \beta - m)}{A^{n+\beta-m}}. \quad (24.8)$$

Truncating the right-hand sum after a few m gives successive inverse- n corrections.

Every symbol in (24.8) depends on normalization conventions:

- shifting the starting power of z changes β ;
- rescaling ϕ_1 rescales $S_{0 \rightarrow 1}$ inversely;
- choosing upper-minus-lower rather than lower-minus-upper changes a sign;
- logarithmic singularities can add derivatives with respect to β and hence $\log n$ terms.

The invariant claim is that the full low-order expansion of one sector governs the full late-order expansion of a neighboring sector.

24.5 A direct contour explanation

Cauchy's coefficient formula says

$$b_n = \frac{1}{2\pi i} \oint \frac{g(\xi)}{\xi^{n+1}} d\xi.$$

For large n , deform the small circle outward. Contributions arise when the contour is caught on singularities. A singularity near A contributes a factor A^{-n} , and its local discontinuity supplies an asymptotic expansion in inverse powers of n . After multiplying by $n!$, one obtains (24.8).

This contour argument is why large-order formulas are not mysterious numerical coincidences. They are coefficient extraction performed on a multiply continued Borel transform.

24.6 Alien derivatives: singularity extractors

Ordinary differentiation measures local change in the original variable. An *alien derivative* measures singular behavior after analytic continuation to a specified Borel location.

We write

$$\Delta_A$$

for an operator associated with the singularity at action A . Very roughly, $\Delta_A \widehat{\phi}$ extracts the minor or discontinuity data of the Borel transform of $\widehat{\phi}$ at $\xi = A$, translated back to the origin. The actual definition includes choices of continuation paths and averages over ways of bypassing earlier singularities.

Alien derivatives are designed to satisfy derivation-like rules:

$$\Delta_A(FG) = (\Delta_A F)G + F(\Delta_A G) \quad (24.9)$$

for the appropriately normalized operator. They interact nontrivially with ordinary differentiation because the exponential weight $e^{-A/z}$ carries the action. A common pointed operator incorporates this weight so that commutation rules become simpler.

A common trap

Alien calculus has several notational normalizations: Δ_A , pointed $\dot{\Delta}_A$, logarithms of Stokes automorphisms, and lateral alien operators. Formulas copied between sources can differ by $e^{-A/z}$, by $2\pi i$, or by a sign. The structural relations are more stable than the typography.

24.7 The Stokes automorphism

The total discontinuity along a ray collects all Borel singularities lying on that ray. Alien derivatives package this collection into a formal automorphism. Schematically,

$$\mathfrak{S}_\theta = \exp \left(\sum_{\substack{A \neq 0 \\ \arg A = \theta}} e^{-A/z} \Delta_A \right). \quad (24.10)$$

With one standard convention, lateral summation satisfies

$$\mathcal{S}_{\theta^+} = \mathcal{S}_{\theta^-} \circ \mathfrak{S}_\theta. \quad (24.11)$$

Thus a Stokes jump is the analytic shadow of an algebraic automorphism acting on formal transseries.

Because the exponent in (24.10) is a derivation, its exponential is an automorphism. This is the mechanism that allows Stokes continuation to respect multiplication and nonlinear equations.

24.8 Bridge equations

For a transseries solution of a differential equation, alien derivatives are often expressible as ordinary derivatives with respect to transseries parameters. In a simple one-parameter normalization, a bridge equation may read

$$e^{-A/z} \Delta_A \widehat{\Phi}(z, \sigma) = S_1 \frac{\partial}{\partial \sigma} \widehat{\Phi}(z, \sigma). \quad (24.12)$$

It is called a bridge because it connects:

- analytic continuation in the Borel plane, on the left;
- motion through the formal family of solutions, on the right.

Exponentiating (24.12) gives a translation of σ and recovers the elementary shift (23.7).

In resonant or multi-parameter problems, the right side is a vector field in parameter space,

$$\sum_j S_{A,j}(\sigma) \frac{\partial}{\partial \sigma_j},$$

and the Stokes map can be triangular or nonlinear.

24.9 The Euler example in alien language

For

$$\widehat{\mathcal{B}y_0}(\xi) = \frac{1}{1 - \xi},$$

the only finite singularity is at $A = 1$. Its singular data are constant, corresponding to the one-instanton sector e^{-x} with no fluctuation series. The alien derivative therefore maps the perturbative sector to a constant multiple of the exponential sector. Exponentiation yields the parameter shift by $2\pi i$ found directly in Chapter 22.

This example is unusually finite. In a nonlinear problem, the singularity at A leads to the $k = 1$ sector, the singularity at $2A$ to the $k = 2$ sector, and analytic continuation of each sector leads both forward and backward through a network of sectors.

24.10 Convolution builds the singularity lattice

Suppose g and h have singularities at A and B . In

$$(g * h)(\xi) = \int_0^\xi g(\eta) h(\xi - \eta) d\eta,$$

the integration contour becomes pinched when η approaches A while $\xi - \eta$ approaches B . This occurs near $\xi = A + B$. Hence nonlinear products generate summed singularities.

Repeated convolution creates

$$A, 2A, 3A, \dots$$

or the multi-action semigroup (24.1). This analytic mechanism is exactly parallel to the sector convolution derived formally in Chapter 19.

24.11 Resonance and logarithmic singularities

If two different paths in action space reach the same point, their singular contributions mix. Integer relations among actions can produce logarithmic branching in the Borel plane. After Laplace transformation, such logarithms lead to powers of $\log z$ and polynomial dependence on sector indices.

This is the analytic counterpart of the formal resonance described in Chapter 21. It also complicates bridge equations: a single alien derivative can connect several monomials with the same total exponential weight.

24.12 Numerically reading Stokes data from coefficients

Suppose the leading large-order law is

$$a_n \sim C A^{-n} \Gamma(n + \beta).$$

Then the ratio

$$\frac{a_n}{a_{n-1}} \sim \frac{n + \beta - 1}{A} \quad (24.13)$$

estimates A from a long coefficient list. After estimating A and β , normalize

$$C_n := \frac{a_n A^{n+\beta}}{\Gamma(n + \beta)}. \quad (24.14)$$

The sequence tends slowly to C , with corrections in powers of $1/n$. Richardson extrapolation or nonlinear sequence transformations can accelerate it. Subtracting the predicted contribution of the nearest singularity then reveals the next one.

A practical large-order investigation therefore proceeds hierarchically:

1. inspect coefficient ratios to guess the closest action;
2. divide out its factorial-over-power law;
3. fit inverse- n corrections from the neighboring sector;
4. subtract this contribution;
5. repeat for more distant actions.

High-precision arithmetic is often essential because subleading singularities are exponentially suppressed in coefficient order.

24.13 What resurgence does not claim

Not every transseries is resurgent. A logarithmic-exponential Hahn series may be perfectly valid as a formal ordered object without any chosen Borel transform. Conversely, a resurgent object may involve fractional powers, logarithms, oscillatory actions, or other ingredients outside a narrow grid-based LE field.

It is useful to reserve three separate adjectives:

formal transseries: an element of a specified generalized-series algebra;

Borel summable: reconstructible in a specified direction by a specified Borel–Laplace procedure;

resurgent: having controlled repeated analytic continuation and singularity relations in the Borel plane.

None of the last two follows merely from the first.

24.14 Exercises

Exercise 24.1. Let $\mathcal{B}\widehat{f}(\xi) = (1 - \xi/A)^{-2}$. Compute its Taylor coefficients and the corresponding coefficients of $\widehat{f}(z)$ under the convention $\mathcal{B}(\sum a_n z^{n+1}) = \sum a_n \xi^n / n!$.

Exercise 24.2. Starting from (24.5), use the generalized binomial theorem to derive (24.6).

Exercise 24.3. Explain geometrically why convolution of singularities at A and B can create a singularity at $A + B$.

Exercise 24.4. Assume (24.13). Construct an estimator for A from a_n/a_{n-1} when β is known, and find its first relative error if one additional c/n correction is present.

Chapter summary

Resurgence turns Borel singularities into channels of communication among transseries sectors. Darboux analysis converts local singular behavior into factorial large-order laws. Alien derivatives extract singular data, the Stokes automorphism combines it along a ray, and bridge equations translate analytic continuation into motion in transseries-parameter space.

Transasymptotics, hyperasymptotics, and movable singularities

A transseries has at least two notions of order. In

$$\sum_{k \geq 0} \sigma^k e^{-kA/z} \sum_{n \geq 0} a_{k,n} z^n,$$

we may move deeper in perturbative order n , or move across exponential sectors k . Ordinary asymptotics expands first in small z at fixed k . Near certain boundaries that ordering becomes inefficient: many exponential sectors have comparable size and should be summed together.

A *transasymptotic expansion* reorganizes the same information by summing across sectors at fixed combinations such as

$$\tau = \sigma z^\beta e^{-A/z}. \quad (25.1)$$

This often reveals nonlinear transitions and movable singularities that are invisible in any finite sector truncation.

25.1 An exactly solvable logistic model

Consider

$$y' = -y + y^2, \quad x \longrightarrow +\infty. \quad (25.2)$$

The zero solution is the perturbative sector. Linearization gives the small exponential e^{-x} . Set

$$\tau = \sigma e^{-x}.$$

Since $\tau' = -\tau$, the function

$$y(x) = \frac{\tau}{1 + \tau} \quad (25.3)$$

satisfies (25.2). Expanding at small τ gives

$$y = \tau - \tau^2 + \tau^3 - \tau^4 + \cdots = \sum_{k \geq 1} (-1)^{k-1} \sigma^k e^{-kx}. \quad (25.4)$$

The infinite tower of nonlinear sectors is simply the Taylor expansion of a rational function of the transasymptotic variable.

The exact solution has a movable pole when

$$1 + \sigma e^{-x} = 0. \quad (25.5)$$

Its location depends on the integration constant σ , hence “movable.” No finite truncation of (25.4) has a pole. The pole emerges only after summing all sectors.

Intuition

Perturbative order describes fine structure inside one sector. Transasymptotic order describes collective behavior of many sectors. A singularity of the summed function can be encoded in perfectly regular coefficients spread over infinitely many sectors.

25.2 Reorganizing a two-dimensional array

Suppose

$$\hat{y}(z, \sigma) = \sum_{k \geq 0} \tau^k \sum_{n \geq 0} a_{k,n} z^n, \quad \tau = \sigma z^\beta e^{-A/z}. \quad (25.6)$$

Instead of summing over n inside each k , collect equal powers of z :

$$\hat{y}(z, \sigma) = \sum_{n \geq 0} z^n F_n(\tau), \quad F_n(\tau) := \sum_{k \geq 0} a_{k,n} \tau^k. \quad (25.7)$$

The functions F_n are called transasymptotic coefficient functions. Substituting (25.7) into the differential equation often yields a hierarchy of algebraic or differential equations in τ :

$$\mathcal{E}_0(F_0) = 0, \quad \mathcal{L}_{F_0} F_1 = R_1(F_0), \quad \dots$$

The leading F_0 frequently has poles or branch points. Those singularities predict where the original solution ceases to be regular.

25.3 The Riccati model revisited

For Equation (19.1), the natural collective variable was

$$\tau = Cx^2 e^{-x}.$$

The leading sector coefficients suggested

$$\phi_k(x) \sim (-1)^{k-1} x^{2k} \quad (k \geq 1).$$

Thus

$$\sum_{k \geq 1} C^k e^{-kx} \phi_k(x) \sim \sum_{k \geq 1} (-1)^{k-1} \tau^k = \frac{\tau}{1 + \tau}.$$

The approximate movable-pole condition

$$1 + Cx^2 e^{-x} = 0 \quad (25.8)$$

can be inverted with Lambert W . Taking a square root formally,

$$xe^{-x/2} = \pm(-C)^{-1/2},$$

so

$$x = -2W_j \left(\mp \frac{1}{2\sqrt{-C}} \right), \quad (25.9)$$

with branch choices j producing arrays of approximate complex poles. Corrections from $F_1, F_2, \dots$ refine these locations.

This is a recurring pattern: a transseries derived at infinity predicts global singularities by solving an equation in the combined exponential variable.

25.4 Painlevé I as a nonlinear prototype

The first Painlevé equation is

$$u'' = 6u^2 + z. \quad (25.10)$$

Its solutions have no movable branch points, but they do have movable poles. For large z in suitable complex sectors, dominant balance gives

$$u_0(z) \sim \pm \sqrt{-\frac{z}{6}}. \quad (25.11)$$

Substitution generates a formal algebraic-power expansion around either branch. Linearizing $u = u_0 + \eta$ gives

$$\eta'' \sim 12u_0\eta. \quad (25.12)$$

A WKB estimate therefore produces exponentials whose actions are proportional to $(-z)^{5/4}$. Nonlinearity generates their integer multiples, yielding a one- or multi-parameter transseries depending on the sector and normalization.

Reorganizing those sectors in a variable

$$\tau = \sigma z^\beta \exp(-A(-z)^{5/4})$$

produces meromorphic coefficient functions $F_n(\tau)$. Their pole conditions give asymptotic arrays of Painlevé poles. This is one of the most striking uses of transasymptotics: data computed from a remote irregular limit predict the location of movable singularities of a nonlinear special function. Detailed normalization depends on the chosen branch of $(-z)^{1/4}$ and on the sector under study [33, 15].

25.5 When the transasymptotic variable is order one

Even if $e^{-A/z}$ is tiny, σ or z^β may compensate it. More importantly, analytic continuation can move into regions where $\Re(A/z)$ decreases. The crossover occurs when

$$|\tau| = |\sigma z^\beta e^{-A/z}| \asymp 1. \quad (25.13)$$

At that point, truncating at $k = 0$ or $k = 1$ is no longer uniform. A rational or meromorphic $F_0(\tau)$ can remain useful across the transition.

This is analogous to a boundary layer. There one introduces a stretched coordinate such as $X = x/\varepsilon$ because ordinary powers of ε cease to be uniform. Here one introduces an exponential coordinate because the hierarchy of sectors ceases to be uniform.

25.6 Hyperasymptotics: expanding the remainder

Optimal truncation of a divergent power series leaves a remainder of the size of the least term, often

$$e^{-A/z}.$$

Instead of stopping, one can expand that remainder using the neighboring saddle or Borel singularity. The new expansion is itself usually divergent and can be optimally truncated, leaving a still smaller remainder such as $e^{-B/z}$ or $e^{-2A/z}$. Repeating this process gives *hyperasymptotics*.

The hierarchy is:

1. truncate the perturbative sector near its least term;
2. represent the remainder by an adjacent exponential sector;
3. truncate that sector near its own least term;
4. continue through the network of actions.

Hyperasymptotics emphasizes exponentially improved numerical approximation. Resurgence emphasizes the analytic relations that make the hierarchy possible. Transseries provide the common formal container.

25.7 Terminants and smooth switching

Near a Stokes ray, the remainder after optimal truncation is often expressed using an incomplete gamma function or a closely related *terminant*. Its uniform asymptotics produce the error-function profile of Berry smoothing. The terminant therefore resolves three descriptions at once:

- far from the ray, one exponential multiplier is nearly constant;
- inside the transition layer, it varies smoothly;
- in the infinite-parameter limit, the smooth profile approaches a step.

This is the analytic mechanism behind the informal phrase “an exponential switches on.”

25.8 Sector summation versus perturbative summation

There are two different infinite sums to control:

$$\sum_{k \geq 0} \sigma^k e^{-kA/z} \underbrace{\left(\sum_{n \geq 0} a_{k,n} z^n \right)}_{\text{usually Borel summed}} \quad \text{and} \quad \underbrace{\sum_{k \geq 0} \tau^k F_k}_{\text{sector sum}}. \quad (25.14)$$

Borel summability of every inner series does not automatically prove convergence of the outer sector sum. Rigorous reconstruction requires uniform estimates in k , or a topology in which the whole transseries summation is continuous. Costin's rank-one theory is designed to supply such control in a broad nonresonant class [12, 13].

25.9 Practical pole prediction

Given a computed transseries:

1. factor each sector into a common variable τ ;
2. compute the leading coefficients $a_{k,0}$ for many k ;
3. identify or Padé-approximate $F_0(\tau) = \sum a_{k,0}\tau^k$;
4. find its nearest singularities $\tau = \tau_*$;
5. solve $\sigma z^\beta e^{-A/z} = \tau_*$ for z ;
6. use $F_1, F_2, \dots$ to perturb the predicted locations;
7. compare against numerical integration of the original equation.

The inversion in step 5 often reduces to Lambert W at leading order, exactly as in (25.9).

25.10 Exercises

Exercise 25.1. Solve (25.2) by separation of variables and match its integration constant to σ in (25.3).

Exercise 25.2. Substitute $y = \sum_{n \geq 0} z^n F_n(\tau)$, with $\tau = \sigma e^{-x}$ and $z = x^{-1}$, into a simple first-order equation of your choice. Derive the leading equation for F_0 .

Exercise 25.3. Starting from (25.8), derive (25.9) and state where branch choices enter.

Exercise 25.4. For Painlevé I, use (25.11)–(25.12) to show that the WKB exponent grows as a constant times $(-z)^{5/4}$.

Chapter summary

Transasymptotics sums across exponential sectors by holding $\tau = \sigma z^\beta e^{-A/z}$ fixed. It can reveal movable poles hidden across infinitely many sectors. Hyperasymptotics instead repeatedly expands optimally truncated remainders. Both exploit the same action network exposed by resurgence.

What analytic reconstruction theorems actually say

Formal algebra is exact within its own rules. Analytic reconstruction is conditional: it depends on the equation, the sector, growth bounds, nonresonance, and the chosen summation method. This chapter separates broadly established conclusions from tempting but invalid shortcuts.

26.1 Four levels of assertion

For a displayed object $\widehat{y}$, ask which of the following is meant.

1. **Formal existence.** The support is admissible and all algebraic operations are defined.
2. **Formal solution.** Substitution into the formal equation gives zero coefficient by coefficient.
3. **Asymptotic realization.** There is an actual function y whose asymptotic expansion is $\widehat{y}$ in a specified sector and scale.
4. **Canonical reconstruction.** A stated summation procedure assigns such a function to $\widehat{y}$ and respects the operations needed by the problem.

Each implication requires a theorem. A formal solution need not automatically possess a function realization, and a realization need not be unique without sectorial conditions.

26.2 A representative rank-one theorem

A broad class studied by Costin consists, after normalization, of analytic systems near infinity of the schematic form

$$\mathbf{y}' = -\Lambda \mathbf{y} - \frac{1}{x} B \mathbf{y} + \mathbf{f}_0(x) + \mathbf{g}(x, \mathbf{y}), \quad (26.1)$$

where Λ is a constant matrix, $\mathbf{f}_0$ is small at infinity, and $\mathbf{g}$ is analytic and at least nonlinear in $\mathbf{y}$ after suitable separation of terms. Under nondegeneracy and nonresonance assumptions, one constructs formal transseries

$$\widehat{\mathbf{y}}(x, \mathbf{C}) = \sum_{\mathbf{k} \in \mathbb{N}^r} \mathbf{C}^{\mathbf{k}} e^{-\mathbf{k} \cdot \lambda x} x^{\mathbf{k} \cdot \beta} \widehat{\mathbf{y}}_{\mathbf{k}}(x). \quad (26.2)$$

The rigorous conclusion, stated informally but faithfully, is:

Theorem 26.1 (Generalized Borel reconstruction, schematic). *For systems in the admissible rank-one class, the sector series in (26.2) are summable by a generalized Borel procedure in nonsingular directions where the selected exponentials decay. The summed transseries are actual solutions. Conversely, solutions with the corresponding decay behavior are represented by such summed transseries. Crossing singular directions changes the transseries parameters by Stokes maps.*

The exact theorem includes hypotheses on eigenvalues, resonance, analyticity, and normalization. Its significance is the two-way correspondence: transseries are not merely guessed approximations; within the theorem's class they coordinatize actual solutions [12, 13, 14].

26.3 Why nonresonance appears

If eigenvalue combinations coincide,

$$\mathbf{k} \cdot \lambda = \ell \cdot \lambda, \quad \mathbf{k} \neq \ell,$$

sector equations couple at the same exponential weight. Denominators used to solve recurrences may vanish, logarithmic terms may appear, and parameter transformations may cease to be simple translations. Resonant theories exist, but the cleanest general theorems state explicit restrictions because the normal form and summability estimates change.

26.4 The role of sector width

The Laplace kernel decays only when

$$\Re(xe^{i\theta}) > K$$

for the growth constant K of the continued Borel transform. Hence a Borel sum naturally lives in a sector of the original variable. Rotating that sector eventually makes the kernel grow or makes the path hit singularities.

Analytic continuation of the actual solution may extend farther, but its asymptotic representation must then change. This is why connection problems require several overlapping sectorial solutions and Stokes data.

26.5 Quasianalyticity and uniqueness

A class is quasianalytic if an element with zero asymptotic expansion must itself vanish. Ordinary Poincaré asymptotics on a narrow sector is not quasianalytic because exponentially flat functions survive. Strong Gevrey bounds on sufficiently wide sectors can be quasianalytic. Resurgent and analyzable classes obtain uniqueness by adding continuation and growth structure, not by ignoring the flat functions.

The lesson is subtle:

A divergent series can determine a function uniquely, but only after one specifies a sufficiently rigid analytic category and a direction or median prescription.

26.6 A comparison table

Formal statement	Desired analytic statement	Extra work required
$\widehat{f}$ has well-based support	Infinite expression defines a germ	Evaluation or summation theorem; uniform control
$P(\widehat{f}) = 0$ formally	$P(f) = 0$ analytically	Summation must commute with operations in P
$\widehat{f}' = (\widehat{f})'$	Differentiated sum has summed derivative	Uniform convergence or Borel operational theorem
Formal inverse exists	Analytic function is nonzero and inverse is represented	Sectorial lower bound and multiplicative summation
$\widehat{f} \circ \widehat{g}$ is formally defined	Composition of reconstructed functions matches it	Domain control and a composition theorem
A formal parameter C is free	Every nearby actual solution has one C	Completeness/converse theorem for the ODE
Borel singularity at A	Stokes jump contains $e^{-A/z}$	Analytic continuation and local singularity analysis
Large-order formula is guessed	Stokes constant is determined	Error estimates and exclusion/subtraction of other singularities

26.7 Formal composition is especially delicate

Addition, multiplication, and differentiation often pass through Borel summation under standard hypotheses. Composition is harder because:

- the inner function changes the asymptotic domain;
- infinitely many Taylor terms may contribute at the same magnitude;
- logarithms and exponentials can move singular directions;
- the formal support conditions and analytic convergence conditions are different.

For standard fields of transseries, composition is a formal operation under precise support hypotheses. Recent work establishes broad Taylor-expansion theorems over generalized power-series fields and monotonicity/Taylor approximation for omega-series [5, 31]. These results strengthen the formal-function dictionary, but they do not turn every arbitrary Hahn series into an analytic germ.

26.8 No single universal summation map

One might hope for a map

$$\Sigma : \{\text{all formal transseries}\} \longrightarrow \{\text{all functions}\}$$

that preserves every operation and agrees with ordinary convergence. Such a map is too much to expect without restricting both sides. Problems include natural boundaries, incompatible branch choices, nonunique flat additions, excessive growth, and set-theoretic size.

Successful theories choose a structured domain:

- multisummable series for systems with finitely many Gevrey levels;
- resurgent transseries with controlled Borel continuation;
- analyzable functions obtained by generalized Borel procedures;
- LE-transseries interpreted as germs in Hardy fields;
- omega-series evaluated at positive infinite surreal numbers.

Each answers a different version of “what function does this formal expression represent?”

26.9 Proof checklist for a claimed transseries solution

When reading or writing a serious argument, check:

1. **Ambient field:** Which monomials and supports are allowed?
2. **Formal recursion:** Why is every coefficient or sector uniquely determined?
3. **Growth:** What Gevrey or coefficient bounds are proved?
4. **Continuation:** Where can the Borel transform be continued?
5. **Summation direction:** Which rays are admissible or singular?
6. **Operations:** Does summation commute with the equation’s products, derivatives, and compositions?
7. **Parameter completeness:** Does every relevant actual solution arise?
8. **Stokes data:** How are lateral representations matched?
9. **Uniformity:** Are sector sums controlled when infinitely many k contribute?
10. **Domain:** In exactly which x - or z -sector is the result valid?

A paper need not prove every item from scratch, but it should invoke a theorem whose hypotheses cover them.

26.10 Exercises

Exercise 26.1. Give an example of two distinct functions with the same inverse-power asymptotic expansion on the positive real axis.

Exercise 26.2. Explain why a summation method that does not preserve multiplication cannot automatically reconstruct solutions of a nonlinear ODE from formal solutions.

Exercise 26.3. For the rank-one template (26.1), explain how the eigenvalues of Λ determine the first exponential actions.

Exercise 26.4. Classify each claim as formal, asymptotic, or summational: “ $\log(e^T) = T$,” “ $f(x) \sim x^{-1}$,” and “the two lateral sums differ by $2\pi i e^{-x}$.”

Chapter summary

Formal solvability, asymptotic realization, and canonical summation are different levels. Rigorous reconstruction theorems prove their connection for specified classes, directions, and hypotheses. The broad rank-one theory shows that generalized Borel-summed transseries can parametrize actual nonlinear ODE solutions, but no unrestricted universal summation principle exists.

Part VI

Transseries in the larger mathematical landscape

Hardy fields: transseries as eventual germs

Formal transseries speak the language of growth. Hardy fields speak about actual functions that eventually have stable behavior. The two theories meet because both carry an order, a derivative, and a comparison relation at infinity.

27.1 Germs at positive infinity

Two functions f and g , each defined on some interval extending arbitrarily far to the right, have the same *germ at $+\infty$* if they agree for all sufficiently large x . We forget every bounded initial segment.

For example, the functions

$$f(x) = \log x \quad \text{for } x > 0$$

and

$$g(x) = \begin{cases} 1729, & 0 < x \leq 100, \\ \log x, & x > 100 \end{cases}$$

define the same germ. Their eventual behavior is identical.

Addition, multiplication, and differentiation are defined using representatives:

$$[f] + [g] = [f + g], \quad [f][g] = [fg], \quad [f]' = [f'].$$

Changing a representative on a bounded interval does not affect these operations.

Definition 27.1 (Hardy field). A Hardy field is a field of germs at $+\infty$ of real-valued differentiable functions that is closed under differentiation.

The word *field* has strong consequences. Every nonzero element must possess a reciprocal, so a representative of a nonzero germ must be nonzero for all sufficiently large x .

27.2 Why oscillation is excluded

The germ of $\sin x$ cannot lie in a Hardy field. It has zeros arbitrarily far to the right, so $1/\sin x$ is not represented by a function defined on any final interval (a, ∞) . Similarly, a nonzero element of a Hardy field has an eventual sign.

This nonoscillation is one reason Hardy fields align so well with ordered transseries. A transseries has a leading monomial and therefore a definite sign when its leading coefficient is nonzero. Persistent oscillations such as $\sin x$ require a different asymptotic language, often involving complex phases, Fourier modes, or transseries over nonordered monomial groups.

27.3 Basic examples

The rational function field

$$\mathbb{R}(x)$$

becomes a Hardy field by ordering x as positive infinite and taking ordinary differentiation. Larger examples include fields generated by suitable collections of

$$x^r, \quad \log x, \quad \log \log x, \quad e^x, \quad e^{\sqrt{x}}, \quad \text{and their algebraic combinations.}$$

The germs definable in many tame, or *o-minimal*, expansions of the real field form Hardy fields. Tameness prevents wild oscillation and guarantees eventual regularity.

Not every ad hoc set of familiar functions is a field or is closed under differentiation. “The Hardy field generated by these functions” means the smallest field of germs containing them and closed under the required derivative operations, when such a field is under discussion.

27.4 Eventual order and dominance

For germs f and g , define

$$f < g \iff f(x) < g(x) \text{ eventually.}$$

Because nonzero Hardy-field germs have eventual sign, this is a total order.

Let C be the constant field, usually identified with $\mathbb{R}$. Define

$$f \preceq g \iff |f| \leq c|g| \text{ eventually for some } c \in C^{>0}, \quad (27.1)$$

$$f \prec g \iff |f| < c|g| \text{ eventually for every } c \in C^{>0}. \quad (27.2)$$

Thus $f \prec g$ means $f/g \rightarrow 0$ whenever the quotient has the expected limit interpretation.

Examples are

$$\log x \prec x^a \prec e^{x^b} \prec e^x \quad (0 < a, b < 1).$$

The convex hull of the constants is

$$O = \{f : f \preceq 1\}, \quad (27.3)$$

the ring of bounded germs. Its maximal ideal

$$\mathfrak{o} = \{f : f \prec 1\} \quad (27.4)$$

consists of infinitesimals. The quotient $O/\mathfrak{o}$ records the eventual constant part when that part exists.

A common trap

A general bounded function need not approach a constant. The axiom $O = C + \mathfrak{o}$ is therefore a genuine tameness requirement: every bounded element must be a constant plus an infinitesimal.

27.5 The definition of an H -field

Definition 27.2 (H -field). An H -field is an ordered differential field K with constant field C such that:

1. $O = C + \mathfrak{o}$;
2. if f is larger than every constant, written $f > C$, then $f' > 0$.

The second axiom says that a positive infinite element is increasing. For actual germs this is a deep eventual regularity property, not true of arbitrary functions tending to infinity. For example, $x + \sin(x^2)$ tends to infinity but its derivative changes sign infinitely often. Such a germ cannot live in a Hardy field containing its derivative and satisfying the H -field conditions.

Hardy fields containing $\mathbb{R}$ are H -fields, and the standard transseries field $\mathbb{T}$ is an H -field. The axioms isolate the order–derivative behavior shared by these apparently different objects [3, 2].

27.6 Leading monomials versus valuation classes

In a transseries, the leading monomial $\text{lm}(T)$ determines its dominance class. In an abstract H -field, one may not have named monomials, but the relation

$$f \asymp g \iff f \preceq g \text{ and } g \preceq f$$

plays the same role. The quotient group of nonzero elements by this relation is the *value group*. Very roughly:

$$\boxed{\text{transmonomial}} \longleftrightarrow \boxed{\text{chosen representative of a valuation class}}.$$

A Hahn field has a distinguished cross-section of monomials. A general Hardy field need not.

27.7 Logarithmic derivatives

For $f \neq 0$, the logarithmic derivative is

$$f^\dagger := \frac{f'}{f}. \tag{27.5}$$

It converts multiplication into addition:

$$(fg)^\dagger = f^\dagger + g^\dagger.$$

For typical scales,

$$(x^a)^\dagger = \frac{a}{x}, \quad (e^{x^b})^\dagger = bx^{b-1}, \quad (e^{e^x})^\dagger = e^x.$$

Thus logarithmic derivatives encode the relation between growth and differentiation. Much of asymptotic differential algebra is organized by how valuation classes transform under $f \mapsto f^\dagger$.

A key phenomenon is that differentiation may reverse naive comparisons. Although $1 \prec \log x \prec x$, their derivatives are

$$0, \quad x^{-1}, \quad 1.$$

The correct theory tracks both valuations and logarithmic derivatives, leading to the structure called an *asymptotic couple*.

27.8 Small derivation

An H -field has *small derivation* if

$$f \prec 1 \implies f' \prec 1. \quad (27.6)$$

For germs this says that the derivative of an infinitesimal is still infinitesimal. It holds in Hardy fields and in the standard transseries field.

The condition depends on the chosen asymptotic variable. Under a change of independent variable, the derivation rescales. In differential-algebra language one often replaces ∂ by $\phi\partial$ to reveal the correct dominant part of an equation.

27.9 Liouville closure

An H -field is *Liouville closed* if it is real closed and, for every f , it contains solutions to

$$g' = f \quad \text{and} \quad h' = fh, \quad h \neq 0. \quad (27.7)$$

The second equation asks for an exponential integral:

$$h = c \exp \left(\int f \right).$$

Thus Liouville closure supplies antiderivatives, exponentials of antiderivatives, and algebraic roots.

The full transseries field has these closure properties. Its formal integration algorithm from Chapter 12 is therefore not merely a convenience; it reflects a structural completeness property.

27.10 Newtonianity and differential equations

Hensel's lemma says that an approximate root of a polynomial over a valued field can often be corrected to an exact root. *Newtonianity* is an analogous property for differential polynomials. After rescaling the derivation and isolating a dominant differential part, a suitable approximate root can be lifted.

Another condition, called ω -freeness, excludes a particular kind of pseudolimit obstruction arising from iterated logarithms. The combination of Liouville closure, newtonianity, and ω -freeness gives a powerful axiomatization of the first-order theory of transseries. At a sophomore level, it is enough to remember the roles:

Liouville closed: elementary integrations and exponential integrations exist;

newtonian: dominant differential equations admit corrected roots;

ω -free: no hidden logarithmic-gap obstruction remains.

27.11 An intermediate-value principle for differential polynomials

A differential polynomial has the form

$$P(Y) = P(Y, Y', \dots, Y^{(r)})$$

with coefficients in the field. An H -field version of the intermediate value property says:

$$f < g, \quad P(f) < 0 < P(g) \implies \text{some } y \text{ between } f \text{ and } g \text{ satisfies } P(y) = 0.$$

For ordinary polynomials this characterizes real closed fields. For differential polynomials it captures a remarkable degree of closure under asymptotic differential equations.

The theory developed by Aschenbrenner, van den Dries, and van der Hoeven gives complete and model-complete axiomatizations in this setting; [Chapter 30](#) explains what those logical words mean [2, 3].

27.12 From transseries to germs

A formal transseries T can sometimes be realized by a germ f so that:

$$\begin{aligned} T + U &\mapsto f + g, \\ TU &\mapsto fg, \\ T' &\mapsto f', \\ T < U &\mapsto f < g \text{ eventually.} \end{aligned}$$

Such an embedding makes formal dominance agree with eventual dominance. For logarithmic-exponential transseries, sophisticated realization results place copies of transseries inside Hardy-field-like structures.

The reverse direction is also important: one asks whether a Hardy germ has a transseries expansion. Many tame germs do, but arbitrary maximal Hardy fields are larger and subtler. Current research aims to understand common extensions and embeddings among Hardy fields, transseries, and surreal numbers.

27.13 A three-way dictionary

Idea	Formal transseries	Hardy-field germ
Large variable	symbol $x > \mathbb{R}$	identity germ $x \mapsto x$
Infinitesimal	$x^{-1} \prec 1$	function tending to 0
Order	sign of leading coefficient	eventual sign
Dominance	leading monomial comparison	bounded quotient
Derivative	formal strong derivation	derivative of representatives
Constant	coefficient in $\mathbb{R}$	constant germ
Composition	formal substitution under support rules	ordinary function composition where defined
Integration	formal antiderivative in a closure	germ of an actual antiderivative

27.14 Exercises

Exercise 27.1. Prove that a nonzero germ in a field of germs must be eventually nonzero.

Exercise 27.2. Order the germs $\log \log x$, $(\log x)^{10}$, $x^{1/100}$, and $e^{\sqrt{\log x}}$ by strict dominance.

Exercise 27.3. Compute the logarithmic derivatives of $x^{\log x}$ and $\exp(x/\log x)$.

Exercise 27.4. Give a function tending to $+\infty$ whose derivative changes sign infinitely often, and explain why this behavior conflicts with the H -field growth axiom.

Chapter summary

Hardy fields are differential fields of nonoscillatory germs at $+\infty$. Their eventual order and dominance mirror leading-term order in transseries. H -fields abstract this common structure; Liouville closure, newtonianity, and related axioms describe their ability to solve asymptotic differential equations.

Surreal numbers: a numerical home for transseries

Transseries treat x as a formal infinitely large quantity. Surreal numbers provide actual canonical numbers that are infinite or infinitesimal. The first infinite ordinal ω can play the role of x , and many transseries can be evaluated at ω or at other positive infinite surreal numbers.

28.1 Numbers born from cuts

A surreal number is constructed recursively as a cut

$$x = \{L \mid R\}, \quad (28.1)$$

where every element of the set L is less than every element of the set R . The value is the *simplest* number lying strictly between the two sides.

The first examples are

$$\begin{aligned} 0 &= \{\emptyset \mid \emptyset\}, \\ 1 &= \{0 \mid \emptyset\}, \\ -1 &= \{\emptyset \mid 0\}, \\ \frac{1}{2} &= \{0 \mid 1\}, \\ 2 &= \{1 \mid \emptyset\}. \end{aligned}$$

At each stage, or “birthday,” new cuts using earlier numbers create new numbers. Continuing through all ordinal birthdays produces a proper class **No** rather than a set.

28.2 The first infinite and infinitesimal surreals

After all finite natural numbers have appeared, the cut

$$\omega = \{0, 1, 2, 3, \dots \mid \emptyset\} \quad (28.2)$$

is greater than every natural number. Arithmetic in **No** gives

$$\omega + 1, \quad \sqrt{\omega}, \quad \omega^\pi, \quad e^\omega,$$

and their reciprocals. In particular,

$$\varepsilon = \frac{1}{\omega}$$

is positive but smaller than every positive real number.

Unlike a nonstandard real obtained from an ultrafilter, ω is canonical inside the surreal construction. No auxiliary choice is used to single it out.

28.3 Conway normal form

Every surreal number has a unique expansion

$$x = \sum_{i < \alpha} r_i \omega^{a_i}, \quad (28.3)$$

where:

- α is an ordinal;
- $r_i \in \mathbb{R} \setminus \{0\}$;
- the surreal exponents satisfy $a_0 > a_1 > a_2 > \dots$.

This is a Hahn-series normal form with monomials ω^a . The support is reverse well ordered, exactly as in Chapter 5.

For example,

$$3\omega^2 - \pi\omega^{1/2} + 7 + \omega^{-1}$$

is a surreal number whose leading monomial is ω^2 . Infinite normal forms can be far longer:

$$\omega + 1 + \omega^{-1} + \omega^{-2} + \dots$$

is also meaningful.

Intuition

A surreal number is simultaneously a number and a generalized series. Conway normal form makes the bridge to transseries visible: replace the formal monomial x^a by the numerical monomial ω^a .

28.4 Summable families

Not every arbitrary family of surreal numbers has a sum. A family of normal forms is summable when:

1. the union of all monomial supports is reverse well ordered;
2. each fixed monomial occurs in only finitely many members of the family.

Then coefficients can be added monomial by monomial. This is the same finiteness principle that made Hahn multiplication safe.

Thus the surreal symbol

$$\sum_{n \geq 0} \omega^{-n}$$

is legitimate, while an attempted sum placing infinitely many nonzero contributions on the same monomial is not a formal surreal sum unless those real coefficients converge through some separate operation.

28.5 Exponentiation and logarithms

The surreal field admits a canonical exponential function extending the real exponential and satisfying the usual laws:

$$\exp(x + y) = \exp(x) \exp(y), \quad \exp(0) = 1.$$

It is an order-preserving isomorphism from the additive group of **No** onto its positive multiplicative group. Hence every positive surreal has a logarithm.

For an infinitesimal ε , familiar series work:

$$e^\varepsilon = 1 + \varepsilon + \frac{\varepsilon^2}{2!} + \cdots,$$

$$\log(1 + \varepsilon) = \varepsilon - \frac{\varepsilon^2}{2} + \frac{\varepsilon^3}{3} - \cdots.$$

These are summable Hahn families. For infinite arguments, the full surreal exponential is defined recursively and is compatible with normal forms in a subtler way.

28.6 Why e^ω is not merely ω^a

Conway monomials have the form ω^a , but exponentiation produces quantities such as e^ω that are much larger than every fixed power ω^a with real a . They are still surreal monomials, because the exponent a in ω^a is itself allowed to be a surreal number. There exists a surreal b with

$$e^\omega = \omega^b.$$

The relation between additive exponents and Conway exponents is encoded by an omega-map and related functions. This flexibility lets the surreal monomial group contain towers comparable to transmonomials.

28.7 Derivations on the surreals

A derivative on all surreal numbers cannot mean an ordinary limit of difference quotients: the surreals are not a conventional real manifold. Instead one asks for a derivation

$$\partial : \mathbf{No} \rightarrow \mathbf{No}$$

satisfying algebraic and asymptotic requirements:

$$\begin{aligned} \partial(x + y) &= \partial x + \partial y, \\ \partial(xy) &= x\partial y + y\partial x, \\ \partial(e^x) &= e^x \partial x, \\ \partial\omega &= 1, \end{aligned}$$

plus compatibility with infinite summation and positivity of derivatives of positive infinite elements.

Berarducci and Mantova constructed a distinguished “simplest” derivation with these properties, making **No** a Liouville-closed H -field [9]. Under it,

$$\partial(\omega^r) = r\omega^{r-1} \quad (r \in \mathbb{R}),$$

just as expected when ω plays the role of an infinite variable.

28.8 Embedding ordinary transseries

There is a natural differential-field embedding

$$\iota : \mathbb{T} \hookrightarrow \mathbf{No} \tag{28.4}$$

that sends the formal variable x to ω , respects order and exponentiation, and is compatible with the Berarducci–Mantova derivation. At the simplest level,

$$\begin{aligned} x^r &\mapsto \omega^r, \\ \log x &\mapsto \log \omega, \\ e^x &\mapsto e^\omega, \\ \sum c_m \mathbf{m} &\mapsto \sum c_m \iota(\mathbf{m}). \end{aligned}$$

The last line is valid because admissible transseries supports map to summable surreal families.

A major theorem strengthens this: the natural embedding of $\mathbb{T}$ into **No** is elementary in the language of ordered differential fields with the relevant structure. In logical terms, every first-order statement about embedded transseries has the same truth value in the ambient surreal field [4].

28.9 Omega-series

Inside **No**, define the field of *omega-series* as the smallest subfield containing $\mathbb{R}$ and ω and closed under:

1. exponential and logarithm;
2. sums of all summable families.

This class contains copies of LE-series and EL-series but is substantially larger. It is a proper class, not merely a set-sized field.

Berarducci and Mantova showed that omega-series admit composition with positive infinite surreal arguments and are compatible with differentiation. Each omega-series f therefore acts as a surreal function

$$x \mapsto f \circ x, \quad x > \mathbb{R},$$

obtained by substituting x for ω in a controlled way [10].

28.10 Evaluation is not naive textual substitution

Consider a nested expression with infinite sums and exponentials. Replacing every printed ω by a surreal x is only a mnemonic. One must prove that:

- each transformed family remains summable;
- different constructions of the same omega-series give the same value;
- composition is associative;
- differentiation obeys a chain rule;
- exponentials and logarithms remain compatible.

The formal tree structure of a transseries is used to establish these facts. This is analogous to proving that a formal power series can be composed: the symbolic instruction is simple, but the support theorem is the real work.

28.11 Why not every surreal has a satisfactory composition

The whole surreal field is vastly larger than the omega-series. It can be viewed as an enormous Hahn field of transseries-like normal forms, but there is no known composition operation on all of **No** that simultaneously extends the desired substitutions and is compatible with the standard derivation. Indeed, structural obstructions show that the omega-series boundary is meaningful in this context [10].

Thus:

every omega-series is surreal	but	not every surreal is an omega-series.
-------------------------------	-----	---------------------------------------

Being a generalized number is weaker than being a composable transseries function.

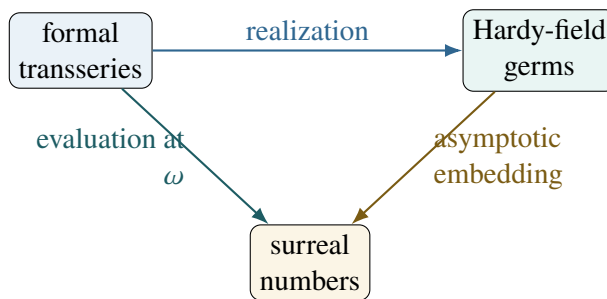
28.12 Surreal evaluation versus real asymptotic realization

If a transseries T maps to a surreal number $T(\omega)$, has it been summed as a real function? Not necessarily.

Surreal evaluation answers an algebraic question: what number results when the infinite variable is interpreted as ω ? Borel summation answers an analytic question: what complex or real function has a given divergent asymptotic expansion in a sector? The two constructions preserve different structures and need not coincide term for term.

For a convergent power series in an infinitesimal, both interpretations often agree with familiar algebra. For factorially divergent resurgent sectors, Borel singularities and Stokes choices carry analytic data not supplied merely by a Hahn normal form.

28.13 A useful conceptual triangle



The arrows require hypotheses and choices; the diagram is a research program, not a claim that all three classes are identical.

28.14 Exercises

Exercise 28.1. Using cuts, verify that $\{0 \mid 1\}$ lies between 0 and 1 and is simpler than every other previously available number in that interval.

Exercise 28.2. Write the leading Conway normal forms of $\omega + \omega^{-1}$, $(\omega + 1)^2$, and $(1 + \omega^{-1})^{-1}$.

Exercise 28.3. Explain why the family $\{\omega^{-n} : n \in \mathbb{N}\}$ is summable, while a family containing the term ω^{-1} with nonzero coefficient infinitely many times fails the finite-contribution condition.

Exercise 28.4. Distinguish in one paragraph the statements “ T can be evaluated at ω ” and “ T is Borel summable along the positive real direction.”

Chapter summary

Surreal numbers form a canonical ordered field containing infinite and infinitesimal numbers. Conway normal form is Hahn-like, and the surreal exponential and Berarducci–Mantova derivation make room for transseries. Omega-series admit controlled infinite sums, composition, and differentiation, while all surreals are too large to carry the same function structure.

Beyond the standard LE field

There is no single maximal object universally called “the transseries.” Different constructions close different starting fields under different kinds of infinite sums, exponentials, logarithms, composition, or transfinite iteration. The standard logarithmic-exponential field $\mathbb{T}$ is extraordinarily useful, but it is one carefully chosen member of a larger ecosystem.

29.1 Finite construction depth is a real restriction

A standard LE-transseries is built by finitely many rounds of operations, even though each round may introduce an infinite Hahn sum. Thus every individual element has finite exponential height and finite logarithmic depth.

Examples such as

$$e^{e^{e^x}} \quad \text{or} \quad \log \log \log x$$

have large but finite depth and cause no problem. The challenge begins with objects requiring unbounded depth in a single support, for example

$$e^{\sqrt{x} + \sqrt{\log x} + \sqrt{\log \log x} + \cdots}. \quad (29.1)$$

No finite construction stage contains all iterated logarithms appearing in the exponent. Larger transseries fields admit such supports while preserving summability, differentiation, and composition.

29.2 LE-series and EL-series

The names reverse the order of two closure ideas.

Logarithmic-exponential (LE) series arise from the construction developed by van den Dries, Macintyre, and Marker, with roots in Dahn–Göring. One adjoins exponentials and logarithms through staged generalized-series extensions.

Exponential-logarithmic (EL) series arise from prelogarithmic Hahn fields followed by exponential closure, as developed in work of Kuhlmann and collaborators.

The resulting fields satisfy many of the same first-order properties as non-Archimedean models of the real exponential field, but the constructions are intrinsically different. Kuhlmann and Tressl proved that the LE field embeds as an ordered exponential field into any of the relevant EL fields, whereas an EL field does not embed into the LE field in the corresponding way [28].

This asymmetry is a useful warning: two fields may contain all familiar finite expressions and satisfy the same broad logical theory while differing on which infinite generalized sums they admit.

29.3 Grid-based versus well-based transseries

A grid-based transseries has support inside a finitely generated ratio grid

$$m_0 \mu^{\mathbf{k}}, \quad \mathbf{k} \in \mathbb{N}^r.$$

A well-based transseries needs only reverse well-order, with no finite ratio basis.

Grid-based fields are smaller but algorithmically friendly:

- monomials can be represented by integer exponent vectors;
- truncation regions are finite-dimensional;
- multiplication is sparse convolution on $\mathbb{Z}^r$;
- fixed-point proofs can use contraction in grid order.

Well-based fields capture supports such as

$$\{e^{-x}, e^{-\sqrt{2}x}, e^{-\pi x}, \dots\}$$

or increasingly complex iterated-log scales that do not lie in one finite grid. They are closer to a maximal formal universe but harder to compute with.

29.4 Schmeling's axiomatic transseries fields

Schmeling's thesis develops broad axioms for fields of transseries, including strong derivations and exponential-logarithmic structure [35]. Rather than defining one field by a single inductive recipe, the axiomatic approach asks which support, summation, exponential, logarithm, and derivation laws a satisfactory transseries field should satisfy.

Typical desiderata include:

1. a Hahn-field presentation with a distinguished monomial group;
2. closure under summable families;
3. exponential and logarithm extending the real functions;
4. a strong derivation preserving admissible sums;
5. compatibility of derivative with exponential and logarithm;
6. enough monomials to integrate broad classes of series;
7. a composition law, when the theory is intended to treat functions.

These properties are not automatically compatible on an arbitrarily large Hahn field.

29.5 Superexponentials and iteration beyond integers

The ordinary exponential tower is defined recursively:

$$E_0(x) = x, \quad E_{n+1}(x) = e^{E_n(x)}.$$

A *superexponential* seeks a function $E_t(x)$ defined for noninteger t such that

$$E_{s+t} = E_s \circ E_t, \quad E_1 = e^x.$$

This is an Abel/iteration problem of the kind introduced in [Chapter 16](#), but now the function grows so quickly that ordinary LE height is shifted by composition.

Its inverse family gives hyperlogarithms. Closing a transseries field under such operators requires monomials indexed not only by finite exponential height but by more general ordinal levels.

29.6 Logarithmic hyperseries

Logarithmic hyperseries extend iterated logarithms into transfinite hierarchies. One introduces symbols behaving like

$$\ell_0 = x, \quad \ell_{n+1} = \log \ell_n,$$

and then continues the indexing beyond finite n in a coherent formal system. At a limit ordinal, no single finite iterate is available, so the construction must use asymptotic relations and generalized supports rather than an ordinary recursive formula.

Van den Dries, van der Hoeven, and Kaplan developed logarithmic hyperseries as a field with composition and derivation, providing a rigorous home for extremely slowly varying scales [29]. These scales arise naturally when repeatedly solving equations whose inverses introduce more logarithms than any fixed finite depth.

29.7 Hyperserial fields

A hyperserial field is designed to support hyperexponentials, hyperlogarithms, Hahn summation, and composition across transfinite levels. Recent work shows that the surreal numbers carry substantial hyperserial structure and can serve as a vast ambient field for these hierarchies [6, 7].

The guiding ladder is

$$\text{power series} \subset \text{LE-transseries} \subset \text{omega-series} \subset \text{hyperseries-like structures} \subset \mathbf{No},$$

but each inclusion suppresses qualifications about the chosen embedding and operations. In particular, a larger ambient field may contain the elements of a smaller one without extending every composition or derivation in the desired way.

29.8 Height, level, and comparability

Two positive infinite elements f, g are logarithmically comparable if repeated logarithms eventually bring them to comparable magnitude. Informally, one studies whether there is an

integer n such that

$$\log_n f \asymp \log_n g.$$

The resulting *levels* distinguish growth like

$$x, \quad e^x, \quad e^{e^x}, \quad \dots$$

LE-transseries can move through finitely many such levels. Hyperexponential structures allow systematic movement through transfinite level indices.

These comparisons matter for composition. Composing with e^x raises levels; composing with $\log x$ lowers them. A field closed under arbitrary repeated composition must have room for every level reached.

29.9 Oscillatory and complex transseries

Ordered real transseries deliberately avoid persistent oscillation, but asymptotic analysis often needs phases such as

$$e^{ix}, \quad e^{iA/z}, \quad \cos(e^x).$$

One can work over complex coefficients and monomial groups that are not totally ordered by magnitude, or separate magnitude from phase:

$$e^{-(a+ib)/z} = e^{-a/z} e^{-ib/z}.$$

The first factor controls exponential size and the second controls oscillation.

Multisummable and resurgent transseries routinely use complex actions. Their sector order is often only partial: two exponentials can have the same magnitude but different phase. This analytic tradition overlaps with, but is not identical to, the ordered Hahn-field tradition.

29.10 Transseries with fractional powers and logarithmic branching

Near singular points, natural ansatzes include

$$z^\alpha (\log z)^m \sum_{n \geq 0} a_n z^{n/q}.$$

Ramification replaces z by w^q , turning fractional powers into integer powers on a covering surface. Logarithms record monodromy. Exponential sectors then take the form

$$e^{-A/z^p} z^\alpha (\log z)^m \widehat{\phi}(z^{1/q}).$$

Such objects fit naturally into resurgent analysis even when they are not presented as elements of one canonical ordered LE field.

29.11 Why there is no final largest field

Suppose a field contains a fast-growing F . One can ask for a compositional inverse, a fractional iterate, or a solution to

$$G' = F, \quad H \circ H = F, \quad K(x+1) = F(K(x)).$$

Adjoining these solutions may create new growth classes, which invite new exponentials, sums, and inverses. Any chosen closure principle suggests a stronger one.

Therefore “the field of all transseries” is best understood relative to a signature of desired operations. The standard $\mathbb{T}$ is a canonical and remarkably complete answer for finite logarithmic-exponential asymptotics and differential algebra, not an absolute terminal object for every possible asymptotic operation.

29.12 A map of common constructions

Construction	Main strength	Typical limitation or question
Grid-based transseries	finite-dimensional support bookkeeping; algorithms	excludes many well-based supports
Well-based LE-series	broad Hahn sums; exp, log, derivative	every element still has finite construction depth
EL-series	large exponential closures of prelogarithmic Hahn fields	not isomorphic to the LE construction
Resurgent transseries	analytic continuation, Stokes data, large order	requires controlled Borel singularities
Omega-series	surreal evaluation, summation, composition	proper class; smaller than all surreals
Logarithmic hyperseries	transfinite slowly varying scales	specialized hierarchy and substantial machinery
Hyperserial fields	hyperexponentials and hyperlogarithms	active structural development
All surreal numbers	universal numerical ambient class	no compatible global composition of the desired kind

29.13 Exercises

Exercise 29.1. Explain why every finite tower $\exp^{on}(x)$ can belong to an LE field even though the family of all heights is unbounded.

Exercise 29.2. Give a support involving all iterated logarithms that cannot be contained in any fixed finite logarithmic depth.

Exercise 29.3. Compare the algorithmic advantages of a two-generator ratio grid with a merely reverse-well-ordered support.

Exercise 29.4. Why does containment of two fields as sets not guarantee that composition on the smaller field extends to the larger one?

Chapter summary

The standard LE field is one disciplined closure of generalized series under finite rounds of exp, log, and summation. EL fields, omega-series, logarithmic hyperseries, and hyperserial fields admit different larger supports or iteration levels. “Transseries” names a family of compatible constructions rather than one uncontested maximal object.

Model theory and asymptotic differential algebra

Model theory studies mathematical structures through statements written in a fixed formal language. For transseries, it asks a surprisingly practical question:

Which systems of algebraic differential equations and inequalities must have solutions merely because the ambient field has the right asymptotic structure?

The answer turns the field $\mathbb{T}$ from a collection of formal expressions into a universal domain for nonoscillatory asymptotic differential algebra.

30.1 Languages, structures, and sentences

A first-order language lists allowed symbols. For ordered differential fields, one may use

$$0, 1, +, -, \cdot, <, \partial,$$

and perhaps a dominance relation $\preceq$. A structure interprets these symbols: in $\mathbb{T}$, ∂ is formal differentiation and $<$ is leading-term order.

A first-order formula can quantify over field elements:

$$\exists y (y' > 0 \wedge y^2 = x).$$

It cannot directly quantify over arbitrary subsets, over all sequences, or over all supports unless those are encoded as field elements.

The *theory* of a structure is the collection of all first-order sentences true in it.

30.2 Elementary equivalence

Two structures are elementarily equivalent if they satisfy exactly the same first-order sentences. They need not be isomorphic or have the same cardinality.

For example, different real closed fields satisfy the same first-order theory in the language of ordered fields. One may be Archimedean and another may contain infinitesimals, yet every

sentence involving only field operations, order, and quantification over elements has the same truth value.

Likewise, different H -closed fields can share the first-order theory of transseries even if their elements have very different concrete representations.

30.3 Elementary embeddings

An embedding $K \hookrightarrow L$ is elementary if every first-order formula with parameters from K has the same truth value before and after embedding. It preserves not only equations and inequalities but every finite logical combination and quantifier pattern.

The natural embedding

$$\mathbb{T} \hookrightarrow \mathbf{No}$$

with the Berarducci–Mantova derivation is elementary in the appropriate H -field language [4]. Thus $\mathbf{No}$ contains $\mathbb{T}$ without introducing a new first-order solution to a finite differential-algebraic problem over $\mathbb{T}$ that was absent from $\mathbb{T}$ itself.

30.4 Existential closure

An existential statement has the form

$$\exists y_1 \cdots \exists y_n \Phi(\mathbf{y}, \mathbf{a}),$$

where Φ has no further quantifiers. It can express a finite system of differential equations and inequalities.

A field K is existentially closed in a class if every such system over K that has a solution in an extension from the class already has a solution in K .

For transseries, this means roughly:

If a finite nonoscillatory asymptotic differential problem is compatible with the H -field axioms, then $\mathbb{T}$ already contains a solution.

This is a structural existence theorem, not an algorithm that immediately prints the solution's coefficients.

30.5 Model completeness

A theory is model complete if every embedding between its models is elementary. Once two H -closed fields sit inside one another, no new first-order facts about the smaller field appear in the larger.

The theory of H -closed fields is model complete. In one expository axiomatization, the theory of $\mathbb{T}$ is specified by:

1. being an H -field with small derivation;
2. being Liouville closed;
3. satisfying the differential intermediate value property.

Equivalent formulations use the more structural conditions of newtonianity and ω -freeness [2, 3].

30.6 Quantifier elimination

Quantifier elimination means that every first-order formula is equivalent, in the theory, to one without quantifiers in an enriched language. For real closed fields, quantifier elimination underlies exact algorithms for polynomial inequalities.

For H -closed fields, quantifier elimination requires additional predicates and functions encoding certain asymptotic cuts. The result says that complicated quantified questions ultimately reduce to explicit algebraic, differential, order, and valuation conditions in that richer language.

Consequences include:

- completeness: all models satisfy the same sentences;
 - decidability: there is an algorithm that eventually answers every sentence in the formal language;
 - tameness of definable sets, including strong restrictions on eventual one-variable behavior.
- “Decidable” here concerns exact symbolic sentences, not numerical convergence or the practical complexity of computing a transseries solution.

30.7 A simple logical example

Consider the sentence

$$\forall f \exists g (g' = f). \quad (30.1)$$

This is one part of Liouville closure. The sentence

$$\forall f \exists h (h \neq 0 \wedge h' = fh) \quad (30.2)$$

asks for exponential integrals.

A differential intermediate-value statement has more structure:

$$\forall f \forall g (f < g \wedge P(f) < 0 < P(g) \Rightarrow \exists y (f < y < g \wedge P(y) = 0)).$$

Once P has fixed finite order and degree, this is first order.

30.8 Differential algebraicity

An element y is differentially algebraic over K if it satisfies a nonzero differential polynomial equation

$$P(y, y', \dots, y^{(r)}) = 0 \quad (30.3)$$

with coefficients in K . Otherwise it is differentially transcendental.

Solutions of algebraic ODEs are differentially algebraic by definition. Many familiar special functions are differentially algebraic over rational-function fields; others, such as the gamma function, satisfy difference equations but are differentially transcendental over $\mathbb{C}(x)$.

The transseries field contains both differential-algebraic and differential-transcendental elements. Its model theory describes finite differential-polynomial constraints without reducing every element to a closed form.

30.9 Newton degree

For an ordinary polynomial, the lowest significant terms after a valuation rescaling determine whether Newton's method can lift a root. For a differential polynomial, one rescales both the unknown and the derivation to expose a stable dominant differential polynomial called the Newton polynomial.

The *Newton degree* is its degree in the differential indeterminate. If the Newton degree is 1, newtonianity guarantees a root in the valuation ring. This abstractly justifies many transseries correction procedures:

$$\begin{array}{ccc} \text{dominant balance} & \longrightarrow & \text{linearized leading equation} \\ & \Downarrow & \\ \text{smaller correction} & \longrightarrow & \text{exact formal root.} \end{array}$$

The concrete fixed-point algorithms of Chapter 17 are computational shadows of this valuation-theoretic principle.

30.10 Why oscillation is the boundary

The order in an H -field is total and compatible with eventual growth. Oscillatory solutions can defeat this framework. For example,

$$y'' + y = 0$$

has nonzero real solutions that change sign forever. No nonzero solution has an eventual sign, so it cannot be an element of a real Hardy field.

One may still study the equation with complex transseries, exponential phases $e^{\pm ix}$, and resurgent methods. What fails is not asymptotic analysis itself, but the ordered H -field interpretation.

This explains the common phrase that $\mathbb{T}$ is a universal domain for *nonoscillatory* asymptotic differential algebra.

30.11 Definability versus explicit construction

Model theory can prove

$$\exists y \, P(y) = 0$$

without giving a support-by-support recipe for y . Conversely, a constructive transseries algorithm may compute thousands of coefficients without clarifying whether the solution is unique under every definable condition.

The two viewpoints complement one another:

constructive asymptotics finds leading monomials, recurrences, truncations, and numerical values;

model theory proves closure, transfer, uniqueness patterns, and impossibility of certain definable pathologies.

30.12 Decidability does not mean easy computation

A decidable theory has an algorithm that terminates on every sentence. It may be far too slow for practical use. Moreover, questions about a specific infinitely described coefficient sequence may not be first-order sentences in the field language.

For example, the following are different tasks:

1. decide whether every solution of a fixed differential equation eventually exceeds $\log x$;
2. compute the first million coefficients of one formal solution;
3. locate all Borel singularities of that solution;
4. prove that a numerical Stokes constant is nonzero.

Only the first is immediately in the style of first-order H -field logic. The others require algorithms or analytic information outside bare decidability.

30.13 Current bridges among germs, transseries, and surreals

The common H -field language makes it possible to compare:

- formal LE-transseries;
- Hardy fields of actual germs;
- the surreal field with its distinguished derivation.

Major embedding and elementary-equivalence results now connect these worlds. Questions about maximal Hardy fields, hyperserial extensions, and canonical analytic realizations remain active. The goal is not to collapse the three worlds into one, but to determine exactly which algebraic, differential, order, and composition properties transfer among them.

30.14 Exercises

Exercise 30.1. Write a first-order sentence expressing that every positive element has a square root.

Exercise 30.2. Explain the difference between an embedding of ordered differential fields and an elementary embedding.

Exercise 30.3. Why can the property “every sequence has a least upper bound” not be expressed directly by one first-order sentence in the language of ordered fields?

Exercise 30.4. Classify e^x , a solution of $y' = y$, and a generic formal series with algebraically independent coefficients as differentially algebraic or potentially differentially transcendental over $\mathbb{R}(x)$.

Chapter summary

Model theory describes $\mathbb{T}$ through finite statements about field operations, order, dominance, and differentiation. H -closed fields form a model-complete theory, and an enriched language admits quantifier elimination and decidability. These results make transseries a universal domain for broad nonoscillatory differential-algebraic problems, while remaining distinct from coefficient computation and Borel analysis.

How to represent and compute with transseries

A computer algebra system cannot store a completed infinite transseries. It stores finite descriptions, lazy generators, and truncation rules. The mathematical support conditions tell us which operations are legitimate; the software design must make those conditions explicit enough to enforce.

This chapter describes an implementation architecture rather than a particular programming language. The central rule is:

Never represent a transseries as an unstructured syntax tree plus a hope that simplification will sort it out. Separate monomials, coefficients, ordered support, construction level, and truncation policy.

31.1 Three layers of representation

A practical system benefits from three distinct layers.

1. **Expression layer.** User syntax such as

$$e^{x+1/x} \left(1 + \frac{2}{x} \right) + \log x.$$

2. **Canonical transseries layer.** A sparse ordered map

$$\mathfrak{m} \longmapsto c_{\mathfrak{m}}$$

with normalized transmonomials and admissible support.

3. **Truncated computation layer.** A finite projection determined by a cutoff, ratio grid, requested precision, or observable.

The expression layer preserves convenient structure. The canonical layer decides equality and leading terms. The truncated layer makes algorithms terminate.

31.2 A monomial data type

At one finite logarithmic-exponential level, a monomial may be represented conceptually as

$$\mathfrak{m} = x^a (\log x)^b \exp(L), \quad (31.1)$$

where $a, b \in \mathbb{R}$ and L is a purely large transseries of lower or controlled height. More generally, iterated logarithms can be stored as a finite exponent vector:

$$x^{a_0} (\log x)^{a_1} (\log_2 x)^{a_2} \cdots \exp(L).$$

A possible immutable record is:

```
Monomial:
  logPowers : finite map depth -> exact real exponent
  exponent  : PurelyLargeSeries
  height    : nonnegative integer
  cachedHash, cachedLeadingData
```

Multiplication adds logarithmic-power vectors and exponent series. Inversion negates them. Equality is structural after normalization.

31.3 Canonical normalization

Expressions with different syntax can denote the same monomial:

$$e^{2 \log x} = x^2, \quad x^a e^L x^b e^M = x^{a+b} e^{L+M}.$$

A normalizer should:

1. flatten products and exponentials where formal identities permit;
2. collect real multiples of iterated logarithms into power coordinates;
3. normalize L as a purely large series with no constant or small part;
4. remove zero exponents;
5. canonicalize exact real coefficients and branch conventions.

Over the real ordered field, $x > 0$ and principal real logarithms remove many branch ambiguities. Over complex transseries, the normal form must record branches or work on a chosen logarithmic covering.

31.4 Comparing monomials

To decide whether

$$\mathfrak{m} = e^L x^a \ell_1^{a_1} \cdots \quad \text{is larger than} \quad \mathfrak{n} = e^M x^b \ell_1^{b_1} \cdots,$$

compare their logarithms:

$$\log \frac{\mathfrak{m}}{\mathfrak{n}} = (L - M) + (a - b) \log x + (a_1 - b_1) \log_2 x + \cdots. \quad (31.2)$$

The sign of the leading term of (31.2) determines the order.

Thus monomial comparison is recursive in exponential height. Memoizing leading data is essential: sorting a support can otherwise recompute the same lower-level differences many times.

```
CompareMonomials(m, n):
  if m is n: return Equal
  D := Normalize(LogMonomial(m) - LogMonomial(n))
  return SignOfLeadingTerm(D)
```

A production implementation avoids constructing the full difference when a lexicographic leading signature already distinguishes the monomials.

31.5 Series as sparse ordered maps

A finite truncation can use a balanced search tree keyed by monomial order:

```
Series:
  terms : OrderedMap<Monomial, Coefficient>
  ambientLevel : LevelDescriptor
  supportProof : SupportCertificate
```

The map should enumerate from largest to smallest monomial. Then

$$\text{lm}(T) = \text{first key}, \quad \text{lc}(T) = \text{its value}.$$

Zero coefficients are removed immediately.

For lazy infinite series, replace the complete map by a stream of decreasing terms plus a certificate that the stream is well based. Not every generator that emits decreasing terms is safe for multiplication; one also needs local finiteness guarantees.

31.6 Ratio-grid representation

If the support lies in

$$\mathfrak{m}_0 \mu_1^{k_1} \cdots \mu_r^{k_r}, \quad \mathbf{k} \in \mathbb{N}^r,$$

store each term by its exponent vector $\mathbf{k}$. Multiplication becomes addition of integer vectors.

```
GridSeries:
  base      : Monomial
  generators : array<SmallMonomial>
  coeffs    : SparseMap<IntegerVector, Coefficient>
  cone      : admissible affine submonoid
```

The monomial order is induced by the generators, but it need not be ordinary lexicographic order on $\mathbb{N}^r$. One can precompute a comparison oracle from leading logarithms of the μ_j .

Grid representation makes truncation geometric. A cutoff corresponds to a finite set of lattice points in a cone.

31.7 Truncation is part of the semantics

A request such as “expand to order x^{-10} ” is insufficient when exponentials and logarithms coexist. Possible cutoffs include:

monomial cutoff: keep every term $m \succcurlyeq \tau$;

grid box: keep $0 \leq k_j \leq N_j$;

sector cutoff: keep instanton degree $|\mathbf{k}| \leq K$ and perturbative order $n \leq N$;

term budget: keep the first M terms in dominance order;

adaptive observable: keep enough terms to determine a requested leading residual or numerical tolerance.

A mathematically reliable API passes a `TruncationContext` through every operation. Silent global defaults invite errors, because an intermediate product may require terms below the final cutoff to produce retained terms after cancellation or division.

31.8 Addition

Addition is a merge of two decreasing support streams.

```
Add(A, B, cutoff):
  i := A.leadingIterator()
  j := B.leadingIterator()
  while i or j has term above cutoff:
    compare i.monomial and j.monomial
    emit larger term
    if equal: emit summed coefficient if nonzero
```

The cost is linear in the number of retained terms, apart from monomial comparisons.

Cancellation can expose a much smaller leading term. Therefore a lazy iterator must be able to request further terms when equal leading coefficients cancel.

31.9 Multiplication

For finite truncations, multiplication is sparse convolution:

$$[m](AB) = \sum_{n+p=m} [n]A [p]B. \quad (31.3)$$

The Neumann support theorem guarantees finite coefficient sums for well-based supports. The algorithm still needs a way to enumerate only products above the cutoff.

A naive double loop over N and M retained terms costs $O(NM)$. A priority-queue algorithm can enumerate products in descending monomial order:

```

MultiplyStreams(A, B, cutoff):
  insert candidate (0, 0) with monomial A[0].m * B[0].m
  while queue maximum is above cutoff:
    pop all candidates with the same product monomial
    sum their coefficient products
    emit if nonzero
    insert unseen neighboring index pairs

```

For grid series, hash convolution on integer vectors is often simpler and faster. Fast polynomial multiplication can be used when a finite rectangular grid is dense.

31.10 Geometric inversion

Given nonzero

$$T = am(1 + S), \quad S \prec 1,$$

compute

$$T^{-1} = a^{-1}m^{-1} \sum_{n \geq 0} (-S)^n. \quad (31.4)$$

```

Inverse(T, cutoff):
  (a, m) := LeadingTerm(T)
  S := T/(a*m) - 1
  result := 1
  power := 1
  n := 1
  while power can still contribute above cutoff:
    power := Truncate(power * S, cutoffRelativeTo(m))
    result += (-1)^n * power
    n += 1
  return Truncate(result/(a*m), cutoff)

```

The termination proof is asymptotic: successive powers of S become smaller. A coarse cutoff can often bound in advance how many powers are needed.

31.11 Exponentiation

Use the decomposition

$$T = L + c + S$$

with L purely large, c constant, and S small. Then

$$e^T = e^L e^c \sum_{n \geq 0} \frac{S^n}{n!}. \quad (31.5)$$

The factor e^L is a new monomial at the next height. Only the small part is expanded.

```

Exp(T, cutoff):
  (L, c, S) := SplitLargeConstantSmall(T)

```

```

m := InternMonomial(ExpOfPurelyLarge(L))
P := TruncatedPowerSeriesExp(S, cutoff / m)
return exp(c) * m * P

```

Interning canonical monomials ensures that identical e^L nodes share storage and compare quickly.

31.12 Logarithms and real powers

For positive

$$T = am(1 + S), \quad a > 0,$$

use

$$\log T = \log a + \log m + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} S^n. \quad (31.6)$$

Then

$$T^r = \exp(r \log T).$$

The positivity test is formal: $T > 0$ exactly when its leading coefficient is positive in the real ordered field. For rational powers of a negative series, a separate real-domain policy is needed.

31.13 Differentiation

Differentiate termwise:

$$(cm)' = cm (\log m)'.$$

A single monomial may differentiate into a series because $(\log m)'$ can have several terms.

```

Differentiate(T, cutoff):
  result := 0
  for (m, c) in terms of T that may contribute:
    result += c * m * Differentiate(LogMonomial(m))
  return NormalizeAndTruncate(result, cutoff)

```

A strong derivation must preserve admissible infinite sums. For a lazy implementation, this requires a support theorem showing that derivative contributions remain locally finite above each cutoff.

31.14 Formal integration

Integration is not simply monomial-by-monomial division. For

$$\int e^L dx,$$

start with

$$\frac{e^L}{L'}$$

and correct the residual. The fixed-point algorithm of [Chapter 12](#) can be implemented as:

```

Integrate(F, cutoff):
  G := LeadingAntiderivativeGuess(F)
  repeat:
    R := Truncate(F - Differentiate(G), residualCutoff)
    if R has no term above cutoff: return G
  G := Truncate(G + LeadingAntiderivativeGuess(R), cutoff)

```

Each correction removes the current leading residual. Termination at finite cutoff follows if residual leading monomials strictly decrease.

Logarithmic resonance must be detected. Integrating x^{-1} creates $\log x$ rather than $x^0/0$.

31.15 Composition

Composition is the most delicate routine. For $T \circ S$:

1. verify that S is positive infinite in the domain expected by T ;
2. recursively substitute in logarithms and exponentials;
3. preserve summability of every support family;
4. use Taylor expansion only when the increment is small in the correct relative sense;
5. propagate a cutoff backward through the substitution.

For a Taylor step,

$$F(S + U) = \sum_{n \geq 0} \frac{F^{(n)}(S)}{n!} U^n,$$

the stopping criterion is not merely $n = N$. Stop when no later $U^n F^{(n)}(S)$ can reach the requested monomial cutoff.

A common trap

A syntactically small U need not be small enough for Taylor composition. The relevant comparison depends on the derivative scale of F . For $F = e^{e^x}$, an increment $U = e^{-x/2}$ tends to zero, yet $e^x U = e^{x/2}$ is large, so the first exponential response is not a small perturbation.

31.16 Fixed points by coefficient stabilization

Many equations can be written

$$Y = \Phi(Y)$$

with Φ contracting in the dominance topology. A lazy solver iterates

$$Y_{n+1} = \Phi(Y_n)$$

and records which leading coefficients have stabilized.

```

SolveFixedPoint(Phi, cutoff):
  Y := InitialApproximation()
  stablePrefix := empty
  loop:
    Z := Truncate(Phi(Y), cutoff)
    stablePrefix := CommonLeadingPrefix(Y, Z)
    if stablePrefix reaches cutoff: return Z
  Y := Z

```

Equality of finite truncations in two consecutive iterations is not by itself a proof if omitted lower terms can feed upward through a noncontractive operation. The solver should carry a contraction certificate or use an equation-specific theorem.

31.17 Solving algebraic equations

A transseries Newton method uses

$$Y_{n+1} = Y_n - \frac{P(Y_n)}{P'(Y_n)}. \quad (31.7)$$

Dominance analysis chooses the first approximation and verifies that $P'(Y_n)$ is invertible. Truncation must preserve enough residual terms to compute the next correction.

For multiple roots, Newton's method may not improve valuation quadratically. A Newton-polygon phase should first detect balances among monomials of P and split branches.

31.18 Sector-indexed resurgent data

A resurgent transseries is naturally a sparse map from sector multi-indices to perturbative series:

```

ResurgentTransseries:
  actions      : array<Complex>
  parameters   : symbolic variables
  sectors      : map<MultiIndex, FormalPowerSeries>
  prefactors   : map<MultiIndex, (power, logPolynomial)>
  stokesData   : directed labeled graph

```

Products convolve multi-indices. Differential equations generate sector recurrences. The Borel transform acts inside each sector. Alien derivatives move along labeled edges of the Stokes graph.

For numerical work, coefficient arrays are finite and often high precision. Store exact rationals or symbolic parameters as long as possible, then convert to arbitrary-precision complex numbers for Borel–Padé analysis.

31.19 Borel–Padé reconstruction

Given coefficients $a_0, \dots, a_N$:

1. form the truncated Borel series

$$g_N(\xi) = \sum_{n=0}^N \frac{a_n}{n!} \xi^n;$$

2. compute a near-diagonal Padé approximant $P(\xi)/Q(\xi)$;
3. inspect poles and pole strings as approximations to Borel singularities and cuts;
4. integrate the rational approximant along a deformed ray;
5. vary N , precision, Padé order, and contour to estimate stability.

Spurious pole-zero pairs, called Froissart doublets, can mimic singularities. True singularity estimates should persist as the order and precision change.

For a singular direction, choose upper and lower contours and compute the difference. Comparing it with a separately summed neighboring sector estimates the Stokes constant.

31.20 Exact coefficients and coefficient domains

Coefficient arithmetic may involve:

- rational numbers;
- algebraic numbers;
- symbolic constants such as π or Stokes parameters;
- formal power series in auxiliary parameters;
- arbitrary-precision complex approximations.

Do not identify coefficients merely because their floating-point approximations are close. Canonical equality belongs to an exact coefficient domain; numerical equality belongs to an error model.

A tower of coefficient domains can be useful:

$$\mathbb{Q} \subset \overline{\mathbb{Q}} \subset \mathbb{Q}(\pi, S_1, \dots) \subset \mathbb{C}_{\text{arb}}.$$

Coercions upward should be explicit.

31.21 Support certificates

A *support certificate* records why a lazy family is admissible. Examples:

- finite support;
- support inside a certified ratio grid;
- decreasing sequence with a proven order type;
- image of a well-based support under a monotone monomial map;
- finite union, Minkowski sum, or derivative image of certified supports.

The certificate need not be a formal proof object in every implementation, but the architecture should preserve enough provenance to reject obviously unsafe operations.

31.22 Testing invariants

Property-based tests are especially valuable. Generate random finite grid truncations and verify:

$$\begin{aligned}
 A(B + C) &= AB + AC, \\
 (AB)' &= A'B + AB', \\
 \exp(A + B) &= \exp(A) \exp(B) \quad \text{when the operation is admitted,} \\
 \log(AB) &= \log A + \log B \quad (A, B > 0), \\
 A \cdot A^{-1} &= 1 \quad (\text{mod cutoff}), \\
 (F \circ G)' &= (F' \circ G)G' \quad \text{under composition hypotheses.}
 \end{aligned}$$

Also test ordering laws and support invariants:

- every term stream is strictly decreasing;
- no zero coefficient is stored;
- multiplication produces only finitely many contributions per retained monomial;
- increasing the cutoff refines rather than changes already stable leading coefficients.

31.23 A minimal end-to-end example

To expand

$$T = \frac{e^{x+1/x}}{x^2(1-3/x)}$$

through $e^x x^{-6}$:

1. split $e^{x+1/x} = e^x e^{1/x}$;
2. create the monomial $e^x x^{-2}$;
3. expand

$$e^{1/x} = 1 + x^{-1} + \frac{1}{2}x^{-2} + \frac{1}{6}x^{-3} + \frac{1}{24}x^{-4} + O(x^{-5});$$

4. expand

$$(1 - 3/x)^{-1} = 1 + 3x^{-1} + 9x^{-2} + 27x^{-3} + 81x^{-4} + O(x^{-5});$$

5. convolve the two grid coefficient arrays through relative order x^{-4} ;
6. multiply by $e^x x^{-2}$.

The result is

$$T = e^x x^{-2} \left(1 + 4x^{-1} + \frac{25}{2}x^{-2} + \frac{113}{3}x^{-3} + \frac{2713}{24}x^{-4} + O(x^{-5}) \right). \quad (31.8)$$

This simple calculation exercises monomial extraction, two small-series expansions, sparse convolution, and an absolute cutoff.

31.24 Exercises

Exercise 31.1. Design a canonical record for monomials of the form $x^a(\log x)^b e^L$ and specify multiplication, inversion, and comparison.

Exercise 31.2. Explain why a global “keep 20 terms” policy can fail when leading terms cancel during addition.

Exercise 31.3. Implement on paper the geometric inversion of $1 + x^{-1} + e^{-x}$ through the cutoff $e^{-x}x^{-2}$, carefully ordering all retained terms.

Exercise 31.4. Recompute (31.8) and verify its coefficients.

Chapter summary

Reliable transseries software separates syntax, canonical Hahn data, and finite truncation. Monomials require recursive canonical comparison; series require ordered sparse supports and explicit cutoffs. Algebra uses geometric expansions and sparse convolution, while composition and lazy fixed points demand support and contraction certificates. Resurgent numerics add sector arrays, Borel transforms, Padé continuation, and contour integration.

A practical workflow and the most common traps

Transseries calculations become manageable when performed in a disciplined order. Most mistakes come from doing a correct local manipulation in the wrong ambient scale, or from silently changing between formal and analytic meanings.

32.1 The end-to-end workflow

Given an equation and an asymptotic limit:

1. **Specify the limit and sector.** State whether $x \rightarrow +\infty$, $z \rightarrow 0$ in a complex sector, or $\varepsilon \rightarrow 0^+$ with other variables fixed.
2. **List candidate scales.** Include powers, logarithms, exponentials, and any scales suggested by homogeneous or linearized equations.
3. **Find dominant balances.** Compare terms before expanding anything deeply.
4. **Choose a leading branch.** Algebraic equations and nonlinear ODEs usually have several.
5. **Compute the perturbative sector.** Derive a recurrence and inspect coefficient growth.
6. **Linearize about it.** Integrate the linearized exponent to discover actions and power prefactors.
7. **Introduce one parameter per independent mode.** Let nonlinear products generate higher sectors.
8. **Derive sector equations.** Use index convolution rather than expanding manually sector by sector.
9. **Check support admissibility.** Verify a ratio grid or well-based support and identify resonances.
10. **Determine analytic status.** Prove or invoke Gevrey bounds, Borel continuation, and summability if actual functions are claimed.
11. **Compute Stokes geometry.** Locate action rays and define lateral sums.

12. **Match global data.** Use initial conditions, boundary conditions, reality, monodromy, or pole-free requirements to fix parameters.
13. **Validate.** Substitute truncated results, estimate residuals, and compare with high-precision numerical solutions.

Each stage answers a different question. Skipping directly from a guessed ansatz to numerical summation often hides a wrong branch or missing scale.

32.2 Trap 1: confusing asymptotic equality with equality

The statement

$$f(x) \sim \sum_{n \geq 0} a_n x^{-n}$$

does not mean that the infinite sum converges to $f(x)$. It means a sequence of remainder estimates after finite truncation. Write equality only inside a formal field or after naming a summation operation.

Good notation is:

$\widehat{f} = \sum a_n x^{-n}$	formal object,
$f \sim \sum a_n x^{-n}$	asymptotic relation,
$f = S_\theta \widehat{f}$	summation theorem.

32.3 Trap 2: believing a zero power series means a zero function

The function e^{-x} has zero expansion in powers of x^{-1} . A boundary condition may therefore be encoded entirely in an exponentially small sector. Before declaring uniqueness, identify the kernel of the linearized operator and the width of the analytic sector.

32.4 Trap 3: expanding the large part of an exponential

For

$$e^{x+1/x},$$

do not write a Taylor series in $x + 1/x$. Keep e^x as a monomial and expand only $e^{1/x}$:

$$e^{x+1/x} = e^x \left(1 + x^{-1} + \frac{1}{2}x^{-2} + \cdots \right).$$

The canonical split $T = L + c + S$ exists precisely to enforce this rule.

32.5 Trap 4: using a geometric series without a small ratio

The identity

$$(1 + S)^{-1} = 1 - S + S^2 - \cdots$$

is valid formally because $S \prec 1$, not because the symbol visually resembles a correction. If $S = x$, powers grow and the support runs in the wrong direction.

Always factor by the leading monomial first.

32.6 Trap 5: treating support as decoration

A generalized sum with no largest term can make multiplication undefined. For example,

$$1 + x + x^2 + x^3 + \cdots$$

is not a transseries at $x \rightarrow +\infty$ in the usual Hahn orientation. Its support climbs forever.

Support conditions are the analogue of convergence conditions for formal algebra. They guarantee that each product coefficient receives finitely many contributions.

32.7 Trap 6: assuming every decreasing family is a finite grid

A reverse-well-ordered support can require infinitely many independent ratios. Algorithms designed for $\mathbb{N}^r$ may silently omit terms. State whether a result is grid-based or merely well-based.

32.8 Trap 7: forgetting the power prefactor

Linearization may give

$$\eta' + \left(A - \frac{\beta}{x} + \cdots \right) \eta = 0.$$

Integrating yields

$$\eta \sim e^{-Ax} x^\beta (1 + \cdots),$$

not just e^{-Ax} . The logarithmic integral of the x^{-1} term creates the power.

32.9 Trap 8: mistaking a free parameter for a Stokes constant

The parameter σ labels solutions. A Stokes constant S belongs to the equation and chosen normalization. Crossing a ray changes σ by a map involving S .

Initial data determine σ ; analytic continuation of the equation determines S .

32.10 Trap 9: counting generated sectors as new constants

A first-order nonlinear ODE can generate sectors e^{-kx} for every $k \geq 1$ while still having only one free parameter. Higher sector coefficients are polynomially generated from that parameter.

Parameter count follows the dimension of the solution space, not the number of sector labels.

32.11 Trap 10: missing resonance

If two action combinations coincide, sector recurrences may become singular. A vanished denominator often signals a required factor of x , $\log x$, or a polynomial in parameters. Do not regularize it away numerically before checking resonance.

32.12 Trap 11: assuming factorial divergence proves resurgence

Growth like $n!$ guarantees only local convergence of the first Borel transform. Resurgence additionally requires controlled analytic continuation. A Borel transform may have a natural boundary or uncontrolled singular set.

32.13 Trap 12: reading every Padé pole as real

Finite Padé approximants contain spurious pole-zero pairs. A trustworthy Borel singularity persists under changes of order and precision and fits the predicted action geometry. Branch cuts appear as strings of poles, not necessarily as isolated true poles.

32.14 Trap 13: ignoring summation direction

A Borel sum is directional. The same formal series can have different lateral sums across a singular ray. Always attach θ , θ^+ , θ^- , or a median prescription.

32.15 Trap 14: using one asymptotic expansion globally

An expansion valid in one complex sector may fail after rotation because an exponential becomes large. Cover the plane by sectors and use connection formulas. “As $|z| \rightarrow \infty$ ” is incomplete without angular restrictions.

32.16 Trap 15: interchanging infinite sums without uniform control

In

$$\sum_k \tau^k \sum_n a_{k,n} z^n,$$

Borel summing every n -series does not justify summing over k . Transasymptotic reorganization, sector convergence, and uniform estimates are separate issues.

32.17 Trap 16: Taylor expanding composition under the wrong smallness condition

The increment U in $F(S + U)$ must be small relative to the variation scale of F . For rapidly growing F , even $U \rightarrow 0$ may produce a large exponent change. Examine $F'(S)U/F(S)$ and higher derivative scales.

32.18 Trap 17: ignoring branches

Powers, logarithms, Lambert W , and action integrals are multivalued in the complex plane. Record branch cuts and continuation paths. Two formulas with different branches may be different sectors of one global solution rather than contradictions.

32.19 Trap 18: trusting a formal residual without ordering it

After substituting a truncation, do not report merely that the residual is “small.” Compute its leading monomial. A successful N -term approximation should push the residual below the declared cutoff in the ambient dominance order.

32.20 Trap 19: truncating by printed term count

Ten terms can mean ten power coefficients, ten monomials across several sectors, or ten grid lattice points. Cancellations and mixed scales make printed length a poor precision measure. Use a monomial or sector/order cutoff.

32.21 Trap 20: overclaiming analytic meaning

A formal transseries solution is already valuable. There is no need to call it a convergent function unless a theorem supports that statement. Precise modesty is stronger mathematics:

“This is a formal solution in the grid-based LE field; in the stated non-Stokes sector, the rank-one summability theorem provides an analytic realization.”

32.22 A complete miniature example

Consider

$$\varepsilon y' + y = x, \quad y(0) = 0, \quad \varepsilon \rightarrow 0^+. \quad (32.1)$$

Step 1: outer balance

Set $\varepsilon = 0$:

$$y_0 = x.$$

A regular expansion gives

$$y_{\text{out}} = x - \varepsilon,$$

with no further power corrections because differentiating $x - \varepsilon$ gives 1 exactly.

Step 2: detect the boundary mismatch

At $x = 0$,

$$y_{\text{out}}(0) = -\varepsilon,$$

so the expansion satisfies the boundary condition only at leading order, not exactly.

Step 3: homogeneous scale

The homogeneous equation

$$\varepsilon h' + h = 0$$

gives

$$h = C e^{-x/\varepsilon}.$$

Step 4: impose the boundary condition

The transseries ansatz is

$$y = x - \varepsilon + C e^{-x/\varepsilon}.$$

At $x = 0$, $C = \varepsilon$, so

$$y = x - \varepsilon + \varepsilon e^{-x/\varepsilon}. \quad (32.2)$$

Step 5: verify

Direct substitution gives zero residual and $y(0) = 0$. For every fixed $x > 0$, the exponential is beyond all algebraic orders in ε . In the inner variable $X = x/\varepsilon$ it becomes order ε :

$$y = \varepsilon(X - 1 + e^{-X}).$$

One formula connects the outer and boundary-layer descriptions.

This elementary problem already displays dominant reduction, an invisible homogeneous mode, a transseries parameter fixed by boundary data, and nonuniformity repaired by a stretched variable.

32.23 A diagnostic decision tree

Power coefficients grow geometrically: the series may converge; use ordinary analytic methods first.

Power coefficients grow like $n!$: try an order-one Borel transform and inspect its nearest singularities.

A boundary condition is missing: solve the homogeneous or linearized problem for exponential scales.

Nonlinearity is present: expect powers and combinations of fundamental actions.

A recurrence denominator vanishes: test for resonance and logarithmic sectors.

Many sectors become comparable: introduce a transasymptotic variable τ .

Repeated logarithmic depth grows without bound: move beyond a fixed LE level toward omega-series or hyperseries.

Persistent oscillation appears: leave the real ordered H -field setting and use complex phases or another asymptotic framework.

32.24 Exercises

Exercise 32.1. For $\varepsilon y' + (1+x)y = 1$ with $y(0) = 0$, find the outer expansion and the leading boundary-layer exponential.

Exercise 32.2. Construct an example in which $U \rightarrow 0$ but $e^{F(x+U)}/e^{F(x)}$ does not tend to 1.

Exercise 32.3. A coefficient ratio satisfies $a_n/a_{n-1} \sim (n + 3/2)/A$. What action and branch exponent does this suggest?

Exercise 32.4. Write a checklist distinguishing what must be shown for a formal solution, an asymptotic realization, and a Borel-summed solution.

Chapter summary

A successful transseries calculation proceeds from limit and scale selection through dominant balance, perturbative recursion, linearized actions, sector equations, support checks, and only then analytic reconstruction. Most errors are category errors: confusing equality with asymptotics, parameters with Stokes constants, support with notation, or directional summability with a global function.

A guided reading map

The literature uses several dialects. This final chapter of the main text explains what to read for each purpose and how the sources fit together.

33.1 For a first formal introduction

Edgar’s *Transseries for Beginners* develops grid-based logarithmic-exponential transseries with unusually concrete examples, ratio sets, derivatives, and fixed-point methods [22]. It is the closest source in spirit to the formal middle of this tutorial.

Edgar’s paper on composition, recursion, and convergence treats composition and fixed-point phenomena in greater depth [23]. His paper on fractional iteration is a natural continuation of Chapter 16 [24].

33.2 For formal differential algebra

Van der Hoeven’s *Transseries and Real Differential Algebra* develops a broad computational and differential-algebraic theory, including operators, asymptotic equations, and algorithmic viewpoints [38]. It is more advanced and denser than an introductory text, but it supplies much of the structural backbone.

The book by Aschenbrenner, van den Dries, and van der Hoeven gives the definitive asymptotic differential algebra and model theory of $\mathbb{T}$ [2]. Their ICM survey *On Numbers, Germs, and Transseries* is a shorter conceptual bridge among Hardy fields, transseries, and surreals [3].

33.3 For Borel summation and resurgence

Costin’s *Asymptotics and Borel Summability* develops Borel methods with rigorous differential-equation applications [15]. His rank-one papers establish the correspondence between generalised Borel-summed transseries and actual nonlinear ODE solutions [12, 13, 14].

Dorigoni’s review is a compact modern introduction to resurgence, alien calculus, and physical examples [19]. The long primer by Aniceto, Başar, and Schiappa develops resurgent transseries, large-order relations, and many worked examples from mathematics and physics [1].

Écalle's original works are foundational and far more extensive. They introduce resurgent functions, alien calculus, acceleration, analyzable functions, and the machinery used in the proof of Dulac's conjecture [20, 21].

33.4 For Stokes phenomena and exponential improvement

Classical asymptotics texts by Dingle, Berry, Howls, Olver, and Olde Daalhuis explain optimal truncation, Stokes multipliers, hyperasymptotics, and terminants. The supplied note by Olde Daalhuis gives a particularly direct route from transseries to Stokes phenomena [33]. Costin–Kruskal treats optimal truncation and Berry smoothing for nonlinear systems [16].

33.5 For the ordered exponential-field constructions

The papers of van den Dries, Macintyre, and Marker construct logarithmic-exponential series and connect them with the model theory of the real exponential field [36, 37]. Dahn–Göring is an early source for exponential-logarithmic terms [17]. Kuhlmann–Tressl compares LE and EL constructions directly [28]. Schmeling's thesis provides a broad axiomatic framework [35].

33.6 For surreal numbers and omega-series

Conway's *On Numbers and Games* and Gonshor's monograph develop surreal arithmetic and normal forms [11, 25]. Berarducci–Mantova constructs derivations and transseries structure on the surreals [9]. Their later paper develops omega-series, surreal composition, and the interpretation of transseries as surreal functions [10].

The survey by Mantova and Matusinski is a gentle bridge among surreal derivations, Hardy fields, and transseries [30]. More recent work on logarithmic hyperseries and hyperserial fields continues beyond finite LE depth [29, 6, 7].

33.7 For recent Taylor and composition results

Formal Taylor expansion over generalized power-series fields has received renewed attention. Bagayoko and Mantova give broad existence and convergence results for Taylor expansions in exponential differential fields of generalized series [5]. Mantova proves monotonicity of composition and a Taylor approximation theorem for omega-series with a maximal validity radius [31].

These papers are appropriate after the reader is comfortable with [Chapters 13 and 28](#); they clarify exactly when a familiar Taylor heuristic is a theorem in large transseries fields.

33.8 Suggested routes through this book

Asymptotics route: [Chapters 1 to 3, 17 to 19 and 22 to 24.](#)

Formal algebra route: [Chapters 4 to 9, 11 and 13.](#)

Differential-equation route: Chapters 2, 11, 12, 17 to 20 and 26.

Logic and surreal route: Chapters 5, 10 and 27 to 30.

Computational route: Chapters 6, 8, 15, 17, 31 and 32.

33.9 What a reader should now be able to do

After working through the tutorial, a reader should be able to:

1. distinguish formal, asymptotic, Borel-summed, and resurgent claims;
2. order transmonomials and verify simple support conditions;
3. compute with large/constant/small decompositions;
4. differentiate, integrate, invert, exponentiate, and compose basic transseries;
5. derive perturbative and exponential sectors of elementary ODEs;
6. recognize factorial divergence and perform a basic Borel–Laplace reconstruction;
7. explain lateral sums, Stokes parameter changes, and large-order relations;
8. understand how Hardy fields and surreal numbers supply two different meanings for transseries;
9. identify when a problem requires a larger field, resonance terms, or complex oscillatory scales;
10. design a finite truncated representation suitable for symbolic computation.

The subject extends far beyond this list, but these skills are enough to read introductory research papers without treating every infinite exponential expression as magic.

Chapter summary

The formal, analytic, logical, and surreal literatures emphasize different structures. Edgar and van der Hoeven lead into formal computation; Costin and Écalle into summation and resurgence; Aschenbrenner–van den Dries–van der Hoeven into H -fields and model theory; Berarducci–Mantova into surreal evaluation and omega-series. The best next source depends on which meaning of transseries the reader needs.

A compact prerequisite toolkit

This appendix collects facts used repeatedly in the main text. It is not a replacement for courses in real analysis, complex analysis, or differential equations, but it makes the tutorial more self-contained.

A.1 Asymptotic notation

Let $g(x) > 0$ eventually as $x \rightarrow \infty$.

$$\begin{aligned} f(x) = O(g(x)) &\iff |f(x)| \leq Cg(x) \text{ eventually for some } C, \\ f(x) = o(g(x)) &\iff \frac{f(x)}{g(x)} \longrightarrow 0, \\ f(x) \sim g(x) &\iff \frac{f(x)}{g(x)} \longrightarrow 1. \end{aligned}$$

The relation $f \sim g$ is much stronger than $f = O(g)$, but neither means $f = g$.
If

$$f(x) \sim \sum_{n \geq 0} a_n \phi_n(x)$$

with $\phi_{n+1} = o(\phi_n)$, then for every fixed N ,

$$f(x) - \sum_{n=0}^{N-1} a_n \phi_n(x) = o(\phi_{N-1}(x)).$$

An equivalent common convention uses $O(\phi_N)$ when the scale has additional regularity. Always inspect the stated remainder convention.

A.2 Uniqueness of coefficients in an asymptotic scale

Suppose

$$\sum_{n=0}^N c_n \phi_n = o(\phi_N)$$

and $\phi_{n+1} = o(\phi_n)$. Divide by ϕ_0 and let $x \rightarrow \infty$ to obtain $c_0 = 0$. Repeat after removing the first term. Thus all $c_n = 0$.

Consequently, the coefficients of a Poincaré asymptotic expansion in a fixed scale are unique, even though the function realizing them need not be.

A.3 Geometric and binomial series

For an ordinary number $|u| < 1$,

$$\begin{aligned}\frac{1}{1-u} &= \sum_{n \geq 0} u^n, \\ \log(1+u) &= \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} u^n, \\ e^u &= \sum_{n \geq 0} \frac{u^n}{n!}, \\ (1+u)^\alpha &= \sum_{n \geq 0} \binom{\alpha}{n} u^n,\end{aligned}$$

where

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!}.$$

For transseries, replace $|u| < 1$ by $u \prec 1$. The support of u^n moves toward smaller monomials, so each retained monomial receives only finitely many contributions.

A.4 Formal coefficient extraction

Write $[z^n]F(z)$ for the coefficient of z^n . Then

$$\begin{aligned}[z^n](FG) &= \sum_{k=0}^n [z^k]F [z^{n-k}]G, \\ [z^n]F' &= (n+1)[z^{n+1}]F.\end{aligned}$$

These identities produce most perturbative recurrences after substitution into an ODE.

For a sector variable σ , use

$$[\sigma^k](FG) = \sum_{i+j=k} [\sigma^i]F [\sigma^j]G.$$

This is why nonlinear sector labels add.

A.5 The gamma and beta integrals

The gamma function is

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt, \quad \Re s > 0. \quad (\text{A.1})$$

Integration by parts gives

$$\Gamma(s+1) = s\Gamma(s), \quad \Gamma(n+1) = n!.$$

Scaling $t = x\xi$ yields

$$\int_0^\infty e^{-x\xi} \xi^{s-1} d\xi = \Gamma(s)x^{-s}, \quad \Re x > 0.$$

This is the continuous form of the Borel–Laplace moment identity.

The beta integral is

$$\int_0^1 t^{a-1} (1-t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \quad (\text{A.2})$$

After $\eta = \xi t$, it proves the convolution rule for Borel transforms.

A.6 Stirling's approximation

As $n \rightarrow \infty$,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n. \quad (\text{A.3})$$

For a term

$$t_n = \frac{n!}{x^{n+1}},$$

the ratio is $(n+1)/x$, so the least term occurs near $n = x$. Stirling then gives

$$t_n|_{n \approx x} \asymp x^{-1/2} e^{-x}$$

up to constant factors. Thus the optimally truncated remainder has the scale of the first missing exponential.

A.7 First-order linear ODEs

For

$$y' + p(x)y = q(x), \quad (\text{A.4})$$

define the integrating factor

$$\mu(x) = \exp\left(\int p(x) dx\right).$$

Then

$$(\mu y)' = \mu q,$$

so

$$y = \mu^{-1} \left(C + \int \mu q dx \right). \quad (\text{A.5})$$

The homogeneous scale μ^{-1} is exactly the beyond-all-orders sector that a power expansion can miss.

A.8 Riccati reduction

A second-order linear equation

$$y'' = Q(x)y$$

can be reduced by

$$R = \frac{y'}{y}.$$

Then

$$R' + R^2 = Q.$$

A formal expansion for R can be integrated to obtain

$$y = \exp \left(\int R \, dx \right).$$

The two signs of the leading square root $R \sim \pm\sqrt{Q}$ produce the two WKB exponentials.

A.9 Leading WKB terms

Take

$$y'' = Q(x)y$$

with slowly varying positive Q . Write

$$y = e^S, \quad S'' + (S')^2 = Q.$$

At leading order, $S' \sim \pm\sqrt{Q}$, hence

$$S \sim \pm \int \sqrt{Q} \, dx.$$

Keeping the first derivative correction gives

$$y \sim Q^{-1/4} \exp \left(\pm \int^x \sqrt{Q(t)} \, dt \right).$$

The factor $Q^{-1/4}$ follows by substituting $S' = \pm\sqrt{Q} + r$ and solving the first linear correction

$$2(\pm\sqrt{Q})r \sim -\frac{Q'}{2\sqrt{Q}}.$$

A.10 Complex sectors and branches

A sector at the origin is

$$S(\theta_0, \alpha, R) = \{0 < |z| < R : |\arg z - \theta_0| < \alpha/2\}.$$

Its opening is α . A branch of $\log z$ on the sector determines

$$z^\beta = e^{\beta \log z}.$$

If the sector winds around the origin, no single-valued logarithm exists without passing to a covering surface.

For asymptotics at infinity, replace $0 < |z| < R$ by $|z| > R$.

A.11 Cauchy's coefficient formula

If g is analytic near 0,

$$[\xi^n]g(\xi) = \frac{1}{2\pi i} \oint \frac{g(\xi)}{\xi^{n+1}} d\xi.$$

Moving the contour outward makes nearby singularities control large n . This is the analytic basis of Darboux's principle and resurgent large-order relations.

A.12 Residues and lateral jumps

If a Laplace path crosses a simple pole of g at A , the difference between paths bypassing it on opposite sides is a small closed contour. Thus

$$\int_+ e^{-\xi/z} g(\xi) d\xi - \int_- e^{-\xi/z} g(\xi) d\xi = \pm 2\pi i e^{-A/z} \text{Res}_{\xi=A} g(\xi).$$

The sign depends on path orientation. This one-line residue calculation is the simplest Stokes jump.

A.13 Well orders and reverse well orders

A totally ordered set is well ordered if every nonempty subset has a least element. It is reverse well ordered if every nonempty subset has a greatest element.

A support of transmonomials is reverse well ordered because terms are written from largest to smaller. This rules out an infinite strictly increasing sequence

$$\mathfrak{m}_0 \prec \mathfrak{m}_1 \prec \mathfrak{m}_2 \prec \cdots$$

inside the support.

A.14 Why Hahn products have finite coefficients

Let A and B be reverse-well-ordered subsets of an ordered abelian group. Fix a monomial $\mathfrak{m}$. Suppose infinitely many pairs $(a_i, b_i) \in A \times B$ satisfy $a_i b_i = \mathfrak{m}$. Choose the a_i in strictly decreasing order after passing to a subsequence. Then

$$b_i = \mathfrak{m} a_i^{-1}$$

forms a strictly increasing sequence in B , contradicting reverse well-order. Hence only finitely many pairs contribute.

This is the core local-finiteness fact behind Hahn multiplication.

A.15 Contraction in dominance order

A map Φ is contracting near a candidate fixed point if

$$U - V \preccurlyeq \mathfrak{m} \implies \Phi(U) - \Phi(V) \prec \mathfrak{m}.$$

Then each iteration determines at least one additional layer of smaller monomials. The sequence

$$U_{n+1} = \Phi(U_n)$$

need not converge numerically, but its coefficients stabilize from large to small. This is the formal topology used by many transseries fixed-point proofs.

A.16 Optimal truncation in one formula

Suppose

$$a_n \sim CA^{-n}\Gamma(n + \beta).$$

The n th term of $\sum a_n z^n$ has ratio approximately

$$\left| \frac{a_{n+1}z^{n+1}}{a_n z^n} \right| \sim \frac{|z|}{|A|} (n + \beta).$$

The least term is near

$$n_* \approx \frac{|A|}{|z|} - \beta.$$

Its exponential size is approximately $e^{-|A|/|z|}$, modified by angular dependence and powers. This is why the distance to the nearest Borel singularity controls both coefficient growth and optimal error.

Chapter summary

The prerequisite toolkit consists of asymptotic scales, coefficient extraction, gamma and beta integrals, linear ODEs, WKB reduction, complex sectors, residues, and well-ordered support. Each has a direct transseries role: factorials lead to Borel transforms, kernels lead to exponentials, and reverse well orders make infinite algebra finite coefficient by coefficient.

Hints and selected solutions

The exercises in the main text are intended for active calculation. This appendix gives representative solutions and hints, emphasizing methods that recur elsewhere.

B.1 Scales and ordinary asymptotics

Ordering logarithmic and exponential scales

For every $a > 0$ and every $b > 0$,

$$\log x = o(x^a), \quad x^a = o(e^{x^b}).$$

For the first, apply l'Hôpital's rule to $\log x/x^a$. For the second, take logarithms:

$$\log \frac{x^a}{e^{x^b}} = a \log x - x^b \longrightarrow -\infty.$$

Repeated logarithms give the extended chain

$$1 \prec \log_3 x \prec \log_2 x \prec \log x \prec x^a \prec e^{x^b}.$$

An invisible function

For every $N > 0$,

$$x^N e^{-x} \longrightarrow 0.$$

Repeated l'Hôpital or comparison of logarithms proves this. Therefore $e^{-x} = o(x^{-N})$ for every N , so its complete inverse-power asymptotic expansion is zero.

B.2 The Euler equation

Insert

$$\widehat{y} = \sum_{n \geq 0} a_n x^{-n-1}$$

into $y' + y = x^{-1}$. Since

$$y' = - \sum_{n \geq 0} (n+1) a_n x^{-n-2},$$

the x^{-1} coefficient gives $a_0 = 1$, and the coefficient of x^{-n-2} gives

$$a_{n+1} = (n+1)a_n.$$

Thus $a_n = n!$.

The Borel transform is

$$\sum_{n \geq 0} \frac{n!}{n!} \xi^n = \frac{1}{1-\xi}.$$

The pole at 1 predicts the missing e^{-x} sector.

B.3 Hahn supports

Why $1 + x + x^2 + \dots$ is wrong at infinity

The support $\{1, x, x^2, \dots\}$ has no largest element. Hence it is not reverse well ordered under the dominance order for $x \rightarrow +\infty$. In contrast,

$$1 + x^{-1} + x^{-2} + \dots$$

has largest monomial 1 and every subset has a largest exponent closest to zero.

A product

For

$$A = 1 + x^{-1} + x^{-2} + \dots,$$

formal multiplication gives

$$A^2 = 1 + 2x^{-1} + 3x^{-2} + 4x^{-3} + \dots.$$

The coefficient of x^{-n} receives exactly $n+1$ contributions. Local finiteness holds despite both supports being infinite.

B.4 Grids

Let $\mu_1 = x^{-1}$ and $\mu_2 = e^{-x}$. The monomial

$$x^{-3}e^{-2x}$$

corresponds to $(3, 2) \in \mathbb{N}^2$. Multiplying by $x^{-4}e^{-x}$ adds vectors:

$$(3, 2) + (4, 1) = (7, 3),$$

which represents $x^{-7}e^{-3x}$.

A cutoff keeping sector degree $k \leq 2$ and power order $n \leq 4$ is a rectangle in this grid. A dominance cutoff is generally a different shape because every e^{-x} is smaller than every fixed power x^{-N} .

B.5 Exponentials and logarithms

To expand

$$\log \left(3e^{x^2}x^{-5}(1 + 2/x + e^{-x}) \right),$$

use the canonical factorization:

$$\begin{aligned} \log T &= \log 3 + x^2 - 5 \log x + \log(1 + 2/x + e^{-x}) \\ &= x^2 - 5 \log x + \log 3 + (2/x + e^{-x}) - \frac{1}{2}(2/x + e^{-x})^2 + \dots \end{aligned}$$

The first mixed terms are

$$\log T = x^2 - 5 \log x + \log 3 + 2x^{-1} - 2x^{-2} + \frac{8}{3}x^{-3} + \dots + e^{-x}(1 - 2x^{-1} + 4x^{-2} - \dots) - \frac{1}{2}e^{-2x} + \dots$$

One must state a cutoff before deciding how far each sector is expanded.

B.6 Formal integration

Seek

$$\int e^{x^2} dx \asymp e^{x^2} \sum_{n \geq 0} c_n x^{-2n-1}.$$

Differentiate one term:

$$\frac{d}{dx} \left(e^{x^2} c_n x^{-2n-1} \right) = e^{x^2} \left(2c_n x^{-2n} - (2n+1)c_n x^{-2n-2} \right).$$

Matching the constant term gives $2c_0 = 1$, so $c_0 = 1/2$. Cancellation at x^{-2n-2} gives

$$2c_{n+1} = (2n+1)c_n.$$

Hence

$$c_n = \frac{(2n-1)!!}{2^{n+1}},$$

and

$$\int e^{x^2} dx \asymp \frac{e^{x^2}}{2x} \left(1 + \frac{1}{2x^2} + \frac{3}{(2x^2)^2} + \frac{15}{(2x^2)^3} + \dots \right).$$

B.7 Lambert W

For large positive x , write

$$W(x) = L_1 - L_2 + u, \quad L_1 = \log x, \quad L_2 = \log \log x.$$

The equation $W + \log W = L_1$ becomes

$$u + \log \left(1 - \frac{L_2 - u}{L_1} \right) = 0.$$

Expanding the logarithm and solving recursively gives

$$u = \frac{L_2}{L_1} + \frac{L_2(-2 + L_2)}{2L_1^2} + \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} + \dots$$

The large term is L_1 , the next correction is $-L_2$, and powers of L_1^{-1} carry polynomial coefficients in L_2 .

B.8 Fixed points

For

$$Y = 1 + tY^2,$$

write $Y = \sum_{n \geq 0} c_n t^n$. Then $c_0 = 1$ and

$$c_n = \sum_{i+j=n-1} c_i c_j.$$

This gives

$$1, 1, 2, 5, 14, 42, \dots,$$

the Catalan numbers

$$c_n = \frac{1}{n+1} \binom{2n}{n}.$$

The exact branch satisfying $Y(0) = 1$ is

$$Y = \frac{1 - \sqrt{1 - 4t}}{2t}.$$

Formal iteration stabilizes coefficients because multiplication by t pushes every new error to higher order.

B.9 Dominant balance in an exponential cubic

For

$$Y^3 - xY - e^x = 0,$$

the exponentially large branch balances Y^3 with e^x :

$$Y \sim e^{x/3}.$$

Set $Y = e^{x/3}(1 + u)$. Then

$$(1 + u)^3 - 1 - xe^{-2x/3}(1 + u) = 0.$$

The small parameter is $q = xe^{-2x/3}$. Solving

$$3u + 3u^2 + u^3 = q(1 + u)$$

recursively gives $u = q/3 + O(q^2)$.

The other two complex branches use $c = \omega$ and $c = \omega^2$, where $\omega^3 = 1$ and $\omega \neq 1$. Setting

$$Y = c e^{x/3} (1 + u)$$

gives

$$(1 + u)^3 - 1 = cq(1 + u), \quad q = x e^{-2x/3},$$

so that $u = cq/3 + O(q^2)$. Thus all three branches have magnitude $e^{x/3}$; only the branch with $c = 1$ is real for large positive x . The lesson is to test every neglected term after proposing a balance.

B.10 Airy WKB

For $y'' - xy = 0$, take

$$y = e^S.$$

Then

$$S'' + (S')^2 = x.$$

The leading equation gives $S' \sim \pm x^{1/2}$, hence

$$S \sim \pm \frac{2}{3} x^{3/2}.$$

The first correction is $S' \sim \pm x^{1/2} - 1/(4x)$, so

$$y \sim x^{-1/4} e^{\pm 2x^{3/2}/3}.$$

Successive inverse powers of $x^{3/2}$ produce the standard Airy coefficient series.

B.11 The nonlinear Riccati recurrence

From

$$a_m = (m-1)a_{m-1} + \sum_{i=1}^{m-1} a_i a_{m-i}$$

and $a_1 = 1$:

$$a_2 = 1 + 1 = 2,$$

$$a_3 = 2 \cdot 2 + (1 \cdot 2 + 2 \cdot 1) = 8,$$

$$a_4 = 3 \cdot 8 + (1 \cdot 8 + 2 \cdot 2 + 8 \cdot 1) = 44,$$

$$a_5 = 4 \cdot 44 + (1 \cdot 44 + 2 \cdot 8 + 8 \cdot 2 + 44 \cdot 1) = 296.$$

The convolution term is the coefficient of x^{-m} in ϕ_0^2 .

B.12 A boundary layer

For

$$\varepsilon y' + (1+x)y = 1, \quad y(0) = 0,$$

set $y \sim y_0 + \varepsilon y_1 + \dots$. Then

$$y_0 = \frac{1}{1+x}, \quad y_1 = -\frac{y_0'}{1+x} = \frac{1}{(1+x)^3}.$$

The leading outer value at 0 is 1, so the boundary mismatch is order one. The homogeneous equation gives

$$h = C \exp\left(-\frac{1}{\varepsilon} \int_0^x (1+t) dt\right) = C \exp\left(-\frac{x+x^2/2}{\varepsilon}\right).$$

At leading order $C = -1$. Near $x = 0$, the inner variable $X = x/\varepsilon$ recovers e^{-X} .

B.13 Borel transforms

For

$$\widehat{f}(x) = \sum_{n \geq 0} (-1)^n n! x^{-n-1},$$

we have

$$\mathcal{B}\widehat{f}(\xi) = \frac{1}{1+\xi}.$$

Its nearest singularity is at $\xi = -1$, so positive-direction Borel summation is unobstructed while the negative direction is singular.

For

$$\sum_{n \geq 0} A^{-n-1} n! x^{-n-1},$$

we obtain

$$\frac{1}{A-\xi},$$

with a singularity at $\xi = A$.

B.14 A simple Stokes jump

If

$$g(\xi) = \frac{r}{\xi - A},$$

then upper-minus-lower lateral contours differ by an oriented loop around A . In the small- z convention,

$$\text{Disc} \int e^{-\xi/z} g(\xi) \frac{d\xi}{z} = \pm \frac{2\pi i r}{z} e^{-A/z}.$$

If the Borel convention assigns the pole with an extra factor of z or uses $r/(A-\xi)$, the displayed prefactor and sign change. This is why conventions must be fixed before quoting a Stokes constant.

B.15 Branch singularities and large order

The binomial expansion gives

$$(1 - w)^{-\beta} = \sum_{n \geq 0} \frac{\Gamma(n + \beta)}{\Gamma(\beta)\Gamma(n + 1)} w^n.$$

With $w = \xi/A$ and $a_n = n![\xi^n]g$, one gets

$$a_n = \frac{\Gamma(n + \beta)}{\Gamma(\beta)A^n}.$$

This exactly realizes the factorial-over-power law rather than merely approximating it.

B.16 Hardy-field ordering

Order

$$\log \log x, \quad (\log x)^{10}, \quad e^{\sqrt{\log x}}, \quad x^{1/100}.$$

Taking logarithms where helpful:

$$\log \log x \prec (\log x)^{10} \prec e^{\sqrt{\log x}} \prec x^{1/100}.$$

For the last comparison,

$$\log \left(\frac{e^{\sqrt{\log x}}}{x^{1/100}} \right) = \sqrt{\log x} - \frac{1}{100} \log x \rightarrow -\infty.$$

The logarithmic derivatives are

$$\begin{aligned} (x^{\log x})^\dagger &= \frac{d}{dx} ((\log x)^2) = \frac{2 \log x}{x}, \\ (e^{x/\log x})^\dagger &= \left(\frac{x}{\log x} \right)' = \frac{\log x - 1}{(\log x)^2}. \end{aligned}$$

B.17 Surreal normal forms

Using formal Hahn arithmetic,

$$\begin{aligned} (\omega + 1)^2 &= \omega^2 + 2\omega + 1, \\ (1 + \omega^{-1})^{-1} &= 1 - \omega^{-1} + \omega^{-2} - \omega^{-3} + \dots \end{aligned}$$

The latter family is summable because its monomials strictly decrease.

B.18 Computational inversion with mixed scales

Let

$$T = 1 + x^{-1} + e^{-x}.$$

Set $S = x^{-1} + e^{-x}$. Then

$$T^{-1} = 1 - S + S^2 - S^3 + \dots.$$

Ordering by dominance gives, through algebraic order x^{-2} and exponential order $e^{-x}x^{-2}$,

$$\begin{aligned} T^{-1} &= 1 - x^{-1} + x^{-2} + O(x^{-3}) \\ &\quad - e^{-x}(1 - 2x^{-1} + 3x^{-2} + O(x^{-3})) + e^{-2x}(1 + O(x^{-1})) + \dots. \end{aligned}$$

Whether the e^{-2x} term is retained depends on the precise cutoff: it is smaller than every $e^{-x}x^{-N}$, so a cutoff at $e^{-x}x^{-2}$ normally discards it.

B.19 A composition counterexample

Let

$$F(x) = e^{e^x}, \quad U(x) = e^{-x/2}.$$

Then $U \rightarrow 0$, but

$$\frac{F(x+U)}{F(x)} = \exp\left(e^x(e^U - 1)\right).$$

Since $e^U - 1 \sim U$,

$$e^x(e^U - 1) \sim e^{x/2} \rightarrow \infty.$$

Therefore the ratio tends to infinity, not to 1. Absolute smallness of the increment is insufficient for Taylor-small composition.

Chapter summary

The selected solutions repeatedly use the same moves: compare logarithms, factor the leading monomial, derive coefficient convolutions, linearize for exponential scales, and inspect neglected terms. These are more important than memorizing any one coefficient list.

Glossary of transseries language

The same word can carry slightly different emphases in formal differential algebra, exponential asymptotics, resurgence, model theory, and surreal analysis. This glossary records the meaning intended in this tutorial and, where useful, points out common variations.

Term	Meaning in this tutorial
Action	A quantity A appearing in an exponential scale $e^{-A/z}$ or e^{-Ax} . In differential equations it is normally obtained by integrating an appropriate eigenvalue of the linearized or WKB problem. Borel singularities frequently occur at the same values A .
Alien derivative	An operator Δ_A that extracts, in a normalized way, the resurgent singularity located at Borel position A . It does not mean an ordinary derivative with respect to the independent variable.
Analyzable function	In Ècalle’s theory, roughly, a function represented by a transseries that can be reconstructed through generalized Borel–Laplace procedures, possibly including acceleration. Precise definitions require more structure than formal LE transseries alone.
Anti-Stokes direction	A direction on which two relevant exponentials have equal magnitude. Some authors interchange the names “Stokes” and “anti-Stokes,” so the defining equation should always be stated.
Asymptotic expansion	A sequence of finite approximation statements relative to a chosen ordered scale. It is not an assertion that the corresponding infinite sum converges.
Asymptotic scale	A sequence (ϕ_n) with $\phi_{n+1} = o(\phi_n)$ in the relevant limit. The order may combine powers, logarithms, exponentials, and iterated exponentials rather than only powers of one variable.
Beyond all orders	Smaller than every term in a specified asymptotic scale. For example, e^{-x} is beyond all orders of x^{-1} as $x \rightarrow +\infty$. The phrase is relative to a scale, not absolute.

Continued on the next page

Term	Meaning in this tutorial
Borel plane	The complex ξ -plane in which the Borel transform lives. Singularities there encode factorial growth, Stokes jumps, and often relations among transseries sectors.
Borel sum	A function obtained by Borel transforming a divergent formal series, analytically continuing the transform, and applying a Laplace integral along a specified direction. Direction and normalization are part of the data.
Borel transform	A transform that removes factorial growth, typically sending $\sum_{n \geq 0} a_n z^{n+1}$ to $\sum_{n \geq 0} a_n \xi^n / n!$. Several shifted normalizations are standard.
Bridge equation	A relation connecting alien derivatives to ordinary variation of transseries parameters, schematically $\Delta_A \Phi = C_A(\sigma) \partial_\sigma \Phi$. It turns Borel singularity data into transformations of the full transseries.
Coefficient stabilization	The phenomenon that successive formal iterates agree through progressively smaller monomials. This is the correct replacement for numerical convergence in many transseries fixed-point constructions.
Constant part	In the decomposition $T = L + c + S$, the real constant c . The terms L and S are respectively purely large and purely small.
Depth	A measure of how many iterated logarithms are needed to express a transseries after an appropriate change of variable. Conventions vary among constructions; it should not be confused with exponential height.
Dominance	The order of asymptotic size. We write $f \prec g$ when $f/g \rightarrow 0$ in a functional realization, or when the corresponding ordered monomials have that formal relation.
Dominant balance	A leading-order cancellation among the largest candidate terms of an equation. Every proposed balance must be checked by substituting its implied scale into all neglected terms.
Écalle acceleration	A generalized summation step used when ordinary Borel–Laplace summation has insufficient growth control. It changes the effective time scale in the Borel plane and permits treatment of more rapidly divergent levels.
EL series	“Exponential-logarithmic” series constructed by a particular ordering of field, exponential, logarithmic, and Hahn-series extensions. EL and LE constructions are closely related but are not identical by definition.
Exponential height	The number of genuinely large exponential-extension stages needed to build a monomial or transseries. Expanding a small exponential in an already constructed field need not increase height.

Continued on the next page

Term	Meaning in this tutorial
Formal equality	Equality of coefficients attached to every monomial in the chosen transseries field. It does not by itself assert convergence, summability, or equality of analytic functions.
Gevrey order	A quantitative factorial-growth class. In a common convention, Gevrey-1 coefficients satisfy $ a_n \leq CK^n n!$. Different authors index Gevrey classes differently, so the bound is safer than the label alone.
Grid	A finitely generated multiplicative family of monomials, usually generated by small ratio monomials. Grid-based transseries are especially convenient for algorithms but form a proper subclass of all well-based transseries.
Hahn field	A field of generalized series $\sum a_m m$ with coefficients in a field, monomials in an ordered abelian group, and suitably well-ordered support. Transseries fields are built from iterated Hahn-field ideas plus exponential, logarithmic, and differential structure.
Hardy field	A field of germs at $+\infty$ closed under differentiation. Ordering is by eventual sign. It provides a functional interpretation of asymptotic comparison but does not naturally contain persistent oscillation such as $\sin x$.
H-field	An ordered differential field satisfying axioms abstracting the behavior of Hardy fields and transseries, including compatibility between order and derivation for infinitely large elements.
Hyperasymptotics	A hierarchy of exponentially improved approximations obtained by expanding the remainder after optimal truncation, then repeating the process. It is an analytic approximation method rather than merely a larger formal support.
Hyperseries	Generalized transseries allowing levels beyond the standard finite logarithmic-exponential hierarchy, designed to accommodate transfinite iteration of exponential and logarithmic behavior. Exact definitions depend on the chosen hyperserial framework.
Instanton sector	Physics terminology for a component multiplied by an exponential such as $e^{-kA/z}$. In this tutorial it is synonymous with an exponential transseries sector; no physical instanton interpretation is required.
Lateral sum	A Borel–Laplace sum taken just above or just below a singular direction. The two lateral sums may differ by exponentially small terms.
Large part	The sum L of monomials larger than 1 in the canonical decomposition $T = L + c + S$. “Purely large” means that all nonzero monomials are larger than 1.
LE series	“Logarithmic-exponential” generalized power series obtained by iterating appropriate exponential and logarithmic extensions. The standard field $\mathbb{T}$ is one canonical LE-type construction.

Continued on the next page

Term	Meaning in this tutorial
Logarithmic derivative	The quotient $f^\dagger = f'/f$. It converts products into sums and is often the fastest way to compare derivatives of transmonomials or to derive WKB corrections.
Median summation	A balanced prescription, built from lateral sums and part of the Stokes automorphism, that can preserve reality on a Stokes direction when the problem has the appropriate conjugation symmetry.
Monomial	A multiplicative unit of asymptotic size, such as x^a , $e^{-x}x^b$, or $\exp(-e^x + x^2)$. A transseries is a well-organized formal linear combination of such monomials.
Newton polygon	A geometric device for detecting balances from exponents or valuations of terms. In transseries problems it may be generalized to ordered monomial values rather than only integer powers.
Normal form	A canonical representation in which like monomials are collected and the support is ordered. Normalization is essential for reliable equality testing, truncation, and coefficient extraction.
Omega-series	A large field of surreal expressions generated from real numbers and the simplest infinite surreal ω using summation, exponential, logarithm, and composition subject to summability conditions. It extends the ordinary LE transseries viewpoint.
Optimal truncation	Stopping a divergent asymptotic series near its least term. The resulting error is often exponentially small and reveals the scale of the next transseries sector.
Perturbative sector	The sector with no exponentially small factor, usually denoted Φ_0 . It is often an ordinary formal power series and may be factorially divergent.
Ratio set	A finite set of small monomials whose nonnegative products generate a grid. Ratio sets provide a concrete calculus for witnessing and manipulating relative smallness.
Resonance	An integer relation among actions, exponents, or linearized eigenvalues that causes sectors to collide. Resonance may force powers of logarithms, mixed parameters, or modified recurrences.
Resurgence	The principle, made precise through endlessly continuable Borel transforms and singularity calculus, that different asymptotic sectors encode one another. Factorial large-order growth is one visible consequence.
Sector	Either a region of the complex independent-variable plane or a component of a transseries carrying a fixed exponential weight. Context normally makes the two meanings clear; “angular sector” and “instanton sector” remove the ambiguity.

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Term	Meaning in this tutorial
Small part	The sum S of monomials smaller than 1 in $T = L + c + S$. Formal Taylor, geometric, logarithmic, and binomial expansions are valid around $S = 0$ when the resulting support is summable.
Stokes automorphism	The formal transformation relating the two lateral resummations across a singular direction. In resurgent notation it is built from exponentials of alien derivatives, with convention-dependent normalization.
Stokes constant	A numerical invariant measuring a normalized Borel singularity or the associated jump between sectors. It is not the same thing as an arbitrary integration constant or transseries parameter.
Stokes direction	A direction along which a Borel singularity lies on the Laplace ray, or, in WKB language, a curve/direction where the relevant exponential phases have a specified alignment. Naming conventions vary, so the defining condition is primary.
Support	The set of monomials with nonzero coefficients in a generalized series. Its order-theoretic properties determine whether infinite addition, multiplication, composition, and differentiation are legitimate.
Summable family	A family of generalized series whose union of supports is suitably well-based and for which each monomial occurs in only finitely many members. This is the formal condition that allows the family to be added.
Surreal number	An element of Conway's proper class No . Every surreal has a Hahn-like normal form $\sum r_i \omega^{a_i}$. Suitable transseries can be evaluated at infinite surreal arguments, but this is different from Borel summation.
Transasymptotics	Reorganization of a transseries when infinitely many ordinary sectors become comparable, often by keeping a composite variable such as $\tau = \sigma e^{-A/z} z^\beta$ unexpanded and summing its dependence first.
Transmonomial	Another name for a transseries monomial, emphasizing that its scale may involve nested exponentials and logarithms rather than a single power.
Transseries	A formal generalized series on an ordered family of transmonomials, equipped with enough support control to permit algebra and often calculus. In analytic asymptotics the word also denotes a sectorial formal object intended for Borel–Laplace reconstruction. The surrounding structure must be specified.
Transseries parameter	A formal constant such as σ multiplying a homogeneous exponential sector. Boundary or initial data select its value; Stokes continuation may change its coordinate description.
Well-based series	A generalized series whose support is reverse well ordered when monomials are listed from largest to smaller. Well-based supports are more general than finitely generated grids.

Continued on the next page

Term	Meaning in this tutorial
Well ordering	The order condition guaranteeing an extremal element in every nonempty subset. Hahn-series authors often use well-ordered support with the valuation order; this tutorial uses reverse well ordering with the dominance order. The two conventions encode the same idea after reversing the order.
WKB transseries	A transseries built from exponential phases $\exp(\pm \int \sqrt{Q})$, algebraic prefactors, and divergent correction series for a differential equation with a large parameter or asymptotic independent variable.

Chapter summary

When terminology is uncertain, ask four questions: What is the independent limit? What is the ordered monomial group? What support condition is imposed? What analytic reconstruction, if any, is claimed? Those questions usually resolve apparent contradictions between different transseries dialects.

Notation and a cross-dialect concordance

D.1 Core notation used in this book

Notation	Meaning
$x \rightarrow +\infty$	Default large-variable limit in the formal LE chapters.
$z \rightarrow 0$	Default small-parameter limit in the Borel and resurgence chapters.
$f \prec g$	f is asymptotically smaller than g ; functionally, $f/g \rightarrow 0$.
$f \succ g$	f is asymptotically larger than g .
$f \asymp g$	f and g have comparable magnitude; neither ratio is forced to tend to zero.
$f \sim g$	Usually asymptotic equivalence, $f/g \rightarrow 1$. In a few cited sources $\sim$ may denote comparability, which is why this tutorial uses $\asymp$ for the latter.
$f \sim \sum a_n \phi_n$	f has the displayed asymptotic expansion. The symbol does not mean convergence of the infinite sum.
$\mathfrak{M}$ or $\mathfrak{G}$	An ordered multiplicative group of monomials.
$\mathfrak{m}, \mathfrak{n}$	Individual monomials.
$\text{supp } T$	The support of T : all monomials with nonzero coefficient.
$\text{lm}(T)$	Leading, hence largest, monomial of a nonzero transseries T .
$\text{lc}(T)$	Coefficient of the leading monomial.
$\mathbb{T}$	The standard logarithmic-exponential transseries field in the formal chapters. The exact construction is specified in Chapter 7 .
$T = L + c + S$	Canonical decomposition into purely large part, real constant, and purely small part.
$T = a\mathfrak{m}(1 + S)$	Leading-monomial factorization, with $a \neq 0$ and $S \prec 1$.
$f^\dagger$	Logarithmic derivative f'/f .

Continued on the next page

Notation	Meaning
$\mathcal{B}$	A Borel transform, with normalization stated locally.
$\mathcal{L}_\theta$	Laplace transform along the ray of angle θ .
$\mathcal{S}_{\theta^\pm}$	Upper or lower lateral Borel–Laplace sum along a singular direction.
Disc_θ	Difference of the two lateral values across direction θ , with order stated in context.
$\mathfrak{S}_\theta$	Stokes automorphism in direction θ .
Δ_A	Alien derivative associated with Borel singularity/action A .
A	An action, so that an exponential sector contains $e^{-A/z}$ or an equivalent large-variable scale.
σ	A transseries parameter, normally indexing a homogeneous sector.
Φ_k	The coefficient series in the k th exponential sector.
Γ	Gamma function; $\Gamma(n+1) = n!$.
Res	Complex residue.
No	The proper class of surreal numbers.
ω	The simplest infinite positive surreal number; also occasionally a root of unity in local algebraic examples, where this is stated explicitly.

D.2 Large-variable and small-parameter forms

Many formulas occur in two equivalent-looking conventions. Setting

$$x = \frac{1}{z}$$

translates the elementary scales as follows:

$$\begin{array}{ccc}
 x^{-n} & \longleftrightarrow & z^n, \\
 e^{-Ax} & \longleftrightarrow & e^{-A/z}, \\
 x^\beta e^{-Ax} & \longleftrightarrow & z^{-\beta} e^{-A/z}, \\
 \log x & \longleftrightarrow & -\log z.
 \end{array}$$

The translation is simple algebraically but not harmless analytically: angular sectors at $z = 0$ become sectors at $x = \infty$ with reversed angles, and branch choices for $\log z$ must be tracked.

D.3 Three common Borel normalizations

Suppose

$$\widehat{f}(z) = \sum_{n \geq 0} a_n z^{n+1}.$$

A convenient normalization in this book is

$$\mathcal{B}\widehat{f}(\xi) = \sum_{n \geq 0} \frac{a_n}{n!} \xi^n, \quad \mathcal{S}_\theta \widehat{f}(z) = \int_0^{e^{i\theta}\infty} e^{-\xi/z} \mathcal{B}\widehat{f}(\xi) \, d\xi.$$

Equally common are:

$$\widehat{g}(z) = \sum_{n \geq 0} b_n z^n, \quad \mathcal{B}_1 \widehat{g}(\xi) = b_0 \delta(\xi) + \sum_{n \geq 1} \frac{b_n}{(n-1)!} \xi^{n-1},$$

or a Laplace integral with $d\xi/z$. These conventions relocate factors of z , signs, and residues. Consequently, a bare formula for a Stokes jump or alien derivative is incomplete unless its Borel and Laplace normalizations are known.

D.4 Formal, asymptotic, and analytic assertions

The following statements are logically different:

$\mathcal{E}(\widehat{F}) = 0$	in a formal transseries field,
$f(z) \sim \widehat{F}(z)$	in a specified sector,
$f(z) = \mathcal{S}_\theta \widehat{F}(z)$	after a specified summation procedure,
$f_+(z) = f_-(z) \circ \mathfrak{S}_\theta$	as a resurgent continuation relation.

A proof of the first line does not automatically prove any later line. A rigorous theorem must supply hypotheses that connect them.

D.5 A dictionary among five viewpoints

Viewpoint	Primary object	Question it answers
Formal Hahn/LE	A well-based sum of ordered monomials	Are algebra, differentiation, composition, and recursive construction well defined coefficient by coefficient?
Asymptotic analysis	A hierarchy of finite remainder estimates	How accurately does a finite truncation approximate a function in a limit or sector?
Borel–resurgent	Analytic continuation and singularities of Borel transforms	Can a divergent transseries be reconstructed, and how do its sectors jump and determine large-order growth?
Hardy-field	A germ of a real function at $+\infty$	Which formal order and differential identities have an eventual functional realization?
Surreal	A value or function at infinite surreal arguments	Can the transseries be interpreted inside a vast ordered exponential field with canonical infinite and infinitesimal numbers?

D.6 A minimum data sheet for a transseries claim

Before trusting or communicating a transseries formula, record:

1. the limit and angular domain;

2. the coefficient field and monomial group;
3. the support condition (grid-based, well-based, multisummable, and so on);
4. the normalization of logarithms, powers, Borel transforms, and Laplace transforms;
5. the meaning of equality: formal, asymptotic, summed, germ equality, or surreal equality;
6. the status of free parameters and how boundary data choose them;
7. singular directions and the convention for Stokes jumps;
8. the theorem, if any, that realizes the formal object analytically.

This data sheet is deliberately repetitive: most avoidable mistakes in the subject arise from silently changing one of these items.

Chapter summary

Notation in transseries theory is not fully standardized. The safest practice is to define dominance, support orientation, Borel normalization, Stokes-jump orientation, and the intended notion of equality before using compact formulas. Once those choices are fixed, the different dialects fit together cleanly.

Bibliography

- [1] I. Aniceto, G. Başar, and R. Schiappa, *A primer on resurgent transseries and their asymptotics*, Physics Reports **809** (2019), 1–135.
- [2] M. Aschenbrenner, L. van den Dries, and J. van der Hoeven, *Asymptotic Differential Algebra and Model Theory of Transseries*, Annals of Mathematics Studies, vol. 195, Princeton University Press, 2017.
- [3] M. Aschenbrenner, L. van den Dries, and J. van der Hoeven, *On numbers, germs, and transseries*, in *Proceedings of the International Congress of Mathematicians 2018*, vol. 2, 19–42.
- [4] M. Aschenbrenner, L. van den Dries, and J. van der Hoeven, *The surreal numbers as a universal H -field*, Journal of the European Mathematical Society **21** (2019), no. 4, 1179–1199.
- [5] V. Bagayoko and V. Mantova, *Taylor expansions over generalised power series*, arXiv:2509.08473 (2025).
- [6] V. Bagayoko and J. van der Hoeven, *The hyperserial field of surreal numbers*, arXiv:2310.14873 (2023).
- [7] V. Bagayoko and J. van der Hoeven, *Surreal numbers as hyperseries*, arXiv:2310.14879 (2023).
- [8] W. Balser, *From Divergent Power Series to Analytic Functions: Theory and Application of Multisummable Power Series*, Lecture Notes in Mathematics, vol. 1582, Springer, 1994.
- [9] A. Berarducci and V. Mantova, *Surreal numbers, derivations and transseries*, Journal of the European Mathematical Society **20** (2018), no. 2, 339–390.
- [10] A. Berarducci and V. Mantova, *Transseries as germs of surreal functions*, Transactions of the American Mathematical Society **371** (2019), no. 5, 3549–3592.
- [11] J. H. Conway, *On Numbers and Games*, 2nd ed., A K Peters, 2001.
- [12] O. Costin, *Exponential asymptotics, transseries, and generalized Borel summation for analytic nonlinear rank-one systems of ordinary differential equations*, International Mathematics Research Notices (1995), no. 8, 377–417.

-
- [13] O. Costin, *On Borel summation and Stokes phenomena for rank-one nonlinear systems of ordinary differential equations*, Duke Mathematical Journal **93** (1998), no. 2, 289–344.
- [14] O. Costin, *On Borel summation and Stokes phenomena of nonlinear differential systems*, arXiv:math/0608408 (2006).
- [15] O. Costin, *Asymptotics and Borel Summability*, Chapman & Hall/CRC Monographs and Surveys in Pure and Applied Mathematics, vol. 141, 2009.
- [16] O. Costin and M. D. Kruskal, *On optimal truncation of divergent series solutions of nonlinear differential systems; Berry smoothing*, Proceedings of the Royal Society of London A **455** (1999), 1931–1956.
- [17] B. Dahn and P. Göring, *Notes on exponential-logarithmic terms*, Fundamenta Mathematicae **127** (1987), no. 1, 45–50.
- [18] R. B. Dingle, *Asymptotic Expansions: Their Derivation and Interpretation*, Academic Press, 1973.
- [19] D. Dorigoni, *An introduction to resurgence, trans-series and alien calculus*, Annals of Physics **409** (2019), article 167914.
- [20] J. Écalle, *Les Fonctions Résurgentes*, 3 vols., Publications Mathématiques d’Orsay, 1981.
- [21] J. Écalle, *Introduction aux fonctions analysables et preuve constructive de la conjecture de Dulac*, Hermann, 1992.
- [22] G. A. Edgar, *Transseries for beginners*, Real Analysis Exchange **35** (2009/2010), no. 2, 253–310; arXiv:0801.4877.
- [23] G. A. Edgar, *Transseries: composition, recursion, and convergence*, arXiv:0909.1259 (2009).
- [24] G. A. Edgar, *Fractional iteration of series and transseries*, arXiv:1002.2378 (2010).
- [25] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series, vol. 110, Cambridge University Press, 1986.
- [26] H. Hahn, *Über die nichtarchimedischen Größensysteme*, Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften, Mathematisch-Naturwissenschaftliche Klasse, Abteilung IIa, **116** (1907), 601–655.
- [27] G. H. Hardy, *Orders of Infinity: The “Infinitärcalcul” of Paul du Bois-Reymond*, Cambridge Tracts in Mathematics and Mathematical Physics, no. 12, Cambridge University Press, 1910.
- [28] S. Kuhlmann and M. Tressl, *Comparison of exponential-logarithmic and logarithmic-exponential series*, Mathematical Logic Quarterly **58** (2012), no. 6, 434–448.
- [29] L. van den Dries, J. van der Hoeven, and E. Kaplan, *Logarithmic hyperseries*, Transactions of the American Mathematical Society **372** (2019), no. 7, 5199–5241.

- [30] V. Mantova and M. Matusinski, *Surreal numbers with derivation, Hardy fields and transseries: a survey*, in *New Paths in Algebraic Geometry*, Contemporary Mathematics, vol. 697, American Mathematical Society, 2017, 265–290.
- [31] V. Mantova, *Monotonicity and a Taylor approximation theorem for transseries*, arXiv:2601.07747v2 (8 May 2026).
- [32] C. Mitschi and D. Sauzin, *Divergent Series, Summability and Resurgence I: Monodromy and Resurgence*, Lecture Notes in Mathematics, vol. 2153, Springer, 2016.
- [33] A. B. Olde Daalhuis, *Transseries analysis and Stokes phenomena*, unpublished lecture/project note, 2018.
- [34] F. W. J. Olver, *Asymptotics and Special Functions*, Academic Press, 1974; reprinted by A K Peters, 1997.
- [35] M. C. Schmeling, *Corps de transséries*, Ph.D. thesis, Université Paris VII, 2001.
- [36] L. van den Dries, A. Macintyre, and D. Marker, *Logarithmic-exponential power series*, Journal of the London Mathematical Society **56** (1997), no. 3, 417–434.
- [37] L. van den Dries, A. Macintyre, and D. Marker, *Logarithmic-exponential series*, Annals of Pure and Applied Logic **111** (2001), 61–113.
- [38] J. van der Hoeven, *Transseries and Real Differential Algebra*, Lecture Notes in Mathematics, vol. 1888, Springer, 2006.

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